Acceptance statistics (statistics)

Contract, version 0.10.0

This module builds, before the run and from a declared honest model, the acceptance set against which a protocol's observed outcome frequencies are tested, and states how often an honest run passes that test; the key-rate engine of the meat-core module reads the set as the acceptance set of the protocol declaration it solves. Its core fixes the frequency vector of a run and the two admissible geometries, a single frequency vector or a rational polytope in the probability simplex, the polytope possibly in canonical box form with recorded bounds per outcome. The core's main results are that every geometry passing its validation predicate is a non-empty, compact and convex subset of the simplex, as the engine's acceptance assumption requires (Theorem 1.5), and that over a non-empty canonical box, filling the coordinates of smallest coefficient first minimises a rational linear functional exactly, with a dual witness of the same value in the format the engine's certificate expects, while the outcomes with a positive upper bound include every outcome on which a point of the box puts weight (Theorem 3.1). The core also supplies the binomial tails in closed form and the honest envelope, a per-outcome range of honest probabilities of which one honest law is the degenerate case. The binomial-box submodule constructs the box and bounds the probability that an honest run fails the test; the singleton submodule evaluates a one-point test exactly. The declared premises are the honest-round model, an acceptance value recorded in every round, an honest law that some feasible state of the engine's declaration produces, and the output-state model of the test statistics (A-TSM), under which the acceptance outcome carries every per-round public choice of known bias made at a measuring station, a letter-side choice being a component of the letter drawn as declared (A-COIN); that the alphabet and round count are those of that declaration, and that the data are exact and the tails conservatively evaluated, are conditions the engine decides. The honest-round model and the premise on that feasible state bear on completeness, rate and usability, never on the soundness of a key.

Grey text, here and throughout, is non-normative: it explains, and nothing in it is assumed or proved.

The design notes (non-normative) explain why the module is built as it is and draw its processing flowchart. The dependencies between the sections are drawn in the section dependency matrix, and the graph index opens every section's dependency graph. The statements digest holds every definition, assumption, result and requirement without the proofs. How to read a contract explains this layout and its links.

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Main results

The module's most important results, those its overview names as what it establishes, with every assumption each rests on, directly or through the claims it uses. Each links to its statement and proof; the statements digest holds them in full.

Theorem 1.5 Validity of the admissible geometries

An acceptance set whose data pass validation (a probability vector, or a feasible linear system) is a non-empty, compact, admissible acceptance set (Definition "Admissible acceptance set" (meat-core/def:admissible-acceptance-set)). For a box of per-outcome bounds, validation means the lower sum to at most one and the upper to at least one.

Rests on A-ALPHABET

Theorem 3.1 Acceptance support and the exact box oracle

Over a non-empty canonical box, filling the coordinates of smallest coefficient first minimises a rational linear functional exactly, with a dual witness of the same value. The outcomes with a positive upper bound include every outcome on which a point of the box puts weight, and can include more.

Rests on A-ALPHABET

Theorem 9.2 Properties of the constructed box

For any admissible endpoint rule (count bounds meeting binomial tail budgets), the acceptance box has exact rational bounds and contains every law within the honest envelope's bounds. It is non-empty and compact and, by construction, an admissible acceptance set (Definition "Admissible acceptance set" (meat-core/def:admissible-acceptance-set)).

Rests on A-ALPHABET

Theorem 10.2 The completeness bound

If each honest round's conditional outcome law stays inside the envelope, an honest run fails the acceptance test with probability at most ε^*, the sum of the binomial tails at the envelope's ends beyond the box's count bounds, and ε^* is at most the completeness budget. No independence is needed.

Rests on A-ALPHABET, A-HON-ENV

Theorem 18.1 Completeness of a singleton acceptance set

Under i.i.d. honest rounds a singleton acceptance set accepts with its point's multinomial probability. That is 0, with no completeness budget below one, if n times a coordinate is no integer or the point leaves the law's support; else the least budget is one minus it, zero only for one point mass.

Rests on A-ALPHABET, A-HON

Assumptions at a glance

Every assumption this contract states or its claims assume, with its category and the claims of this contract that assume it directly. An assumption of another contract links into that contract's PDF.

Assumption	Category	Title	Stated as	Assumed directly by
A-ALPHABET	protocol structure	An acceptance value recorded in every round	Assumption 5.3	Thm. 1.5, Thm. 3.1, Rem. 8.4, Thm. 9.2, Thm. 10.2, Thm. 11.1, Thm. 12.4, Thm. 18.1
A-TSM	protocol structure	Output-state test statistics	reductions/ass: tsm	Rem. 8.4, Rem. 10.4
A-HON	honest model	Honest i.i.d. rounds	Assumption 5.1	Rem. 10.4, Thm. 18.1
A-HON-ENV	honest model	Honest rounds inside a declared envelope	Assumption 5.2	Thm. 10.2, Rem. 10.4
A-REACH	honest model	An honest law produced by a feasible state	Assumption 5.4	Thm. 11.1, Rem. 11.2

Notation

The symbols of the repository's notation catalogue (spec/notation.md) that this contract uses, in the catalogue's order, with their meaning.

Symbol	Meaning
$\mathfrak{D}$	Protocol declaration $\big(A, \tau_A;\, Q, \mathbb{S};\, \{M_x\};\, \{F_v\};\, \mathfrak{P};\, a_A, a_B;\, \phi;\, r, \mathsf{h};\, \mathfrak{L}\big)$ — the tuple of meat-core/def:declaration: the party partition $\mathfrak{P}$ is a slot, and the marginal slot τ_A takes a state or the no-marginal value (trusted- vs untrusted-source mode)
$\widehat{C}$, $\widehat{\mathcal{C}}$	The acceptance-statistic register $\widehat{C}$ and its alphabet $\widehat{\mathcal{C}}$
$\mathcal{S}_{\mathrm{acc}}$	Acceptance set
$\mathbf{\Phi}$	The test-statistics map $\mathcal{L}(AQ) \to \mathbb{R}^{\widehat{\mathcal{C}}}$
$\mathcal{W}$	Feasible set of sector-block-diagonal marginals: $\operatorname{tr}\omega = 1$ always; $\omega_A = \tau_A$ in trusted-source mode only; $\mathfrak{L}$ if declared (meat-core/def:feasible)
$\underline{h}_\alpha$	Computable relaxation: lower bound on $h_\alpha^{\uparrow,\mathrm{QKD}}$; written `h_low` in plain text
Ω_{AT}	Acceptance event $\mathrm{freq}_{\hat{C}_1^n} \in \mathcal{S}_{\mathrm{acc}}$
D	Classical KL divergence, base 2; also the quantum relative entropy on operators
T	Transpose, X^{T}

Terms

The phrases this contract defines, and the terms of other contracts it uses, each with a one-line gloss and the definition that fixes it. In the text, a linked term opens its definition.

Term	Meaning	Defined in
admissible endpoint rule	A map from a round count, a success probability and two tail budgets to two count endpoints whose binomial tails meet those budgets, with fixed values at a zero budget and at a degenerate mass.	Definition 7.1
admissible geometry	A tag, singleton or polytope, together with the data of that tag; an acceptance set is declared as one.	Definition 1.4
admissible input	The data the construction accepts: alphabet, round count, honest model, budget, constrained outcomes, allocation, and choice of rule and evaluation, the numbers exact and the alphabet and round count those of the consuming declaration.	Definition 7.1

Term	Meaning	Defined in
canonical box form	A polytope geometry that records a lower and an upper bound per outcome and whose rows are exactly the two bounds of each outcome.	Definition 1.4
conservative evaluation	An evaluation of the acceptance-set construction on exact rational data that decides every tail comparison exactly or against a bound rounding the tail upward.	Definition 4.3
construction map	The map that turns an admissible input and an admissible endpoint rule into the acceptance box: the guarded count endpoints over n on the constrained outcomes, 0 and 1 elsewhere, intersected with the simplex.	Definition 9.1
consumer	a module other than a supplier that reads declared data	foundations/def: declared-data
count	The number of rounds of an outcome string in which a given outcome occurs.	Definition 1.1
degenerate envelope	An honest envelope whose lower and upper vectors are equal, hence one honest law.	Definition 4.1
envelope set	The laws on the simplex that lie between the lower and the upper vector of an honest envelope.	Definition 4.1
exact binomial quantile rule	The endpoint rule that returns the extremal endpoints, the extremal binomial quantiles of the two tail budgets.	Definition 8.1

Term	Meaning	Defined in
exact island	the computations of the engine, its instances' declarations included, by which a consumer of data or the declarer about to declare them decides whether they satisfy the conditions their declaration states, a declarer computes and decides the reference of a measurement family, a consumer's replacement of an emitted scalar bound is decided to lie on the bound's weakening side, a producer constructs the operator data it emits, the acceptance side of the optimisation is constructed and certified, or a lower bound on a linear minimum over the feasible side of the optimisation is certified	foundations/def: exact-island
extremal endpoints	The largest lower count and the smallest upper count in 0 to n whose binomial tails meet their allocated budgets.	Definition 8.1
frequency vector	The counts of an outcome string divided by its length n, one coordinate per outcome.	Definition 1.1
guarded endpoints	A rule's two count endpoints after the guard: the lower one capped at the floor of n times the envelope's lower value, the upper one raised to at least the ceiling of n times its upper value.	Definition 7.1
honest envelope	A lower and an upper exact rational probability per outcome, the declared range of honest probabilities.	Definition 4.1
honest-containment guard	The widening of a rule's count interval that makes it contain n times the whole envelope interval of the outcome.	Definition 7.1
instance	the party that fixes one protocol's device data, protocol data and run configuration	foundations/def: declared-data
matched alphabet and round count	An acceptance alphabet, with a fixed enumeration, and a round count that are those of the declaration the consumer solves.	Definition 1.3

Term	Meaning	Defined in
monotone allocation rule	A map assigning to each completeness budget an allocation within it, every allocated tail budget non-decreasing in the budget.	Definition 12.1
monotone in the allocation	Said of an endpoint rule whose lower endpoint never falls and whose upper endpoint never rises as its own tail budget grows.	Definition 7.1
outcome alphabet	The finite ordered set of device outcomes that a private setting is joined to.	reductions/def: joint-family
point form	The honest model given as one law rather than as an envelope.	Definition 4.1
round-homogeneous	one fixed list of objects describes every round, and every round behaves as they describe whatever the other rounds do	foundations/def: round-homogeneity
side-wise reading	The convention that a single-value claim of this submodule is read at the envelope's lower value for the lower endpoint and at its upper value for the upper endpoint.	Definition 7.1
uniform rule	The allocation rule that splits the completeness budget equally over the two tails of every constrained outcome.	Definition 12.1
validation predicate	The condition on a geometry's data that validation checks: a probability vector for a singleton, a feasible linear system for a polytope.	Definition 1.4

Part I
Mathematics

1 Frequencies and acceptance geometries

A run's outcome counts divided by the round count give a frequency vector in the probability simplex, and the acceptance test asks whether that vector lies in a declared set. The alphabet and the round count are those of the declaration the consuming engine solves. The set is an admissible geometry, a single frequency vector or a rational polytope in the simplex, the polytope possibly in canonical box form with recorded lower and upper bounds per outcome. The main result, Theorem 1.5, is that a geometry passing its validation predicate is a non-empty, compact and convex subset of the simplex; for the canonical box form the predicate is a closed-form condition on the sums of the bounds.

An outcome's count is the number of rounds in which it occurs, and the frequency vector of a run of n rounds is its counts divided by n. The acceptance test asks whether this vector lies in the acceptance set.

The event $\Omega_{\mathrm{AT}} = \{\mathrm{freq}_{\hat{c}_1^n} \in \mathcal{S}_{\mathrm{acc}}\}$ that a run of n rounds passes the acceptance test, the acceptance event of the key-length post-processing (Definition "Fixed-length post-processing model" (meat-core/def:postprocessing-model); the symbols are listed in `spec/notation.md#35-optimization-and-key-length-objects`), is defined on the frequency vector fixed here.

Definition 1.1 (Frequency vector and counts)**.** For an outcome string $\hat{c}_1^n = (\hat{c}_1, \ldots, \hat{c}_n) \in \widehat{\mathcal{C}}^n$ and $\hat{c} \in \widehat{\mathcal{C}}$ put

$$\mathsf{N}_{\hat{c}}(\hat{c}_1^n) \coloneqq \big|\{\, j \in \{1, \ldots, n\} : \hat{c}_j = \hat{c} \,\}\big|, \qquad \mathrm{freq}_{\hat{c}_1^n}(\hat{c}) \coloneqq \tfrac{1}{n}\, \mathsf{N}_{\hat{c}}(\hat{c}_1^n) : \tag{1.1}$$

the *count* of the outcome $\hat{c}$ and the *frequency vector* of the string. Then $\mathrm{freq}_{\hat{c}_1^n} \in \Delta(\widehat{\mathcal{C}})$ for every string, and its coordinates lie in $\frac{1}{n}\{0, 1, \ldots, n\}$.

Used by Lem. 1.2, Def. 7.1, Thm. 10.2, Lem. 12.2, Lem. 14.1, Thm. 18.1 · newly derived · accepted

Every frequency vector has non-negative coordinates summing to one, since each round contributes exactly one outcome. Intersecting a constraint set with the probability simplex therefore never changes the probability that a run passes the test, so simplex constraints cost no completeness.

Lemma 1.2 (Frequency vectors in the probability simplex)**.** Let $n \geq 1$, and write $\Delta(\widehat{\mathcal{C}})$ for the set of $\mathbf{q} \in \mathbb{R}^{\widehat{\mathcal{C}}}$ with non-negative coordinates summing to one.

(i) For every outcome string $\hat{c}_1^n \in \widehat{\mathcal{C}}^n$, every count $\mathsf{N}_{\hat{c}}(\hat{c}_1^n)$ of Definition 1.1 is a non-negative integer, at most n of them are non-zero, and $\sum_{\hat{c}\in\widehat{\mathcal{C}}} \mathsf{N}_{\hat{c}}(\hat{c}_1^n) = n$. Hence $\mathrm{freq}_{\hat{c}_1^n} \in \Delta(\widehat{\mathcal{C}})$.

(ii) For every set $S \subseteq \mathbb{R}^{\widehat{\mathcal{C}}}$ and every random string $\widehat{C}_1^n$ with values in $\widehat{\mathcal{C}}^n$, the sets $\{\mathrm{freq}_{\widehat{C}_1^n} \in S \cap \Delta(\widehat{\mathcal{C}})\}$ and $\{\mathrm{freq}_{\widehat{C}_1^n} \in S\}$ of sample points are equal. In particular, whenever either is an event,

$$\Pr\left[\mathrm{freq}_{\widehat{C}_1^n} \in S \cap \Delta(\widehat{\mathcal{C}})\right] = \Pr\left[\mathrm{freq}_{\widehat{C}_1^n} \in S\right]. \tag{1.2}$$

Uses Def. 1.1 · *Used by* Thm. 10.2 · newly derived · accepted

Proof. *(i).* Fix $\hat{c}_1^n$. By (1.1), $\mathsf{N}_{\hat{c}}(\hat{c}_1^n)$ is the cardinality of the set $J_{\hat{c}} \coloneqq \{j \in \{1,\dots,n\} : \hat{c}_j = \hat{c}\}$, a non-negative integer. Each round $j \in \{1,\dots,n\}$ lies in $J_{\hat{c}}$ for exactly one $\hat{c}$, namely $\hat{c} = \hat{c}_j$, so the sets $J_{\hat{c}}$ for $\hat{c} \in \widehat{\mathcal{C}}$ are pairwise disjoint with union $\{1,\dots,n\}$. Hence at most n of them are non-empty, and summing their cardinalities gives $\sum_{\hat{c}\in\widehat{\mathcal{C}}} \mathsf{N}_{\hat{c}}(\hat{c}_1^n) = n$. Dividing by $n \geq 1$, the coordinates $\mathrm{freq}_{\hat{c}_1^n}(\hat{c}) = \frac{1}{n}\mathsf{N}_{\hat{c}}(\hat{c}_1^n)$ are non-negative and sum to one, which is $\mathrm{freq}_{\hat{c}_1^n} \in \Delta(\widehat{\mathcal{C}})$.

(ii). The inclusion $\{\mathrm{freq}_{\widehat{C}_1^n} \in S \cap \Delta(\widehat{\mathcal{C}})\} \subseteq \{\mathrm{freq}_{\widehat{C}_1^n} \in S\}$ holds because $S \cap \Delta(\widehat{\mathcal{C}}) \subseteq S$. Conversely, at every sample point the value of $\widehat{C}_1^n$ is a string in $\widehat{\mathcal{C}}^n$, so by (i) the frequency vector there lies in $\Delta(\widehat{\mathcal{C}})$; if it also lies in S, it lies in $S \cap \Delta(\widehat{\mathcal{C}})$. The two sets are therefore equal, and equal events have equal probability, which is (1.2). □

An alphabet and a round count are matched when the alphabet, with a fixed enumeration, is the acceptance alphabet of the declaration the consumer solves and the round count is the same. The acceptance set then lives on the consumer's coordinates and is normalised by its round count.

The definition relates this module's inputs to the declaration the consumer solves, so no input of this module decides it; the agreement of outcome alphabets across artefacts is checked where the consumer validates its inputs (`spec/architecture.md#36-bundle-validation-is-input-validation-not-a-module`; Requirement 6.4), and the round count is one run parameter that this module and the consumer both read.

Definition 1.3 (Alphabet and round count matched to the consuming declaration)**.** An acceptance alphabet $\widehat{\mathcal{C}}$ and a round count n are a *matched alphabet and round count* when $\widehat{\mathcal{C}}$ is the finite, non-empty acceptance alphabet of the same declaration $\mathfrak{D}$ (Definition "Protocol declaration" (protocol-compiler/def:declaration)) the consumer solves, the codomain of ϕ, with a fixed enumeration, and n is the same total round count.

Used by Thm. 1.5, Thm. 3.1, Req. 6.4, Def. 7.1, Rem. 8.4, Thm. 11.1, Thm. 18.1 · newly derived · accepted

An admissible geometry is a singleton, one rational vector, or a polytope, rational linear inequalities intersected with the probability simplex, each with its validation predicate. A polytope in canonical box form is given by recorded lower and upper bounds per outcome.

The acceptance test asks whether the frequency vector of a run lies in a declared set. The set is declared in one of two forms, and a polytope may in addition record per-outcome bounds from which its rows are built.

Definition 1.4 (Admissible geometries and the canonical box form)**.** An *admissible geometry* is a pair consisting of a *tag* and its *data*, drawn from exactly the following two cases, each with its data, the set Γ the data denote and its *validation predicate*. There is no third tag.

singleton**.** Data: a vector $\mathbf{q}^* \in \mathbb{Q}^{\widehat{\mathcal{C}}}$. Denotation $\Gamma \coloneqq \{\mathbf{q}^*\}$. *Validation predicate:* $q^*(\hat{c}) \geq 0$ for every $\hat{c}$ and $\sum_{\hat{c}} q^*(\hat{c}) = 1$.

polytope**.** Data: an integer row count $m_{\mathrm{row}} \geq 0$, a matrix $B \in \mathbb{Q}^{m_{\mathrm{row}} \times \widehat{\mathcal{C}}}$ and a vector $\mathbf{b} \in \mathbb{Q}^{m_{\mathrm{row}}}$. Denotation

$$\Gamma \coloneqq \{\mathbf{q} \in \Delta(\widehat{\mathcal{C}}) : \ B\mathbf{q} \leq \mathbf{b}\}, \tag{1.3}$$

the inequality componentwise. *Validation predicate:* feasibility of the linear system $\{\mathbf{q} \geq 0, \ \sum_{\hat{c}} q(\hat{c}) = 1, \ B\mathbf{q} \leq \mathbf{b}\}$ over the rationals.

A polytope geometry is in *canonical box form* when it carries, in addition, a pair $\mathbf{q}^-, \mathbf{q}^+ \in \mathbb{Q}^{\widehat{\mathcal{C}}}$ with $0 \leq \mathbf{q}^- \leq \mathbf{q}^+$ componentwise, and its $(B, \mathbf{b})$ are exactly

$$m_{\mathrm{row}} = 2m, \qquad \begin{array}{ll} \text{row } i: & \delta_{\hat{c}^{(i)}}^{\mathsf{T}}, \qquad \text{right-hand side } q^+(\hat{c}^{(i)}), \\ \text{row } m+i: & -\delta_{\hat{c}^{(i)}}^{\mathsf{T}}, \quad \text{right-hand side } -q^-(\hat{c}^{(i)}), \end{array} \qquad i = 1, \ldots, m, \tag{1.4}$$

where $m \coloneqq |\widehat{\mathcal{C}}|$, $\hat{c}^{(1)}, \ldots, \hat{c}^{(m)}$ is the declaration's enumeration of $\widehat{\mathcal{C}}$ and $\delta_{\hat{c}}$ is the standard basis vector of $\mathbb{R}^{\widehat{\mathcal{C}}}$ at $\hat{c}$. Row indices are integers, never alphabet elements; equivalently

$$\Gamma = \{\mathbf{q} \in \Delta(\widehat{\mathcal{C}}) : \ q^-(\hat{c}) \leq q(\hat{c}) \leq q^+(\hat{c}) \ \ \forall \hat{c} \in \widehat{\mathcal{C}}\}. \tag{1.5}$$

Used by Thm. 1.5, Thm. 3.1, Def. 4.1, Req. 6.1, Req. 6.2, Def. 9.1, Thm. 9.2, Lem. 12.2 · newly derived · implemented · *Code* certiqs_key.statistics.taxonomy:GeometryTag

An acceptance set whose data pass validation (a probability vector, or a feasible linear system) is a non-empty, compact, admissible acceptance set (Definition "Admissible acceptance set" (meat-core/def:admissible-acceptance-set)). For a box of per-outcome bounds, validation means the lower sum to at most one and the upper to at least one.

The consuming engine assumes that its acceptance set is a closed convex subset of the probability simplex. The alphabet is the finite, enumerated one of Definition 1.3, so the simplex is compact; a polytope cuts it with finitely many closed half-spaces, and along the segment from the lower to the upper bound vector of a canonical box the coordinate sum moves continuously from the sum of the lower bounds to that of the upper bounds, which gives the predicate in closed form.

Theorem 1.5 (Validity of the admissible geometries). Let $\widehat{\mathcal{C}}$ and n be a matched alphabet and round count (Definition 1.3), and let (tag, data) be an admissible geometry with denotation Γ.

(i) If the validation predicate holds, Γ is a non-empty, closed, convex and compact subset of $\Delta(\widehat{\mathcal{C}})$. In particular Definition "Admissible acceptance set" (meat-core/def:admissible-acceptance-set), that $\mathcal{S}_{\text{acc}}$ is a closed convex subset of the probability simplex on $\widehat{\mathcal{C}}$, holds by construction for every admissible geometry whose validation predicate holds.

(ii) For the tag singleton, the validation predicate is necessary and sufficient for Γ to be a non-empty subset of $\Delta(\widehat{\mathcal{C}})$.

(iii) For the tag polytope in canonical box form with $0 \le \mathbf{q}^- \le \mathbf{q}^+$, the validation predicate, the condition $\Gamma \neq \varnothing$ and the closed-form condition

$$\sum_{\hat{c}\in\widehat{\mathcal{C}}} q^-(\hat{c}) \;\le\; 1 \;\le\; \sum_{\hat{c}\in\widehat{\mathcal{C}}} q^+(\hat{c}) \tag{1.6}$$

are equivalent.

Uses Def. 1.3, Def. 1.4 · *Assumes* A-ALPHABET · *Used by* Thm. 3.1, Thm. 9.2, Thm. 11.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.statistics.taxonomy:validate_geometry · *Oracles* registry_statistics_acceptance.py

Proof. (i). $\Delta(\widehat{\mathcal{C}})$ is the intersection of the closed half-spaces $\{q(\hat{c}) \ge 0\}$ with the hyperplane $\{\sum_{\hat{c}} q(\hat{c}) = 1\}$, hence closed and convex; it is bounded, since $0 \le q(\hat{c}) \le 1$ for every coordinate, hence compact ($\widehat{\mathcal{C}}$ is finite, so $\mathbb{R}^{\widehat{\mathcal{C}}}$ is finite-dimensional and Heine–Borel applies). For polytope, $\{B\mathbf{q} \le \mathbf{b}\}$ is a finite intersection of closed half-spaces, so closed and convex, and Γ is the intersection of two closed convex sets, hence closed and convex; being a closed subset of the compact $\Delta(\widehat{\mathcal{C}})$, it is compact. For singleton, a one-point set is closed, convex and compact, and lies in $\Delta(\widehat{\mathcal{C}})$ exactly when the validation predicate holds. Non-emptiness follows from the validation

predicate in every case: for singleton by (ii); for a general polytope because the predicate is rational feasibility of $\{\mathbf{q} \geq 0,\ \sum_{\hat{c}} q(\hat{c}) = 1,\ B\mathbf{q} \leq \mathbf{b}\}$ and a feasible point is by (1.3) an element of Γ; and in canonical box form equivalently by the closed form (iii).

(ii). $\{\mathbf{q}^*\} \subseteq \Delta(\widehat{\mathcal{C}})$ and $\{\mathbf{q}^*\} \neq \varnothing$ hold if and only if $\mathbf{q}^* \in \Delta(\widehat{\mathcal{C}})$, which is the predicate verbatim.

(iii). We show that the predicate gives $\Gamma \neq \varnothing$, that $\Gamma \neq \varnothing$ gives (1.6), and that (1.6) gives the predicate. By (1.4) the rows $B\mathbf{q} \leq \mathbf{b}$ read $\mathbf{q}^- \leq \mathbf{q} \leq \mathbf{q}^+$ componentwise.

Predicate $\Rightarrow$ *non-emptiness.* A rational solution of the system of the predicate is by (1.3) an element of Γ.

Non-emptiness $\Rightarrow$ (1.6). If $\mathbf{q} \in \Gamma$ then $\mathbf{q}^- \leq \mathbf{q} \leq \mathbf{q}^+$ componentwise and $\sum_{\hat{c}} q(\hat{c}) = 1$; summing the two componentwise inequalities gives (1.6).

(1.6) $\Rightarrow$ *predicate.* Put, for $\theta \in [0,1]$, $\mathbf{q}(\theta) \coloneqq \mathbf{q}^- + \theta(\mathbf{q}^+ - \mathbf{q}^-)$. Since $\mathbf{q}^- \leq \mathbf{q}^+$, every $\mathbf{q}(\theta)$ satisfies $\mathbf{q}^- \leq \mathbf{q}(\theta) \leq \mathbf{q}^+$ and, using $\mathbf{q}^- \geq 0$, also $\mathbf{q}(\theta) \geq 0$. Write $S^- \coloneqq \sum_{\hat{c}} q^-(\hat{c})$ and $S^+ \coloneqq \sum_{\hat{c}} q^+(\hat{c})$, so that $S^- \leq 1 \leq S^+$ by (1.6) and $\sum_{\hat{c}} q(\theta)(\hat{c}) = S^- + \theta(S^+ - S^-)$. Put $\theta^* \coloneqq (1 - S^-)/(S^+ - S^-)$ if $S^+ > S^-$, and $\theta^* \coloneqq 0$ if $S^+ = S^-$, in which case $S^- = S^+ = 1$. In both cases $\theta^* \in [0,1]$ and $\sum_{\hat{c}} q(\theta^*)(\hat{c}) = 1$, and θ^* is rational because $\mathbf{q}^-$ and $\mathbf{q}^+$ are (Definition 1.4). So $\mathbf{q}(\theta^*)$ is a rational vector with $\mathbf{q}(\theta^*) \geq 0$, $\sum_{\hat{c}} q(\theta^*)(\hat{c}) = 1$ and $\mathbf{q}^- \leq \mathbf{q}(\theta^*) \leq \mathbf{q}^+$, a rational solution of the system of the predicate. □

The two tags are exactly the two geometries over which the consuming engine certifies the minimum of a linear functional, and the canonical box form is a sub-form of the polytope, not a third tag. A consumer that ignores the recorded bounds still reads a well-formed polytope.

Remark 1.6 (The two tags and the consumer's certified geometries). The two tags of Definition 1.4 correspond one to one to the two instantiations of Remark "Exact minimisation on the acceptance side" (meat-core/frank-wolfe/rem:acc-side), the geometries for which that remark certifies the acceptance-side minimum, the minimum of a linear functional over the acceptance set: (a) singleton, (b) polytope $\{\mathbf{q} \geq 0,\ \sum_{\hat{c}} q(\hat{c}) = 1,\ B\mathbf{q} \leq \mathbf{b}\}$ for exact B and $\mathbf{b}$. That remark certifies no set of any other form. The canonical box form is a *sub-form* of polytope, not a third tag: it adds a recorded pair $(\mathbf{q}^-, \mathbf{q}^+)$ (Requirement 6.2) that licenses the closed-form oracle of Theorem 3.1, and a consumer that ignores the recorded pair still reads a well-formed polytope.

newly derived · accepted

2 Binomial tails

The count of one outcome over independent rounds with one law is binomial, and the box construction reads its two tails. They are regularised incomplete beta integrals, with every boundary case of the count stated and the upper tail non-decreasing in the success probability (Theorem 2.1). The two strict

tails the construction compares with a budget follow by a shift of the count (Corollary 2.2), and at a rational success probability both can be decided exactly (Lemma 2.3).

The upper and lower tails of a binomial count are regularised incomplete beta integrals, with every boundary case of the count stated. The upper tail is non-decreasing in the success probability.

The count of one outcome over independent rounds with one law is binomial, and the box construction compares its two tails with their budgets. The theorem writes both tails as regularised incomplete beta integrals, deriving the normalisation of the beta integral inside the proof.

Theorem 2.1 (Binomial tails in closed form)**.** Let $n \geq 1$, $p \in [0,1]$, and let N be a binomial random variable with parameters (n,p), i.e. $\Pr[\mathsf{N} = i] = \binom{n}{i} p^i (1-p)^{n-i}$ for $0 \leq i \leq n$. Write

$$\mathcal{I}_x(a,b) \coloneqq \frac{\int_0^x t^{a-1}(1-t)^{b-1}\,\mathrm{d}t}{\int_0^1 t^{a-1}(1-t)^{b-1}\,\mathrm{d}t}, \qquad x \in [0,1],\ a,b \geq 1. \tag{2.1}$$

Then

$$\Pr[\mathsf{N} \geq k] = \frac{n!}{(k-1)!\,(n-k)!} \int_0^p t^{k-1}(1-t)^{n-k}\,\mathrm{d}t = \mathcal{I}_p(k, n-k+1), \quad 1 \leq k \leq n, \tag{2.2}$$

$$\Pr[\mathsf{N} \leq k] = 1 - \Pr[\mathsf{N} \geq k+1] = \mathcal{I}_{1-p}(n-k, k+1), \qquad 0 \leq k \leq n-1, \tag{2.3}$$

together with the boundary values $\Pr[\mathsf{N} \geq k] = 1$ for $k \leq 0$, $\Pr[\mathsf{N} \geq k] = 0$ for $k > n$, $\Pr[\mathsf{N} \leq k] = 0$ for $k < 0$ and $\Pr[\mathsf{N} \leq k] = 1$ for $k \geq n$. Moreover $p \mapsto \Pr[\mathsf{N} \geq k]$ is non-decreasing on $[0,1]$ for every $k \geq 1$.

Used by Cor. 2.2, Lem. 2.3, Rem. 8.4, Lem. 15.1, Lem. 15.2 · newly derived · implemented · *Code* certiqs_key.statistics.tails: binomial_cdf · *Oracles* registry_statistics_acceptance.py

Proof. Idea. The upper tail is a polynomial in p vanishing at $p = 0$ whose derivative telescopes to a single term; integrating that term gives the first form of (2.2), and its value at $p = 1$ gives the normalisation. The lower tail is the upper tail of $n - \mathsf{N}$.

Step 1 (the derivative of the upper tail). Fix k with $1 \leq k \leq n$ and define the polynomial $G(p) \coloneqq \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$, so that $G(p) = \Pr[\mathsf{N} \geq k]$ and $G(0) = 0$ (every summand carries p^i with $i \geq k \geq 1$). Set

$$a_i \coloneqq n\binom{n-1}{i-1} p^{i-1}(1-p)^{n-i}, \qquad 1 \leq i \leq n,$$

a polynomial in p for every such i: the exponent $n - i$ is non-negative throughout, so no negative power of $1-p$ occurs anywhere in this proof and

no division is performed. Differentiate the summands of G separately in the two cases.

For $k \le i \le n-1$, the product rule and the absorption identities $i\binom{n}{i} = n\binom{n-1}{i-1}$ and $(n-i)\binom{n}{i} = n\binom{n-1}{i}$ give

$$\frac{\mathrm{d}}{\mathrm{d}p}\Big[\binom{n}{i}p^i(1-p)^{n-i}\Big] = i\binom{n}{i}p^{i-1}(1-p)^{n-i} - (n-i)\binom{n}{i}p^i(1-p)^{n-i-1} = a_i - a_{i+1},$$

where $i \ge k \ge 1$ makes the first power non-negative, $i \le n-1$ makes the second one non-negative, and $a_{i+1} = n\binom{n-1}{i}p^i(1-p)^{n-i-1}$ by substituting $i+1$ for i in the definition of a_i, which is legitimate because $i+1 \le n$. For $i = n$ the summand is p^n, whose derivative is $np^{n-1} = a_n$; this is the case the general formula would have written with a factor $(1-p)^{-1}$, and it is treated separately for exactly that reason. Summing the two cases, the sum telescopes,

$$G'(p) = \sum_{i=k}^{n-1}\big(a_i - a_{i+1}\big) + a_n = \big(a_k - a_n\big) + a_n = a_k,$$

which is also correct at $k = n$, where the telescoping sum is empty and $G'(p) = a_n = a_k$.

Step 2 (the integral form and its normalisation). Spelling a_k out, $G'(p) = n\binom{n-1}{k-1}p^{k-1}(1-p)^{n-k}$, and by the fundamental theorem of calculus with $G(0) = 0$,

$$G(p) = n\binom{n-1}{k-1}\int_0^p t^{k-1}(1-t)^{n-k}\,\mathrm{d}t, \qquad n\binom{n-1}{k-1} = \frac{n!}{(k-1)!\,(n-k)!}. \tag{2.4}$$

This is the first equality of (2.2). For the second, evaluate (2.4) at $p = 1$: there $\mathsf{N} = n$ with probability one and $k \le n$, so $G(1) = 1$, whence $\int_0^1 t^{k-1}(1-t)^{n-k}\,\mathrm{d}t = \frac{(k-1)!\,(n-k)!}{n!}$; dividing (2.4) by this value gives exactly $\mathcal{I}_p(k, n-k+1)$ as defined in (2.1), with $a = k \ge 1$ and $b = n-k+1 \ge 1$.

Step 3 (the lower tail). For (2.3), note first that $\{\mathsf{N} \le k\}$ and $\{\mathsf{N} \ge k+1\}$ partition the sample space. Next, $n - \mathsf{N}$ is binomial with parameters $(n, 1-p)$, because $\Pr[n-\mathsf{N} = i] = \Pr[\mathsf{N} = n-i] = \binom{n}{n-i}p^{n-i}(1-p)^i = \binom{n}{i}(1-p)^i p^{n-i}$. Hence, for $0 \le k \le n-1$ (so that $n-k \ge 1$), $\Pr[\mathsf{N} \le k] = \Pr[n-\mathsf{N} \ge n-k] = \mathcal{I}_{1-p}(n-k, k+1)$ by (2.2) applied to $n - \mathsf{N}$.

Step 4 (boundary values and monotonicity). The boundary values are immediate from $0 \le \mathsf{N} \le n$. Monotonicity in p follows from $G'(p) = n\binom{n-1}{k-1}p^{k-1}(1-p)^{n-k} \ge 0$ on $[0,1]$ for $1 \le k \le n$; for $k > n$ the tail is the constant 0 by the boundary values. □

For every count k from 0 to n the strict tails of a binomial count N, the probabilities that N lies below k and above k, are regularised incomplete beta integrals, zero at $k = 0$ and at $k = n$ respectively. These are the two tails the endpoint rules compare with their budgets.

An endpoint rule compares the probability that the count falls strictly below a lower endpoint, or strictly above an upper one, with a budget. Both strict tails follow from Theorem 2.1 by a shift of the count.

Corollary 2.2 (The two strict binomial tails)**.** Let $n \geq 1$, $p \in [0,1]$ and let N be binomial with parameters (n,p), as in Theorem 2.1. Then, for $0 \leq k \leq n$,

$$\begin{aligned} \Pr[\mathsf{N} < k] = \Pr[\mathsf{N} \leq k-1] &= \begin{cases} 0, & k = 0, \\ \mathcal{I}_{1-p}(n-k+1,k), & 1 \leq k \leq n, \end{cases} \\ \Pr[\mathsf{N} > k] = \Pr[\mathsf{N} \geq k+1] &= \begin{cases} \mathcal{I}_p(k+1,n-k), & 0 \leq k \leq n-1, \\ 0, & k = n. \end{cases} \end{aligned} \tag{2.5}$$

Uses Thm. 2.1 · *Used by* Lem. 2.3, Prop. 8.2, Rem. 8.4, Rem. 10.3 · newly derived · implemented · *Code* certiqs_key.statistics.tails:upper_tail

Proof. Both are obtained from (2.2)–(2.3) by substituting $k-1$ and $k+1$ respectively, N being integer-valued. For the first form: at $k = 0$ the boundary value $\Pr[\mathsf{N} \leq k-1] = 0$ for $k-1 < 0$ of Theorem 2.1 applies; for $1 \leq k \leq n$ one has $0 \leq k-1 \leq n-1$, and (2.3) at $k-1$ gives $\mathcal{I}_{1-p}(n-(k-1),(k-1)+1) = \mathcal{I}_{1-p}(n-k+1,k)$. For the second form: for $0 \leq k \leq n-1$ one has $1 \leq k+1 \leq n$, and (2.2) at $k+1$ gives $\mathcal{I}_p(k+1,n-(k+1)+1) = \mathcal{I}_p(k+1,n-k)$; at $k = n$ the boundary value $\Pr[\mathsf{N} \geq k+1] = 0$ for $k+1 > n$ of Theorem 2.1 applies. □

At a rational success probability both strict tails of a binomial count are finite sums of binomial probabilities, hence rational numbers, and each comparison of a tail with a rational budget is decided exactly. The endpoint rules can therefore be evaluated without rounding.

Corollary 2.2 writes the two strict tails as regularised incomplete beta integrals, the form a numerical evaluation of the incomplete beta function uses. At a rational success probability the finite sum of the binomial probabilities gives each tail exactly instead; which of the two routes an evaluation takes is constrained only by the requirement that it be a conservative evaluation (Definition 4.3).

Lemma 2.3 (Exact strict tails at a rational success probability)**.** Let $n \geq 1$, let $p \in [0,1] \cap \mathbb{Q}$ and let N be binomial with parameters (n,p), as in

Theorem 2.1. For $0 \le k \le n$,

$$\Pr[\mathsf{N} < k] = \sum_{i=0}^{k-1} \binom{n}{i} p^i (1-p)^{n-i}, \qquad \Pr[\mathsf{N} > k] = \sum_{i=k+1}^{n} \binom{n}{i} p^i (1-p)^{n-i}, \tag{2.6}$$

an empty sum being 0, and:

(i) both strict tails are rational numbers, obtained from n, k and p by finitely many additions and multiplications of rationals, and they are equal to the closed forms (2.5) of Corollary 2.2;

(ii) for every rational ε, whether $\Pr[\mathsf{N} < k] \le \varepsilon$ holds, and whether $\Pr[\mathsf{N} > k] \le \varepsilon$ holds, is decided exactly by evaluating the rational number (2.6) and comparing two rationals, which is a comparison of two integers.

Uses Thm. 2.1, Cor. 2.2 · *Used by* Prop. 8.2, Rem. 8.4, Rem. 10.3 · newly derived · accepted

Proof. The sums. N takes values in $\{0, \dots, n\}$ with $\Pr[\mathsf{N} = i] = \binom{n}{i} p^i (1-p)^{n-i}$ (Theorem 2.1, with $0^0 = 1$, the convention under which this law is a probability law at $p \in \{0, 1\}$). For $0 \le k \le n$ the event $\{\mathsf{N} < k\}$ is the disjoint union of the events $\{\mathsf{N} = i\}$ for the integers $0 \le i \le k-1$, and $\{\mathsf{N} > k\}$ is the disjoint union of the events $\{\mathsf{N} = i\}$ for $k+1 \le i \le n$; at $k = 0$, respectively $k = n$, the union is empty and the event has probability 0. Finite additivity of the probability gives (2.6).

(i). Since p is rational, so is $1 - p$; every $\binom{n}{i}$ is an integer; and $p^i (1-p)^{n-i}$ is a product of n rational factors (the empty product being 1). Each summand of (2.6) is therefore a product of finitely many rationals, and each tail a sum of at most $n+1$ such summands. The rationals are closed under addition and multiplication, so both tails are rational. They equal the forms (2.5) because Corollary 2.2 states that the same two probabilities $\Pr[\mathsf{N} < k]$ and $\Pr[\mathsf{N} > k]$ equal those forms, for the same n, p and k.

(ii). By (i) each tail is a rational number a/b with integers a and $b > 0$, computed in finitely many exact operations. Write the rational ε as c/d with integers c and $d > 0$. Then $a/b \le c/d$ holds if and only if $ad \le cb$, multiplying by $bd > 0$, which is a comparison of two integers and is decided exactly. □

3 The exact box oracle

The consuming engine minimises linear functionals over the acceptance set and needs to know on which outcomes an accepted frequency can put weight. The main result, Theorem 3.1, solves the minimisation over a non-empty canonical box exactly by filling the coordinates of smallest coefficient first, from their lower towards their upper bounds, supplies a dual witness of the

same value in the format the engine's certificate expects, and shows that the outcomes with a positive upper bound include every outcome on which a point of the box puts weight, and can include more.

Over a non-empty canonical box, filling the coordinates of smallest coefficient first minimises a rational linear functional exactly, with a dual witness of the same value. The outcomes with a positive upper bound include every outcome on which a point of the box puts weight, and can include more.

The consuming engine minimises linear functionals over the acceptance set and needs to know on which outcomes an accepted frequency can put weight. Over a canonical box the minimum has a closed form, and the outcomes with a positive upper bound form a closed-form superset of the outcomes that can carry weight. The dual witness of (iii) has the format of instantiation (b) of Remark "Exact minimisation on the acceptance side" (meat-core/frank-wolfe/rem:acc-side): (3.3) implies, at every outcome, the dual-feasibility inequality that instantiation requires, and the witness attains the minimum, (3.4), which is why the bound it gives is tight. Rounded down, that bound remains a lower bound, the direction of Requirement "Rounding direction of emitted bounds" (foundations/req:rounding-direction), so the remark's certified acceptance-side bound β_{acc} may be taken as any downward-rounded evaluation of $\langle g, \mathbf{q}^{\mathrm{opt}} \rangle$.

Theorem 3.1 (Acceptance support and the exact box oracle). Let $\widehat{\mathcal{C}}$ and n be a matched alphabet and round count (Definition 1.3), let $\mathcal{S}_{\mathrm{acc}}$ be a non-empty canonical box, (1.5), with $0 \le \mathbf{q}^- \le \mathbf{q}^+$, and let $g \in \mathbb{Q}^{\widehat{\mathcal{C}}}$.

(i) *(acceptance support)* The set $\widehat{\mathcal{C}}^+ := \{\hat{c} \in \widehat{\mathcal{C}} : q^+(\hat{c}) > 0\}$ contains $\operatorname{supp} \mathbf{q}$ for every $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$; hence it contains the acceptance support $\widehat{\mathcal{C}}_{\mathrm{acc}} = \bigcup_{\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}} \operatorname{supp} \mathbf{q}$ of Definition "Ambient program, pairings, acceptance support and linear minimisation" (meat-core/frank-wolfe/def:first-order-setting), which it need not equal: the box constraints together with the simplex constraint can force $q(\hat{c}) = 0$ at an outcome with $q^+(\hat{c}) > 0$. For a singleton geometry, which declares no box, the union of the supports is $\operatorname{supp} \mathbf{q}^*$, the acceptance support itself.

(ii) *(exact minimum in closed form)* Enumerate $\widehat{\mathcal{C}} = \{c_1, \dots, c_m\}$ so that $g(c_1) \le \cdots \le g(c_m)$. Put $r := 1 - \sum_{\hat{c}} q^-(\hat{c})$ and $D_i := q^+(c_i) - q^-(c_i) \ge 0$, let j^* be the least index with $\sum_{i \le j^*} D_i \ge r$, and define

$$q^{\mathrm{opt}}(c_i) := \begin{cases} q^+(c_i), & i < j^*, \\ q^-(c_i) + \big(r - \sum_{i' < j^*} D_{i'}\big), & i = j^*, \\ q^-(c_i), & i > j^*. \end{cases} \tag{3.1}$$

Then j^* exists, $\mathbf{q}^{\mathrm{opt}} \in \mathcal{S}_{\mathrm{acc}}$, and

$$\langle g, \mathbf{q}^{\mathrm{opt}} \rangle = \min_{\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}} \langle g, \mathbf{q} \rangle, \tag{3.2}$$

the minimum being attained. All quantities are exact rationals when $g, \mathbf{q}^-, \mathbf{q}^+$ are.

(iii) *(dual witness)* With $B, \mathbf{b}$ as in (1.4), $\lambda := g(c_{j^*})$, $v := -\lambda$ and $\mathbf{u} := (\mathbf{u}^+, \mathbf{u}^-)$ given by $u^+(\hat{c}) := \max\{\lambda - g(\hat{c}), 0\}$ and $u^-(\hat{c}) := \max\{g(\hat{c}) - \lambda, 0\}$, one has $\mathbf{u} \geq 0$,

$$g(\hat{c}) + (B^{\mathsf{T}}\mathbf{u})(\hat{c}) + v = 0 \qquad \text{for every } \hat{c} \in \widehat{\mathcal{C}}, \tag{3.3}$$

in particular the left-hand side is ≥ 0, and

$$-\langle \mathbf{b}, \mathbf{u} \rangle - v \;=\; \langle g, \mathbf{q}^{\mathrm{opt}} \rangle \;=\; \min_{\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}} \langle g, \mathbf{q} \rangle. \tag{3.4}$$

The lower bound $-\langle \mathbf{b}, \mathbf{u} \rangle - v$ on the minimum that this witness gives is therefore *tight* for the canonical box, and every downward-rounded evaluation of $\langle g, \mathbf{q}^{\mathrm{opt}} \rangle$ is a lower bound on $\min_{\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}} \langle g, \mathbf{q} \rangle$.

Uses Def. 1.3, Def. 1.4, Thm. 1.5 · *Assumes* A-ALPHABET · *Used by* Rem. 3.2, Req. 6.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.statistics.minimization:box_minimum · *Oracles* registry_statistics_acceptance.py

Proof. (i). If $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ and $q^+(\hat{c}) = 0$ then $0 \leq q(\hat{c}) \leq q^+(\hat{c}) = 0$, so $\hat{c} \notin \operatorname{supp} \mathbf{q}$; contrapositively $\operatorname{supp} \mathbf{q} \subseteq \widehat{\mathcal{C}}^+$, and taking the union over $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ gives $\widehat{\mathcal{C}}_{\mathrm{acc}} \subseteq \widehat{\mathcal{C}}^+$. That the inclusion can be strict: on $\widehat{\mathcal{C}} = \{c_1, c_2, c_3\}$ with $\mathbf{q}^- = (\frac{1}{2}, \frac{1}{2}, 0)$ and $\mathbf{q}^+ = (\frac{1}{2}, \frac{1}{2}, \frac{1}{10})$ the box is non-empty by (1.6), the simplex constraint forces $q(c_3) = 0$ for every $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$, so $\widehat{\mathcal{C}}_{\mathrm{acc}} = \{c_1, c_2\}$ while $\widehat{\mathcal{C}}^+ = \{c_1, c_2, c_3\}$. For a singleton the union $\bigcup_{\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}} \operatorname{supp} \mathbf{q}$ is $\operatorname{supp} \mathbf{q}^*$ by inspection, which is therefore the acceptance support $\widehat{\mathcal{C}}_{\mathrm{acc}}$.

(ii). Existence of j^*. $\mathcal{S}_{\mathrm{acc}} \neq \varnothing$ gives $\sum_{\hat{c}} q^+(\hat{c}) \geq 1$ by Theorem 1.5 (iii), hence $\sum_{i \leq m} D_i = \sum_{\hat{c}} q^+(\hat{c}) - \sum_{\hat{c}} q^-(\hat{c}) \geq 1 - \sum_{\hat{c}} q^-(\hat{c}) = r$, so the index set is non-empty and has a least element. Also $r \geq 0$, again by Theorem 1.5 (iii).

Feasibility of $\mathbf{q}^{\mathrm{opt}}$. For $i \neq j^*$ the value is one of the two box endpoints. At $i = j^*$ the increment $w := r - \sum_{i' < j^*} D_{i'}$ satisfies $w > 0$ when $j^* > 1$, by minimality of j^*, and $w = r \geq 0$ when $j^* = 1$; and $w \leq D_{j^*}$ by the defining inequality $\sum_{i \leq j^*} D_i \geq r$. Hence $q^-(c_{j^*}) \leq q^{\mathrm{opt}}(c_{j^*}) \leq q^+(c_{j^*})$, and $\mathbf{q}^{\mathrm{opt}} \geq \mathbf{q}^- \geq 0$. Summing, $\sum_i q^{\mathrm{opt}}(c_i) = \sum_i q^-(c_i) + \sum_{i < j^*} D_i + w = \sum_i q^-(c_i) + r = 1$. So $\mathbf{q}^{\mathrm{opt}} \in \mathcal{S}_{\mathrm{acc}}$.

Optimality. Let $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ and $\lambda = g(c_{j^*})$. Since $\sum_i \big(q(c_i) - q^{\mathrm{opt}}(c_i)\big) = 1 - 1 = 0$,

$$\langle g, \mathbf{q} \rangle - \langle g, \mathbf{q}^{\mathrm{opt}} \rangle = \sum_{i=1}^{m} \big(g(c_i) - \lambda\big)\big(q(c_i) - q^{\mathrm{opt}}(c_i)\big).$$

Each summand is non-negative: for $i < j^*$ the ordering gives $g(c_i) - \lambda \leq 0$ while $q(c_i) - q^{\mathrm{opt}}(c_i) = q(c_i) - q^+(c_i) \leq 0$; for $i > j^*$, $g(c_i) - \lambda \geq 0$ while

$q(c_i) - q^{\mathrm{opt}}(c_i) = q(c_i) - q^-(c_i) \geq 0$; and at $i = j^*$ the first factor is 0. Hence $\langle g, \mathbf{q}\rangle \geq \langle g, \mathbf{q}^{\mathrm{opt}}\rangle$ for every feasible $\mathbf{q}$, which is (3.2) with attainment at $\mathbf{q}^{\mathrm{opt}}$.

(iii). Non-negativity of $\mathbf{u}$ is clear. With the row structure (1.4), $B^{\mathsf{T}}\mathbf{u} = \mathbf{u}^+ - \mathbf{u}^-$, and for every real x one has $\max\{x, 0\} - \max\{-x, 0\} = x$; applying this at $x = \lambda - g(\hat{c})$ gives $u^+(\hat{c}) - u^-(\hat{c}) = \lambda - g(\hat{c})$, so $g(\hat{c}) + (B^{\mathsf{T}}\mathbf{u})(\hat{c}) + v = g(\hat{c}) + \lambda - g(\hat{c}) - \lambda = 0$, which is (3.3).

For (3.4), note first that the sorted order and $\lambda = g(c_{j^*})$ give $u^-(c_i) = 0$ for $i \leq j^*$ and $u^+(c_i) = 0$ for $i \geq j^*$. Then

$$\begin{aligned}
-\langle \mathbf{b}, \mathbf{u}\rangle - v &= -\langle \mathbf{q}^+, \mathbf{u}^+\rangle + \langle \mathbf{q}^-, \mathbf{u}^-\rangle + \lambda \\
&= -\sum_{i<j^*} q^+(c_i)\big(\lambda - g(c_i)\big) + \sum_{i>j^*} q^-(c_i)\big(g(c_i) - \lambda\big) + \lambda \\
&= \lambda\Big(1 - \sum_{i<j^*} q^+(c_i) - \sum_{i>j^*} q^-(c_i)\Big) + \sum_{i<j^*} g(c_i) q^+(c_i) + \sum_{i>j^*} g(c_i) q^-(c_i) \\
&= \lambda\, q^{\mathrm{opt}}(c_{j^*}) + \sum_{i\neq j^*} g(c_i) q^{\mathrm{opt}}(c_i) \;=\; \langle g, \mathbf{q}^{\mathrm{opt}}\rangle,
\end{aligned}$$

where the fourth line uses $q^{\mathrm{opt}}(c_i) = q^+(c_i)$ for $i < j^*$, $q^{\mathrm{opt}}(c_i) = q^-(c_i)$ for $i > j^*$, $\sum_i q^{\mathrm{opt}}(c_i) = 1$ and $\lambda = g(c_{j^*})$. Combining with (3.2) gives (3.4). In particular $-\langle \mathbf{b}, \mathbf{u}\rangle - v$ is a lower bound on the minimum that it equals, and a downward-rounded evaluation of $\langle g, \mathbf{q}^{\mathrm{opt}}\rangle$ is at most $\langle g, \mathbf{q}^{\mathrm{opt}}\rangle$, which is the minimum by (3.2). □

Over the canonical box the minimum of a linear functional that the consuming engine certifies is computable exactly in closed form, with no linear-programming solve and a dual witness for re-verification. A linear minimisation oracle over the box can be implemented as a sort plus a prefix sum.

Remark 3.2 (The closed-form box minimum for the consuming engine). Remark "Exact minimisation on the acceptance side" (meat-core/frank-wolfe/rem:acc-side) lets the certificate consume, in place of the pairing with a point attaining the minimum, a number β_{acc} with

$$\beta_{\mathrm{acc}} \;\leq\; \min_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}} \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{acc}}} g(\hat{c})\, q(\hat{c}) \tag{3.5}$$

for a given exact g, and for a polytope it computes the minimum by a linear program solved exactly over the rationals. For the representation this module emits the minimum is computable in closed form, exactly, by (3.1): no dual search and no linear-programming solve is needed, and the dual witness of Theorem 3.1 (iii) is available for independent re-verification in exactly the shape that remark's instantiation (b) expects. Two consequences follow. First, the acceptance-side minimisation is not a source of numerical error for this geometry: besides the consumer's outward rounding of the box bounds when it reads the set (Requirement "Exact decision by every consumer" (foundations/req:consumer-decides-exactly)), which leaves a non-empty canonical box containing the emitted one, to which Theorem 3.1 applies again, the only rounding is the final downward evaluation of a rational number, which keeps a

lower bound a lower bound (Requirement "Rounding direction of emitted bounds" (foundations/req:rounding-direction)). Second, a consumer's linear minimisation oracle over $\mathcal{S}_{\mathrm{acc}}$ may be implemented as a sort plus a prefix sum.

Uses Thm. 3.1 · newly derived · accepted

4 The honest envelope

The honest model is an honest envelope: a lower and an upper honest probability per outcome, whose envelope set holds every law between them. One honest law is the degenerate envelope, and assuming that every round's conditional outcome law lies in a degenerate envelope is the same as assuming independent rounds with that law (Lemma 4.2). A conservative evaluation takes the envelope, the budget and its allocation as exact rationals and decides every tail comparison exactly or against an upward-rounded bound (Definition 4.3).

An honest envelope is a lower and an upper exact rational probability per outcome, and its envelope set holds every law in the simplex between them. A single honest law is the degenerate envelope, whose envelope set is that law alone.

The honest model is a range of honest probabilities per outcome rather than a single law, and a single law is the special case of ranges of width zero.

Definition 4.1 (Honest envelope and its envelope set)**.** An *honest envelope* on $\widehat{\mathcal{C}}$ is a pair $p^{\mathrm{hon}^-}, p^{\mathrm{hon}^+} \in \mathbb{Q}^{\widehat{\mathcal{C}}}$ of exact rational vectors with

$$0 \le p^{\mathrm{hon}^-}(\hat{c}) \le p^{\mathrm{hon}^+}(\hat{c}) \le 1 \quad \forall \hat{c} \in \widehat{\mathcal{C}}, \qquad \sum_{\hat{c}\in\widehat{\mathcal{C}}} p^{\mathrm{hon}^-}(\hat{c}) \le 1 \le \sum_{\hat{c}\in\widehat{\mathcal{C}}} p^{\mathrm{hon}^+}(\hat{c}). \tag{4.1}$$

Its *envelope set* is

$$\mathsf{H} \coloneqq \left\{\mathbf{p} \in \Delta(\widehat{\mathcal{C}}) : \; p^{\mathrm{hon}^-}(\hat{c}) \le p(\hat{c}) \le p^{\mathrm{hon}^+}(\hat{c}) \;\; \forall \hat{c} \in \widehat{\mathcal{C}}\right\}, \tag{4.2}$$

the set of honest laws the envelope admits. An honest envelope with $p^{\mathrm{hon}^-} = p^{\mathrm{hon}^+}$ is a *degenerate envelope*; then (4.1) forces the common value to be a law $p^{\mathrm{hon}} \in \Delta(\widehat{\mathcal{C}}) \cap \mathbb{Q}^{\widehat{\mathcal{C}}}$ and $\mathsf{H} = \{p^{\mathrm{hon}}\}$. A single honest law p^{hon} (the *point form*) is read as the degenerate envelope $(p^{\mathrm{hon}}, p^{\mathrm{hon}})$. The pair $(p^{\mathrm{hon}^-}, p^{\mathrm{hon}^+})$ is a canonical box form in the sense of Definition 1.4 (rational, $0 \le p^{\mathrm{hon}^-} \le p^{\mathrm{hon}^+}$), H is its denotation (1.5), and the second condition of (4.1) is its closed-form feasibility condition (1.6).

Uses Def. 1.4 · *Used by* Lem. 4.2, Def. 4.3, Ass. 5.2, Ass. 5.4, Def. 7.1, Lem. 7.2, Def. 8.1, Def. 9.1, Thm. 9.2, Thm. 10.2, Rem. 10.5, Thm. 11.1, Lem. 12.2, Thm. 12.4, Rem. 18.2 · newly derived · implemented · *Code* certiqs_key.statistics.construction:HonestEnvelope

Every round's conditional honest law lying in a degenerate envelope is the same as independent, identically distributed honest rounds with that one law. The envelope assumption therefore generalises the single-law assumption.

The completeness statements read one of two honest-round premises: independent rounds with one law (Assumption 5.1), or rounds whose conditional laws stay inside a declared envelope (Assumption 5.2). At a degenerate envelope the two premises coincide.

Lemma 4.2 (The envelope assumption at a degenerate envelope)**.** Let $p^{\mathrm{hon}} \in \Delta(\widehat{\mathcal{C}}) \cap \mathbb{Q}^{\widehat{\mathcal{C}}}$ and let $\widehat{C}_1, \dots, \widehat{C}_n$ be $\widehat{\mathcal{C}}$-valued random variables on one probability space. The following are equivalent:

(a) the statement of Assumption 5.2 holds for the degenerate envelope $(p^{\mathrm{hon}}, p^{\mathrm{hon}})$ of Definition 4.1: there is a filtration $\mathcal{F}_0 \subseteq \cdots \subseteq \mathcal{F}_n$ with $\widehat{C}_j$ $\mathcal{F}_j$-measurable and $\Pr\left[\widehat{C}_j = \hat{c} \,|\, \mathcal{F}_{j-1}\right] = p^{\mathrm{hon}}(\hat{c})$ almost surely, for every j and $\hat{c}$;

(b) the statement of Assumption 5.1 holds: $\widehat{C}_1, \dots, \widehat{C}_n$ are independent and identically distributed with law p^{hon}.

Uses Def. 4.1 · newly derived · accepted

Proof. With a degenerate envelope, (5.1) reads $p^{\mathrm{hon}}(\hat{c}) \le \Pr\left[\widehat{C}_j = \hat{c} \,|\, \mathcal{F}_{j-1}\right] \le p^{\mathrm{hon}}(\hat{c})$, i.e. the equality displayed in (a).

(a)⇒(b). Fix j. For every $A \in \mathcal{F}_{j-1}$ and every $\hat{c}$, the defining property of conditional probability gives

$$\Pr\left[\{\widehat{C}_j = \hat{c}\} \cap A\right] = \mathbb{E}\left[\mathbf{1}_A \Pr\left[\widehat{C}_j = \hat{c} \,|\, \mathcal{F}_{j-1}\right]\right] = p^{\mathrm{hon}}(\hat{c}) \Pr[A].$$

Taking A the whole space shows $\Pr\left[\widehat{C}_j = \hat{c}\right] = p^{\mathrm{hon}}(\hat{c})$, so the display says $\Pr\left[\{\widehat{C}_j = \hat{c}\} \cap A\right] = \Pr\left[\widehat{C}_j = \hat{c}\right] \Pr[A]$: $\widehat{C}_j$ is independent of $\mathcal{F}_{j-1}$ and has law p^{hon}. Since $\widehat{C}_i$ is $\mathcal{F}_i$-measurable and $\mathcal{F}_i \subseteq \mathcal{F}_{j-1}$ for $i < j$, the event $A = \{\widehat{C}_1 = \hat{c}_1, \dots, \widehat{C}_{j-1} = \hat{c}_{j-1}\}$ lies in $\mathcal{F}_{j-1}$, whence $\Pr\left[\widehat{C}_1 = \hat{c}_1, \dots, \widehat{C}_j = \hat{c}_j\right] = p^{\mathrm{hon}}(\hat{c}_j) \Pr\left[\widehat{C}_1 = \hat{c}_1, \dots, \widehat{C}_{j-1} = \hat{c}_{j-1}\right]$ for all $\hat{c}_1, \dots, \hat{c}_j \in \widehat{\mathcal{C}}$. Induction on j from 1 to n gives $\Pr\left[\widehat{C}_1 = \hat{c}_1, \dots, \widehat{C}_n = \hat{c}_n\right] = \prod_{j=1}^{n} p^{\mathrm{hon}}(\hat{c}_j)$, the product law: the $\widehat{C}_j$ are independent, each with law p^{hon}.

(b)⇒(a). Take the natural filtration: $\mathcal{F}_0$ the trivial σ-algebra and $\mathcal{F}_j := \sigma(\widehat{C}_1, \dots, \widehat{C}_j)$. Each $\widehat{C}_j$ is $\mathcal{F}_j$-measurable. By independence, $\widehat{C}_j$ is independent of $\mathcal{F}_{j-1}$, so for every $A \in \mathcal{F}_{j-1}$, $\mathbb{E}[\mathbf{1}_A \mathbf{1}\{\widehat{C}_j = \hat{c}\}] = p^{\mathrm{hon}}(\hat{c}) \Pr[A] = \mathbb{E}[\mathbf{1}_A\, p^{\mathrm{hon}}(\hat{c})]$; the constant $p^{\mathrm{hon}}(\hat{c})$ is $\mathcal{F}_{j-1}$-measurable, hence a version of $\Pr\left[\widehat{C}_j = \hat{c} \,|\, \mathcal{F}_{j-1}\right]$, and (a) holds. □

An evaluation of the construction is conservative when the honest envelope, the completeness budget and its allocation are exact rationals and every tail comparison is decided exactly or against an upward-rounded tail. It is the arithmetic under which rounding can only widen the acceptance set.

With exact data the construction of the acceptance set lies on the exact island, the computations the engine must carry out exactly (Definition "The exact island" (foundations/def:exact-island) (ii)). Rounding in the safe direction enlarges $\mathcal{S}_{\mathrm{acc}}$, which lowers the consumer's $\underline{h}_\alpha$, an infimum over a superset being at most the infimum over a subset (Lemma 12.2); it costs rate, never soundness. The consuming engine's outward rounding of the emitted bounds is such a rounding: it is the one alteration of emitted data that Requirement "Exact decision by every consumer" (foundations/req:consumer-decides-exactly) allows a consumer, which still decides every predicate on a replaced bound exactly. The engine refuses inexact data (Requirement 16.3) and decides its tail comparisons in this way on each evaluation (Requirements 16.1 and 17.2).

Definition 4.3 (Exact data and conservative evaluation)**.** An evaluation of the acceptance-set construction is a *conservative evaluation* when the honest envelope's vectors $p^{\mathrm{hon}^-}, p^{\mathrm{hon}^+}$ (Definition 4.1; for a single declared law, p^{hon} itself), $\varepsilon_{\mathrm{comp}}$ and the allocation are exact rationals, and every tail comparison an endpoint rule makes (Definition 7.1; for the exact rule, those of Definition 8.1) is decided either exactly or against a bound that rounds the tail *upward*.

Uses Def. 4.1 · *Used by* Prop. 8.2, Thm. 9.2, Thm. 10.2, Rem. 10.3, Thm. 11.1, Thm. 12.4, Prop. 14.2, Rem. 14.3, Prop. 15.3, Rem. 15.4 · newly derived · implemented · *Code* certiqs_key.statistics.exact:as_exact_rational

5 Assumptions

These are the premises this module declares, and none is evidence about an adversary: honest rounds, independent with one law or conditionally inside a declared envelope; an acceptance value recorded in every round; and a law of the envelope that some feasible state of the consuming declaration produces. A failure of the honest-round premises costs completeness, rate or usability, never the soundness of a key. A failure of that last premise means that the declared devices do not describe what produced the law, and a key length computed for them can grow with the mismatch; the consuming engine checks a run whose honest law is supplied or whose declaration carries a constraint the user declared, unless the run opts out, states the cap a failure it proves puts on the probability that devices as declared pass the acceptance test, and refuses the run where that cap is at most a threshold, one half by default. A round missing from the record normalises the counts against the wrong round count.

On an honest run the acceptance statistics are independent and identically distributed with the declared law. The premise bears on the advertised honest abort probability and restricts no adversary.

The premise fixes one honest law for every round and narrows Assumption 5.2 to the degenerate envelope of that law. It is asserted about the honest channel only, never against an adversary. Physically it says that the honest channel acts identically and independently in every round and that there is no memory across rounds: the sameness and the no-memory clause of a round-homogeneous family, asserted of the honest channel.

Assumption 5.1 [A-HON] (Honest i.i.d. rounds)**.** (i) On an honest run the acceptance statistics $\widehat{C}_1, \ldots, \widehat{C}_n$ are independent and identically distributed with law p^{hon} on $\widehat{\mathcal{C}}$.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Without an eavesdropper, every round's acceptance outcome is drawn independently from the same declared distribution. If you expect drift between or within runs, declare an honest envelope instead. This premise only bears on the advertised probability that an honest run aborts: if it fails, an honest run may abort more or less often than advertised. Security is unaffected.

Category honest model · *Affects* completeness · *Refines* A-HON-ENV · *Assumed by* Rem. 10.4, Thm. 18.1 · *Machine key* `honest_rounds_iid` · newly derived · implemented · *Code* certiqs_key.statistics.construction:HONEST_MODEL_TAG

On an honest run each round's probability of each acceptance outcome, given everything before it, stays inside the declared honest envelope. The premise bears on the advertised honest abort probability and restricts no adversary.

The premise is asserted about the honest channel only, never against an adversary. Under it the honest channel may change from round to round within a run, from run to run, and in a way that depends on past outcomes or on further honest variables recorded in $\mathcal{F}_{j-1}$, as long as each round's conditional law stays inside the envelope.

Assumption 5.2 [A-HON-ENV] (Honest rounds inside a declared envelope)**.** Let $(p^{\mathrm{hon}^-}, p^{\mathrm{hon}^+})$ be the declared honest envelope (Definition 4.1). On an honest run there is a filtration $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}_n$ on the underlying probability space such that (i) $\widehat{C}_j$ is $\mathcal{F}_j$-measurable for every $j \in \{1, \ldots, n\}$; (ii) for every such j and every $\hat{c} \in \widehat{\mathcal{C}}$, almost surely,

$$p^{\mathrm{hon}^-}(\hat{c}) \;\le\; \Pr\left[\widehat{C}_j = \hat{c} \;\middle|\; \mathcal{F}_{j-1}\right] \;\le\; p^{\mathrm{hon}^+}(\hat{c}). \tag{5.1}$$

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: When no attacker is present, each round's probability of each acceptance outcome, given everything that happened before, stays inside the honest model: between the lower and upper bounds of the honest envelope where you declared one, and equal to the honest distribution where you gave a single one instead (an envelope of zero width). With an envelope the channel may drift within and between runs as long as it stays inside it; with a single distribution it may not drift at all. This premise only bears on the advertised

probability that an honest run aborts; it never affects security. If drift can leave the honest model, declare a wider envelope.

Uses Def. 4.1 · *Category* honest model · *Affects* completeness · *Assumed by* Thm. 10.2, Rem. 10.4 · *Machine key* `honest_rounds_within_envelope` · newly derived · implemented · *Code* certiqs_key.statistics.construction: ENVELOPE_MODEL_TAG

In every one of the n rounds of the run an acceptance value of the alphabet is recorded. The counts and the frequency vector then describe the whole run, normalised by its round count.

Assumption 5.3 [A-ALPHABET] (An acceptance value recorded in every round)**.** (i) In every one of the n rounds of the run, an acceptance value in $\widehat{\mathcal{C}}$ is recorded.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Every one of the n rounds of your run records an acceptance value from the declared acceptance alphabet: no round counted in n is missing from the record. If rounds go missing (for example through lost time tags), the counts are normalised against the wrong number of rounds and the acceptance test does not describe the run. A round missing from the record also puts security at stake: the acceptance test must be evaluated on the acceptance values of all n rounds (A-CONFORM), and a measured round the record drops is a sifting step the declaration does not describe (A-SPEC). Where the engine builds the acceptance set, it checks that the alphabet, its order and n are those of your protocol declaration; a set carried in the declaration is read in the declaration's order, so writing it in that order is yours to check.

Category protocol structure · *Affects* secrecy, completeness · *Assumed by* Thm. 1.5, Thm. 3.1, Rem. 8.4, Thm. 9.2, Thm. 10.2, Thm. 11.1, Thm. 12.4, Thm. 18.1 · *Machine key* `acceptance_recorded_every_round` · newly derived · accepted

Some feasible state of the consuming declaration produces a law of the declared honest envelope, for a single declared law that law itself. It rules out an incompatibility of the honest model with the declaration.

The premise is an obligation of whoever declares the honest model, together with the declaration: it is a joint property of the declared honest envelope and of the protocol declaration the consumer solves.

Assumption 5.4 [A-REACH] (An honest law produced by a feasible state)**.** (i) There exists $\omega^{\mathrm{hon}} \in \mathcal{W}$, the feasible set of the consuming declaration (Definition "Feasible set" (protocol-compiler/def:feasible)), with $\mathbf{\Phi}[\omega^{\mathrm{hon}}] \in \mathsf{H}$, the envelope set of the declared honest envelope (Definition 4.1); for a single declared law p^{hon} (the degenerate envelope, $\mathsf{H} = \{p^{\mathrm{hon}}\}$) this reads $\mathbf{\Phi}[\omega^{\mathrm{hon}}] = p^{\mathrm{hon}}$.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Your honest model fits your protocol declaration: a state the declaration admits as feasible produces acceptance statistics in your declared honest envelope, or equal to your single declared honest law. If it fails, the declared devices do not describe the statistics and a key length for them is not warranted. The engine's own honest models are built to match the declared Alice marginal; nothing proves this premise for them. Unless switched off, the engine checks every run whose honest statistics you supply or whose declaration

carries a constraint you declared: from a proven mismatch it states a cap on the probability that devices behaving as the declaration says pass your acceptance test, and refuses the run where that cap is at most one half, a threshold you can set. On an unchecked run a mismatch goes undetected and this premise is yours to vouch for.

Uses Def. 4.1 · *Category* honest model · *Affects* secrecy, completeness · *Assumed by* Thm. 11.1, Rem. 11.2 · *Machine key* `honest_reachable` · newly derived · accepted

Part II
Engine requirements

What an implementation of this module must do, record or refuse; each requirement names the result that motivates it.

6 Requirements: the emitted geometry

The engine emits every acceptance set under one of the two geometry tags with that tag's data, and a consumer refuses a set under any other tag. It records a canonical box's lower and upper bounds beside the rows they determine. As the acceptance support it emits the outcomes with a positive upper bound, a superset never substituted for the support itself, and for a singleton the support of its vector. A set is docked only on the alphabet and round count of the declaration its consumer solves: a built set is compared with it, a set the declaration carries is read in its enumeration, and a mismatch produces no result.

Every acceptance set is emitted under one of the two geometry tags with that tag's data. A consumer presented with any other tag refuses the set rather than guessing what it means.

Definition 1.4 admits two tags and no third, Theorem 1.5 shows what a validated set under either tag is, and Remark "Exact minimisation on the acceptance side" (meat-core/frank-wolfe/rem:acc-side) certifies the minimum of a linear functional over the acceptance set for exactly these two forms.

Requirement 6.1 (Acceptance sets under exactly the two geometry tags). The engine emits every acceptance set under one of the two tags of Definition 1.4 (`geometry_tag`) together with the data of that tag, and a consumer presented with an acceptance set under any other tag refuses it rather than guessing.

Motivated by Def. 1.4, Thm. 1.5, meat-core/frank-wolfe/rem:acc-side · implemented · *Code* certiqs_key.statistics.taxonomy:GeometryTag

A geometry in canonical box form is emitted with its lower and upper bound vectors recorded beside the rows they determine. The recorded pair is what licenses the closed-form box oracle.

A consumer that reads the recorded pair can minimise over the box in closed form (Theorem 3.1); one that ignores it still reads a well-formed polytope (Remark 1.6).

Requirement 6.2 (Recorded bounds of a canonical box). The engine emits a geometry in canonical box form with its pair $(\mathbf{q}^-, \mathbf{q}^+)$ recorded under `box_bounds`, beside the rows (1.4) that the pair determines.

Motivated by Def. 1.4, Rem. 1.6, Thm. 3.1 · implemented · *Code* certiqs_key.statistics.taxonomy:canonical_box

For a canonical box the engine emits the outcomes with a positive upper bound, a superset of the acceptance support that is never substituted for it; for a singleton it emits the support of the vector, which is the acceptance support itself.

Theorem 3.1 (i) shows that the outcomes with a positive upper bound contain the support of every point of the box, and that they can contain more than the union of those supports.

REQUIREMENT 6.3 (The emitted acceptance support). The engine emits under `acceptance_support` the set $\widehat{\mathcal{C}}^{+} = \{\hat{c} \in \widehat{\mathcal{C}} : q^{+}(\hat{c}) > 0\}$ of Theorem 3.1 (i) for a geometry in canonical box form, and the set $\operatorname{supp} \mathbf{q}^{*}$ for a singleton geometry. For a box the emitted set is a superset of the acceptance support $\widehat{\mathcal{C}}_{\mathrm{acc}}$ of Definition "Ambient program, pairings, acceptance support and linear minimisation" (meat-core/frank-wolfe/def:first-order-setting) and is never to be substituted for it; for a singleton the emitted set and $\widehat{\mathcal{C}}_{\mathrm{acc}}$ coincide.

Motivated by Thm. 3.1, meat-core/frank-wolfe/def:first-order-setting · implemented (effective: accepted) · *Code* certiqs_key.statistics.taxonomy:acceptance_support

An acceptance set is docked only on the alphabet, in its declared enumeration, and the round count of the declaration its consumer solves: a built set is compared with the declaration first, a set the declaration carries is read in that enumeration. A disagreement produces no result.

Every result of this module that reads an alphabet and a round count reads them as a matched alphabet and round count (Definition 1.3); the match is a relation between artefacts and not a property of an input, so it is decided where both artefacts are available.

REQUIREMENT 6.4 (Alphabet and round count taken from the consuming declaration). The engine docks an acceptance set into the declaration a consumer solves only on a matched alphabet and round count (Definition 1.3).

(i) A set it builds, it builds on the acceptance alphabet, in its declared enumeration, of the declaration the consumer solves and on the round count of the one run parameter the consumer reads, and before the set is docked it compares the set's alphabet and enumeration with the declaration's. A disagreement is a hard error of the consumer's input validation: it is a refusal code of neither module, nothing is repaired, and no result is produced.

(ii) A set that the declaration itself carries, with no alphabet of its own, it reads positionally: its k-th coordinate is the k-th acceptance value in the declaration's declared enumeration, and its round count is the one run parameter the consumer reads, as in (i). A set whose number of

coordinates differs from the size of the acceptance alphabet is refused as ill formed, and no result is produced.

Motivated
by Def. 1.3 · accepted · *Code* certiqs_key.runs.pipeline:_require_matched_alphabet

Part III
Binomial box construction

This submodule turns an honest envelope, a completeness budget and its allocation over selected outcomes into a box of accepted frequencies. An admissible endpoint rule returns, per outcome, a lower count at the envelope's lower value and an upper count at its upper value whose tails meet their budgets, and a guard widens them to contain the whole envelope interval. Its main results are that for every such rule the construction map emits a canonical box of exact rationals that contains every law of the envelope set and meets the consuming engine's acceptance assumption (Theorem 9.2), and that an honest run whose per-round conditional outcome law stays inside the envelope fails the test with probability at most the sum of the binomial tails at the envelope's ends beyond the guarded endpoints, which the allocation keeps within the budget (Theorem 10.2); neither independent rounds nor independent outcome counts are assumed. The construction never fails on an admissible input, so every refusal comes from the input. Under a monotone allocation rule and an endpoint rule monotone in the allocation, which the enclosure evaluation of the exact rule is not claimed to be, a larger budget gives a smaller or equal box; for the exact rule the box is not always nested as the round count grows. The exact binomial quantile rule is the default; the Bernstein rule gives closed-form endpoints whose unrounded half-width decreases strictly with the round count, and a local evaluation of the exact rule decides its tail comparisons on enclosures in log space. Completeness is a usability guarantee, and soundness never rests on the honest-round assumptions.

7 Admissible inputs, endpoint rules and the guard

An admissible input fixes the alphabet, the round count, the honest envelope, the completeness budget, its allocation over the constrained outcomes, and the choice of rule and evaluation, the numbers exact. An endpoint rule returns, for one binomial count, a lower and an upper endpoint whose tails meet their budgets, the lower taken at the envelope's lower value and the upper at its upper value, and the honest-containment guard widens them to contain the whole envelope interval. Lemma 7.2 proves for every rule that the guarded endpoints are ordered around that interval, still meet their budgets, widen the rule's interval by an exactly known amount, and lie within 1/n of the envelope value wherever the guard acts.

An admissible endpoint rule returns count endpoints whose binomial tails meet their budgets. On an admissible input the lower endpoint is taken at the envelope's lower value, the upper at its upper value, and a guard widens both to contain the envelope interval.

The construction turns the two tail budgets of each constrained outcome into an interval of counts. An endpoint rule does this for one binomial count; the definition fixes what every rule must guarantee, the inputs on which the construction runs, and the guard that widens a rule's interval to contain the honest envelope. The lower endpoint is taken at the envelope's lower value and the upper endpoint at its upper value because, under Assumption 5.2, the binomial laws at those two values bound the count of Definition 1.1 from below and from above (Lemma 10.1). The construction and the theorems about it take the admissible endpoint rule R as a parameter, independent of the choice of rule and evaluation an input carries in clause (viii), and state as a hypothesis any further property of R they need; that choice selects the rule in force (Requirement 16.1), and a choice outside clause (viii) is violation (11) of Theorem 11.1 (iii).

Definition 7.1 (Admissible inputs, endpoint rules and guarded endpoints)**.** Fix $n \geq 1$ and let N be binomial with parameters (n, p), where $p \in [0, 1]$ is rational. An *admissible endpoint rule* R assigns to every $(n, p, \varepsilon^-, \varepsilon^+)$ with rational $\varepsilon^-, \varepsilon^+ \in [0, 1)$ two integers $R^- = R^-(n, p, \varepsilon^-)$ and $R^+ = R^+(n, p, \varepsilon^+)$ such that

(a) *(admissibility)* $R^-, R^+ \in \{0, \ldots, n\}$ and

$$\Pr\left[\mathsf{N} < R^-\right] \leq \varepsilon^-, \qquad \Pr\left[\mathsf{N} > R^+\right] \leq \varepsilon^+; \tag{7.1}$$

(b) *(zero budget)* if $0 < p < 1$, then $R^- = 0$ at $\varepsilon^- = 0$ and $R^+ = n$ at $\varepsilon^+ = 0$;

(c) *(degenerate mass)* $R^- = R^+ = 0$ at $p = 0$ and $R^- = R^+ = n$ at $p = 1$, for every $\varepsilon^{\pm}$.

The rule is *monotone in the allocation* when, the other arguments fixed, $\varepsilon^- \mapsto R^-$ is non-decreasing and $\varepsilon^+ \mapsto R^+$ is non-increasing.

An *admissible input* consists of

(i) a finite, non-empty acceptance alphabet $\widehat{\mathcal{C}}$ with a fixed enumeration;

(ii) a round count, an integer $n \geq 1$;

(iii) an honest input on $\widehat{\mathcal{C}}$: exactly one of a law $p^{\mathrm{hon}} \in \Delta(\widehat{\mathcal{C}})$, the point form, and an honest envelope $(p^{\mathrm{hon}^-}, p^{\mathrm{hon}^+})$ (Definition 4.1); a law in the point form is read as the degenerate envelope $(p^{\mathrm{hon}}, p^{\mathrm{hon}})$;

(iv) a completeness budget $\varepsilon_{\mathrm{comp}} \in [0, 1)$;

(v) a set $\widehat{\mathcal{C}}_{\mathrm{con}} \subseteq \widehat{\mathcal{C}}$ of constrained outcomes;

(vi) an allocation $\varepsilon^-, \varepsilon^+ : \widehat{\mathcal{C}}_{\mathrm{con}} \to \mathbb{Q}_{\geq 0}$ with $\sum_{\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}} (\varepsilon^-(\hat{c}) + \varepsilon^+(\hat{c})) \leq \varepsilon_{\mathrm{comp}}$;

(vii) exactness: every coordinate of the honest input, $\varepsilon_{\mathrm{comp}}$ and every value of the allocation is an exact rational;

(viii) a choice of rule, the exact binomial quantile rule of Definition 8.1 or the Bernstein rule of Proposition 14.2, and a choice of one of two evaluations, by the closed forms or by the enclosures of Proposition 15.3, which on the exact rule decides its tail comparisons: by the enclosures, it gives the rule R_{encl} of that proposition in place of the exact rule;

(ix) consistency with the consumer: $\widehat{\mathcal{C}}$, its enumeration and n are those of the declaration the consumer solves, a matched alphabet and round count (Definition 1.3).

The clauses fix the form of the data — an enumerated set $\widehat{\mathcal{C}}$, an integer n, one honest input that is a vector (the point form) or a pair of vectors (the envelope form), a number $\varepsilon_{\mathrm{comp}}$, a set $\widehat{\mathcal{C}}_{\mathrm{con}}$ with a pair of vectors indexed by it, and a named rule and evaluation — and conditions on data of that form.

Fix an admissible input, with its honest envelope $(p^{\mathrm{hon}-}, p^{\mathrm{hon}+})$. For $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$ write $p^-_{\hat{c}} := p^{\mathrm{hon}-}(\hat{c})$ and $p^+_{\hat{c}} := p^{\mathrm{hon}+}(\hat{c})$, let $\mathsf{N}^-_{\hat{c}}$ and $\mathsf{N}^+_{\hat{c}}$ be binomial with parameters $(n, p^-_{\hat{c}})$ and $(n, p^+_{\hat{c}})$ respectively, and abbreviate $R^- := R^-(n, p^-_{\hat{c}}, \varepsilon^-(\hat{c}))$ and $R^+ := R^+(n, p^+_{\hat{c}}, \varepsilon^+(\hat{c}))$: the lower endpoint is the rule's at the envelope's lower value, the upper endpoint the rule's at its upper value. The *guarded endpoints* of R at $\hat{c}$, after the *honest-containment guard*, are

$$k^-(\hat{c}) := \min\left\{R^-,\ \lfloor n\, p^-_{\hat{c}} \rfloor\right\}, \qquad k^+(\hat{c}) := \max\left\{R^+,\ \lceil n\, p^+_{\hat{c}} \rceil\right\}. \tag{7.2}$$

With the degenerate envelope, $p^-_{\hat{c}} = p^+_{\hat{c}} = p^{\mathrm{hon}}(\hat{c}) =: p_{\hat{c}}$, $\mathsf{N}^-_{\hat{c}}$ and $\mathsf{N}^+_{\hat{c}}$ have the one law of $\mathsf{N}_{\hat{c}}$, binomial $(n, p_{\hat{c}})$, and (7.2) reads $k^-(\hat{c}) = \min\{R^-, \lfloor np_{\hat{c}} \rfloor\}$, $k^+(\hat{c}) = \max\{R^+, \lceil np_{\hat{c}} \rceil\}$ with both rule endpoints at $p_{\hat{c}}$.

The *side-wise reading* is the following. By (7.2), $k^-(\hat{c})$ depends on the envelope only through $p^-_{\hat{c}}$ and $k^+(\hat{c})$ only through $p^+_{\hat{c}}$. A claim of this submodule that fixes one value $p_{\hat{c}}$ and one count $\mathsf{N}_{\hat{c}}$ binomial $(n, p_{\hat{c}})$ and speaks of the guarded endpoints — Definition 8.1 and Propositions 8.2, 14.2 and 15.3 do — states each assertion about the lower endpoint (one naming k^-, R^-, ε^- and the lower tail) with $p_{\hat{c}} := p^-_{\hat{c}}$ and $\mathsf{N}_{\hat{c}} := \mathsf{N}^-_{\hat{c}}$, and each assertion about the upper endpoint with $p_{\hat{c}} := p^+_{\hat{c}}$ and $\mathsf{N}_{\hat{c}} := \mathsf{N}^+_{\hat{c}}$; with the degenerate envelope the two readings coincide and are the claim as written.

Uses Def. 1.1, Def. 1.3, Def. 4.1 · *Used by* Lem. 7.2, Def. 8.1, Prop. 8.2, Rem. 8.3, Rem. 8.4, Def. 9.1, Thm. 9.2, Thm. 10.2, Rem. 10.3, Thm. 11.1, Lem. 12.2, Thm. 12.4, Prop. 14.2, Rem. 14.3, Prop. 15.3, Rem. 15.4, Req. 16.1, Req. 16.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.statistics.rules:EndpointRule

For every admissible endpoint rule the guarded endpoints enclose the envelope interval, still meet both tail budgets and widen the rule's interval by an exactly known number of counts. Where the guard acts, the emitted frequency endpoint lies within $1/n$ of that side's envelope value.

The guard acts on whatever endpoints a rule returns, so its properties hold for every rule at once; what is particular to the exact rule is Proposition 8.2.

Lemma 7.2 (The honest-containment guard for every rule)**.** Let R be an admissible endpoint rule (Definition 7.1) and $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$; write $p^{\mp} := p^{\mp}_{\hat{c}}$ for the envelope's values at $\hat{c}$, $R^- := R^-(n, p^-, \varepsilon^-(\hat{c}))$, $R^+ := R^+(n, p^+, \varepsilon^+(\hat{c}))$, and let $k^{\mp}(\hat{c})$ be the guarded endpoints (7.2). Then:

(i) *(ordering)*

$$0 \;\le\; k^-(\hat{c}) \;\le\; \lfloor np^- \rfloor \;\le\; \lceil np^+ \rceil \;\le\; k^+(\hat{c}) \;\le\; n; \tag{7.3}$$

(ii) *(the budgets are met)* the maps $k \mapsto \Pr\big[\mathsf{N}^-_{\hat{c}} < k\big]$ and $k \mapsto \Pr\big[\mathsf{N}^+_{\hat{c}} > k\big]$ are non-decreasing and non-increasing respectively on the integers; consequently

$$\Pr\big[\mathsf{N}^-_{\hat{c}} < k^-(\hat{c})\big] \le \varepsilon^-(\hat{c}), \qquad \Pr\big[\mathsf{N}^+_{\hat{c}} > k^+(\hat{c})\big] \le \varepsilon^+(\hat{c}), \tag{7.4}$$

and more generally *every* pair $(\tilde{k}^-, \tilde{k}^+)$ with $0 \le \tilde{k}^- \le R^-$ and $R^+ \le \tilde{k}^+ \le n$ satisfies (7.4) in place of $k^{\mp}(\hat{c})$: enlarging the rule's interval is always admissible;

(iii) *(exact cost of the guard)* relative to the rule's own interval $[R^-, R^+]$, the guard (7.2) enlarges the interval by exactly

$$R^- - k^-(\hat{c}) = \max\big\{0,\; R^- - \lfloor np^- \rfloor\big\}, \qquad k^+(\hat{c}) - R^+ = \max\big\{0,\; \lceil np^+ \rceil - R^+\big\} \tag{7.5}$$

counts on the lower and the upper side respectively;

(iv) *(distance from the honest value)* measured from the envelope's value on each side, the guard costs at most $1/n$ *unconditionally*:

$$0 \le p^- - \frac{k^-(\hat{c})}{n} \le \max\Big\{p^- - \frac{R^-}{n},\; \frac{1}{n}\Big\}, \qquad 0 \le \frac{k^+(\hat{c})}{n} - p^+ \le \max\Big\{\frac{R^+}{n} - p^+,\; \frac{1}{n}\Big\}, \tag{7.6}$$

so on whichever side the guard is active, the emitted endpoint lies within $1/n$ of that side's envelope value.

With the degenerate envelope, $p^- = p^+ = p_{\hat{c}}$, both $\mathsf{N}^{\mp}_{\hat{c}}$ are binomial $(n, p_{\hat{c}})$, and (i)–(iv) are statements about that single law.

Uses Def. 4.1, Def. 7.1 · *Used by* Prop. 8.2, Rem. 8.3, Rem. 8.4, Thm. 9.2, Thm. 10.2, Rem. 10.5, Lem. 12.2, Thm. 12.4, Prop. 14.2, Rem. 14.3, Prop. 15.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.statistics.intervals: guarded_endpoints · *Oracles* registry_statistics_acceptance.py

Proof. *(i).* The chain (7.3) is immediate from (7.2): $k^-(\hat{c})$ is a minimum one of whose arguments is $\lfloor np^- \rfloor$, $k^+(\hat{c})$ a maximum one of whose arguments is $\lceil np^+ \rceil$, and $\lfloor np^- \rfloor \le np^- \le np^+ \le \lceil np^+ \rceil$ because $p^- \le p^+$ (Definition 4.1); the outer bounds hold because $R^- \ge 0$, $\lfloor np^- \rfloor \ge 0$, $R^+ \le n$ and $\lceil np^+ \rceil \le n$ (Definition 7.1 (a) and $0 \le p^- \le p^+ \le 1$).

(ii). The function $k \mapsto \Pr\left[\mathsf{N}^-_{\hat{c}} < k\right]$ is non-decreasing, because the events $\{\mathsf{N}^-_{\hat{c}} < k\}$ increase with k; similarly $k \mapsto \Pr\left[\mathsf{N}^+_{\hat{c}} > k\right]$ is non-increasing. Hence for $0 \le \tilde{k}^- \le R^-$, $\Pr\left[\mathsf{N}^-_{\hat{c}} < \tilde{k}^-\right] \le \Pr\left[\mathsf{N}^-_{\hat{c}} < R^-\right] \le \varepsilon^-(\hat{c})$ by (7.1) at $p = p^-$, and for $R^+ \le \tilde{k}^+ \le n$, $\Pr\left[\mathsf{N}^+_{\hat{c}} > \tilde{k}^+\right] \le \Pr\left[\mathsf{N}^+_{\hat{c}} > R^+\right] \le \varepsilon^+(\hat{c})$ by (7.1) at $p = p^+$. Since (7.2) gives $0 \le k^-(\hat{c}) \le R^-$ and $R^+ \le k^+(\hat{c}) \le n$ (the latter by (i)), (7.4) is the case $(\tilde{k}^-, \tilde{k}^+) = (k^-(\hat{c}), k^+(\hat{c}))$.

(iii). The two identities (7.5) are (7.2) together with $\kappa - \min\{\kappa, x\} = \max\{0, \kappa - x\}$ and $\max\{\kappa, x\} - \kappa = \max\{0, x - \kappa\}$, valid for all reals κ, x.

(iv). By (7.2),

$$p^- - \frac{k^-(\hat{c})}{n} = p^- - \frac{\min\{R^-, \lfloor np^- \rfloor\}}{n} = \max\left\{p^- - \frac{R^-}{n},\ \frac{np^- - \lfloor np^- \rfloor}{n}\right\},$$

and $np^- - \lfloor np^- \rfloor \in [0, 1)$, so the second entry lies in $[0, 1/n)$ and the maximum is bounded by $\max\{p^- - R^-/n,\ 1/n\}$; non-negativity holds because the second entry is non-negative. The upper display of (7.6) is the mirror statement at p^+, using $\lceil np^+ \rceil - np^+ \in [0, 1)$. □

8 The exact rule

The default rule takes the extremal binomial quantiles: the largest lower count and the smallest upper count whose tails meet their budgets. Proposition 8.2 proves that this is an admissible endpoint rule, monotone in the allocation, and that the guard may widen its raw interval by far more than one count while the emitted endpoint stays within 1/n of the honest value. Two remarks weigh what the guard buys and costs, and show that it acts at ordinary budgets.

The exact rule takes the largest lower count and the smallest upper count whose binomial tails meet their budgets, the lower at the envelope's lower value and the upper at its upper value. The guard then widens this raw interval to contain the envelope interval.

The exact binomial quantile rule takes the extremal binomial quantiles. It is an admissible endpoint rule, monotone in the allocation, by Proposition 8.2 (ii), and it is the default rule (Requirement 16.1).

Definition 8.1 (The exact binomial quantile rule)**.** Fix an admissible input (Definition 7.1), with its round count $n \ge 1$, its honest envelope $(p^{\mathrm{hon}^-}, p^{\mathrm{hon}^+})$ (Definition 4.1; a single law p^{hon} is the degenerate envelope), its constrained

outcomes $\widehat{\mathcal{C}}_{\mathrm{con}} \subseteq \widehat{\mathcal{C}}$, its completeness budget $\varepsilon_{\mathrm{comp}} \in [0,1)$ and its budget allocation $\varepsilon^-, \varepsilon^+ : \widehat{\mathcal{C}}_{\mathrm{con}} \to \mathbb{Q}_{\geq 0}$ with $\sum_{\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}} (\varepsilon^-(\hat{c}) + \varepsilon^+(\hat{c})) \leq \varepsilon_{\mathrm{comp}}$; in particular $0 \leq \varepsilon^{\pm}(\hat{c}) \leq \varepsilon_{\mathrm{comp}} < 1$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$. For $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$ let $p_{\hat{c}}$ be the rule's argument — by the side-wise reading of Definition 7.1, $p_{\hat{c}} = p^-_{\hat{c}}$ in (8.1) and $p_{\hat{c}} = p^+_{\hat{c}}$ in (8.2), both equal to $p^{\mathrm{hon}}(\hat{c})$ for the degenerate envelope — and let $\mathsf{N}_{\hat{c}}$ be binomial with parameters $(n, p_{\hat{c}})$. Define the *extremal endpoints*

$$\kappa^-(\hat{c}) \coloneqq \max\left\{k \in \{0,\dots,n\} : \Pr[\mathsf{N}_{\hat{c}} < k] \leq \varepsilon^-(\hat{c})\right\}, \tag{8.1}$$

$$\kappa^+(\hat{c}) \coloneqq \min\left\{k \in \{0,\dots,n\} : \Pr[\mathsf{N}_{\hat{c}} > k] \leq \varepsilon^+(\hat{c})\right\}. \tag{8.2}$$

The *exact binomial quantile rule* is the endpoint rule $R^{\mp}_{\mathrm{ex}}(n, p_{\hat{c}}, \varepsilon^{\mp}(\hat{c})) \coloneqq \kappa^{\mp}(\hat{c})$. Its guarded endpoints are those of (7.2) with $R^{\mp} = \kappa^{\mp}(\hat{c})$, $\kappa^-(\hat{c})$ taken at $p^-_{\hat{c}}$ and $\kappa^+(\hat{c})$ at $p^+_{\hat{c}}$, i.e.

$$k^-(\hat{c}) = \min\left\{\kappa^-(\hat{c}),\ \lfloor n\, p^-_{\hat{c}} \rfloor\right\}, \qquad k^+(\hat{c}) = \max\left\{\kappa^+(\hat{c}),\ \lceil n\, p^+_{\hat{c}} \rceil\right\}.$$

Uses Def. 4.1, Def. 7.1 · *Used by* Prop. 8.2, Rem. 8.3, Rem. 8.4, Def. 9.1, Lem. 12.2, Thm. 12.4, Prop. 14.2, Rem. 14.3, Prop. 15.3, Rem. 15.4, Req. 16.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.statistics.intervals:extremal_endpoints

The exact quantiles are the extremal counts meeting their budgets and form an admissible endpoint rule, monotone in the allocation, as do counts decided against upper bounds on the tails. The guard may widen the interval by many counts, yet a moved endpoint stays within 1/n of the honest value.

Proposition 8.2 (Properties of the exact binomial quantile rule)**.** Assume a conservative evaluation (Definition 4.3). With the notation of Definition 8.1, for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$:

(i) the two sets in (8.1)–(8.2) are non-empty subsets of the finite set $\{0,\dots,n\}$, so $\kappa^{\pm}(\hat{c})$ exist and are unique;

(ii) *(the exact rule is an admissible endpoint rule)* the pair $R^{\mp}_{\mathrm{ex}} = \kappa^{\mp}(\hat{c})$ satisfies Definition 7.1 (a)–(c): $\kappa^{\pm}(\hat{c}) \in \{0,\dots,n\}$ and

$$\Pr\left[\mathsf{N}_{\hat{c}} < \kappa^-(\hat{c})\right] \leq \varepsilon^-(\hat{c}), \qquad \Pr\left[\mathsf{N}_{\hat{c}} > \kappa^+(\hat{c})\right] \leq \varepsilon^+(\hat{c}); \tag{8.3}$$

if $0 < p_{\hat{c}} < 1$ then $\kappa^-(\hat{c}) = 0$ at $\varepsilon^-(\hat{c}) = 0$ and $\kappa^+(\hat{c}) = n$ at $\varepsilon^+(\hat{c}) = 0$; and $\kappa^-(\hat{c}) = \kappa^+(\hat{c}) = 0$ at $p_{\hat{c}} = 0$, $\kappa^-(\hat{c}) = \kappa^+(\hat{c}) = n$ at $p_{\hat{c}} = 1$. Moreover the rule is monotone in the allocation: $\kappa^-(\hat{c})$ is non-decreasing in $\varepsilon^-(\hat{c})$ and $\kappa^+(\hat{c})$ is non-increasing in $\varepsilon^+(\hat{c})$. Consequently Lemma 7.2 applies to the exact rule, and its guarded endpoints $k^{\mp}(\hat{c})$ satisfy (7.3) and (7.4);

(iii) *(extremality and admissible relaxation)* $\kappa^-(\hat{c})$ is the largest and $\kappa^+(\hat{c})$ the smallest integer in $\{0,\dots,n\}$ satisfying the respective inequality of

(8.3), and *every* pair $(\tilde{k}^-, \tilde{k}^+)$ with $0 \le \tilde{k}^- \le \kappa^-(\hat{c})$ and $\kappa^+(\hat{c}) \le \tilde{k}^+ \le n$ satisfies (7.4) in place of $k^{\mp}(\hat{c})$. Enlarging the interval is therefore always admissible; shrinking it beyond $[\kappa^-, \kappa^+]$ is not;

(iv) the defining inequalities are the closed forms (2.5) of Corollary 2.2, so $\kappa^{\pm}(\hat{c})$ are computable from n, $p_{\hat{c}}$ and the allocation alone (Lemma 2.3);

(v) *(degenerate honest mass)* if $p_{\hat{c}} = 0$ then $k^-(\hat{c}) = k^+(\hat{c}) = 0$, and if $p_{\hat{c}} = 1$ then $k^-(\hat{c}) = k^+(\hat{c}) = n$; in both cases $\varepsilon^-(\hat{c})$ and $\varepsilon^+(\hat{c})$ are unspent, both tails in (7.4) being exactly zero;

(vi) *(exact cost of the guard for this rule)* write $p = p_{\hat{c}}$. Relative to the raw quantile interval $[\kappa^-(\hat{c}), \kappa^+(\hat{c})]$, the guard (7.2) enlarges the interval by exactly (7.5) with $R^{\mp} = \kappa^{\mp}(\hat{c})$, and neither enlargement is bounded by one count: for every $n \ge 1$ and every $p \in (0,1)$, an allocation with $\varepsilon^-(\hat{c}) \ge 1 - p^n$ — admissible whenever $\varepsilon_{\text{comp}}$ permits, since $1 - p^n < 1$ — gives $\kappa^-(\hat{c}) = n$ and a lower enlargement of $n - \lfloor np \rfloor$ counts, that is, at least $1 - p$ in frequency *uniformly in* n; symmetrically $\varepsilon^+(\hat{c}) \ge 1 - (1-p)^n$ gives $\kappa^+(\hat{c}) = 0$ and an upper enlargement of $\lceil np \rceil$ counts. The exact conditions are

$$\kappa^-(\hat{c}) - k^-(\hat{c}) = 0 \iff \varepsilon^-(\hat{c}) < \Pr\left[\mathsf{N}_{\hat{c}} \le \lfloor np \rfloor\right], \tag{8.4}$$
$$\kappa^-(\hat{c}) - k^-(\hat{c}) \le 1 \iff \varepsilon^-(\hat{c}) < \Pr\left[\mathsf{N}_{\hat{c}} \le \lfloor np \rfloor + 1\right], \tag{8.5}$$
$$k^+(\hat{c}) - \kappa^+(\hat{c}) = 0 \iff \varepsilon^+(\hat{c}) < \Pr\left[\mathsf{N}_{\hat{c}} \ge \lceil np \rceil\right], \tag{8.6}$$
$$k^+(\hat{c}) - \kappa^+(\hat{c}) \le 1 \iff \varepsilon^+(\hat{c}) < \Pr\left[\mathsf{N}_{\hat{c}} \ge \lceil np \rceil - 1\right]. \tag{8.7}$$

Measured instead from the honest value, the guard costs at most $1/n$ unconditionally, (7.6) of Lemma 7.2 (iv);

(vii) *(evaluation against upper bounds)* let U_k^- and U_k^+, for $k \in \{0, \dots, n\}$, be rationals determined by n, $p_{\hat{c}}$ and k alone, with $U_k^- \ge \Pr[\mathsf{N}_{\hat{c}} < k]$, $U_k^+ \ge \Pr[\mathsf{N}_{\hat{c}} > k]$, $U_0^- = 0$ and $U_n^+ = 0$, and with every $U_k^{\mp}$ equal to its tail when $p_{\hat{c}} \in \{0, 1\}$. Put

$$\tilde{\kappa}^-(\hat{c}) \coloneqq \max\left\{k \in \{0, \dots, n\} : U_j^- \le \varepsilon^-(\hat{c}) \text{ for } 1 \le j \le k\right\}, \qquad \tilde{\kappa}^+(\hat{c}) \coloneqq \min\left\{k \in \{0, \dots, n\} : \right.$$

Then $\tilde{R}^{\mp} = \tilde{\kappa}^{\mp}(\hat{c})$ is an admissible endpoint rule (Definition 7.1 (a)–(c)), monotone in the allocation, with $\tilde{\kappa}^-(\hat{c}) \le \kappa^-(\hat{c})$ and $\tilde{\kappa}^+(\hat{c}) \ge \kappa^+(\hat{c})$, and it is the exact rule when every $U_k^{\mp}$ equals its tail. Consequently Lemma 7.2 applies to it.

Uses Cor. 2.2, Lem. 2.3, Def. 4.3, Def. 7.1, Lem. 7.2, Def. 8.1 · *Used by* Rem. 8.3, Rem. 8.4, Rem. 10.3, Prop. 15.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.statistics.rules: SweepExactRule · *Oracles* registry_statistics_acceptance.py

Proof. *(i).* $k = 0$ belongs to the set in (8.1), since $\Pr[\mathsf{N}_{\hat{c}} < 0] = 0 \le \varepsilon^-(\hat{c})$; and $k = n$ belongs to the set in (8.2), since $\Pr[\mathsf{N}_{\hat{c}} > n] = 0 \le \varepsilon^+(\hat{c})$. A non-empty finite set of integers has a unique maximum and a unique minimum.

(ii). Admissibility. $\kappa^-(\hat{c})$ is an element of the set in (8.1) and $\kappa^+(\hat{c})$ one of the set in (8.2), so both lie in $\{0, \dots, n\}$ and satisfy the inequalities that define those sets, which are (8.3). This is Definition 7.1 (a).

Monotonicity in the allocation. Fix every argument but $\varepsilon^-(\hat{c})$. Raising $\varepsilon^-(\hat{c})$ enlarges the set in (8.1), so its maximum does not fall; raising $\varepsilon^+(\hat{c})$ enlarges the set in (8.2), so its minimum does not rise.

Zero budget, Definition 7.1 (b). Let $0 < p_{\hat{c}} < 1$ and write $p = p_{\hat{c}}$. At $\varepsilon^-(\hat{c}) = 0$: for $k \ge 1$, $\Pr[\mathsf{N}_{\hat{c}} < k] \ge \Pr[\mathsf{N}_{\hat{c}} = 0] = (1-p)^n > 0$, so only $k = 0$ lies in the set of (8.1) and $\kappa^-(\hat{c}) = 0$. At $\varepsilon^+(\hat{c}) = 0$: for $k \le n-1$, $\Pr[\mathsf{N}_{\hat{c}} > k] \ge \Pr[\mathsf{N}_{\hat{c}} = n] = p^n > 0$, so only $k = n$ lies in the set of (8.2) and $\kappa^+(\hat{c}) = n$.

Degenerate mass, Definition 7.1 (c). If $p_{\hat{c}} = 0$ then $\mathsf{N}_{\hat{c}} = 0$ with probability one, so $\Pr[\mathsf{N}_{\hat{c}} < k]$ is 0 at $k = 0$ and 1 for $k \ge 1$; since $\varepsilon^-(\hat{c}) < 1$ the set in (8.1) is $\{0\}$ and $\kappa^-(\hat{c}) = 0$; and $\Pr[\mathsf{N}_{\hat{c}} > k] = 0$ for every $k \ge 0$, so the set in (8.2) is all of $\{0, \dots, n\}$ and $\kappa^+(\hat{c}) = 0$. If $p_{\hat{c}} = 1$ then $\mathsf{N}_{\hat{c}} = n$ with probability one, so $\Pr[\mathsf{N}_{\hat{c}} < k] = 0$ for every $k \le n$ and $\kappa^-(\hat{c}) = n$; and $\Pr[\mathsf{N}_{\hat{c}} > k]$ is 1 for $k \le n-1$ and 0 at $k = n$, so with $\varepsilon^+(\hat{c}) < 1$ the set in (8.2) is $\{n\}$ and $\kappa^+(\hat{c}) = n$.

The consequence is Lemma 7.2 applied to $R = R_{\mathrm{ex}}$.

(iii). Extremality is the definition of the maximum and the minimum together with the monotonicity of the two tails in k (Lemma 7.2 (ii)); the relaxation statement is Lemma 7.2 (ii) with $R^{\mp} = \kappa^{\mp}(\hat{c})$.

(iv). By Corollary 2.2, with $p = p_{\hat{c}}$, the tails in the inequalities that define the sets of (8.1) and (8.2) are the closed forms (2.5). Under a conservative evaluation $p_{\hat{c}}$, a coordinate of p^{hon^-} or of p^{hon^+} (Definition 8.1), is rational, the envelope's vectors being exact rationals (Definition 4.3); and $\varepsilon^{\mp}(\hat{c})$ are rational by Definition 8.1, which also gives $n \ge 1$ and $p_{\hat{c}} \in [0, 1]$. By Lemma 2.3 (ii) each inequality, one for each $k \in \{0, \dots, n\}$ in each set, is then decided exactly from n, $p_{\hat{c}}$, k and $\varepsilon^{\mp}(\hat{c})$. So both sets, and with them their maximum $\kappa^-(\hat{c})$ and minimum $\kappa^+(\hat{c})$, are computed exactly from n, $p_{\hat{c}}$ and the allocation alone.

(v). By (ii), $\kappa^-(\hat{c}) = \kappa^+(\hat{c}) = 0$ at $p_{\hat{c}} = 0$, and $\lfloor n \cdot 0 \rfloor = \lceil n \cdot 0 \rceil = 0$, so (7.2) gives $k^-(\hat{c}) = k^+(\hat{c}) = 0$; at $p_{\hat{c}} = 1$, $\kappa^{\pm}(\hat{c}) = n$ and $\lfloor n \rfloor = \lceil n \rceil = n$, so $k^-(\hat{c}) = k^+(\hat{c}) = n$. In the first case $\Pr[\mathsf{N}_{\hat{c}} < 0] = 0$ and $\Pr[\mathsf{N}_{\hat{c}} > 0] = 0$ because $\mathsf{N}_{\hat{c}} = 0$ almost surely; in the second, $\Pr[\mathsf{N}_{\hat{c}} < n] = 0$ and $\Pr[\mathsf{N}_{\hat{c}} > n] = 0$ because $\mathsf{N}_{\hat{c}} = n$ almost surely. So both tails in (7.4) are exactly 0.

(vi). The identities are Lemma 7.2 (iii) with $R^{\mp} = \kappa^{\mp}(\hat{c})$.

Unboundedness. If $\varepsilon^-(\hat{c}) \ge 1 - p^n$ then $\Pr[\mathsf{N}_{\hat{c}} < n] = \Pr[\mathsf{N}_{\hat{c}} \le n-1] = 1 - p^n \le \varepsilon^-(\hat{c})$, so $k = n$ is admissible in (8.1) and $\kappa^-(\hat{c}) = n$; the lower

enlargement is then $n - \lfloor np \rfloor \geq n - np = n(1-p)$ counts, i.e. at least $1-p$ in frequency. Such an $\varepsilon^-(\hat{c})$ is admissible whenever $\varepsilon_{\text{comp}}$ permits, because $1 - p^n < 1$ for $p \in (0,1)$. Symmetrically, if $\varepsilon^+(\hat{c}) \geq 1 - (1-p)^n$ then $\Pr[\mathsf{N}_{\hat{c}} > 0] = 1 - (1-p)^n \leq \varepsilon^+(\hat{c})$, so $k = 0$ is admissible in (8.2) and $\kappa^+(\hat{c}) = 0$, giving an upper enlargement of $\lceil np \rceil$ counts.

The four exact conditions. We use twice that $\varepsilon^{\pm}(\hat{c}) \leq \varepsilon_{\text{comp}} < 1$, which Definition 8.1 records for every admissible input.

First, for every integer $K \geq 0$,

$$\kappa^-(\hat{c}) \leq K \quad \Longleftrightarrow \quad \Pr\left[\mathsf{N}_{\hat{c}} \leq K\right] > \varepsilon^-(\hat{c}).$$

If $K + 1 \leq n$: ($\Rightarrow$) $K+1$ is then not admissible in (8.1), by maximality of $\kappa^-(\hat{c})$, which is exactly $\Pr[\mathsf{N}_{\hat{c}} < K+1] = \Pr[\mathsf{N}_{\hat{c}} \leq K] > \varepsilon^-(\hat{c})$; ($\Leftarrow$) for every integer j with $K+1 \leq j \leq n$, monotonicity of $k \mapsto \Pr[\mathsf{N}_{\hat{c}} < k]$ (Lemma 7.2 (ii)) gives $\Pr[\mathsf{N}_{\hat{c}} < j] \geq \Pr[\mathsf{N}_{\hat{c}} \leq K] > \varepsilon^-(\hat{c})$, so no such j is admissible and the maximum is at most K. If instead $K+1 > n$, both sides hold unconditionally: $\kappa^-(\hat{c}) \leq n \leq K$ by (i), and $\Pr[\mathsf{N}_{\hat{c}} \leq K] = 1 > \varepsilon^-(\hat{c})$. By (7.5) the lower enlargement is 0 iff $\kappa^-(\hat{c}) \leq \lfloor np \rfloor$ and is at most 1 iff $\kappa^-(\hat{c}) \leq \lfloor np \rfloor + 1$, so taking $K = \lfloor np \rfloor$ and $K = \lfloor np \rfloor + 1$ gives (8.4) and (8.5).

Second, and symmetrically, for every integer $K \leq n$,

$$\kappa^+(\hat{c}) \geq K \quad \Longleftrightarrow \quad \Pr\left[\mathsf{N}_{\hat{c}} \geq K\right] > \varepsilon^+(\hat{c}),$$

by the same argument applied to the non-increasing map $k \mapsto \Pr[\mathsf{N}_{\hat{c}} > k]$ and the minimality of $\kappa^+(\hat{c})$: for $K - 1 \geq 0$ the index $K-1$ is inadmissible iff $\Pr[\mathsf{N}_{\hat{c}} > K-1] = \Pr[\mathsf{N}_{\hat{c}} \geq K] > \varepsilon^+(\hat{c})$, and for $K \leq 0$ both sides hold unconditionally. Taking $K = \lceil np \rceil$ and $K = \lceil np \rceil - 1$, both of which are at most n, gives (8.6) and (8.7).

The distance statement is Lemma 7.2 (iv).

(vii). Both sets are non-empty: the condition defining the first is vacuous at $k = 0$, and $k = n$ belongs to the second because $U_n^+ = 0$. Write $\tilde{k}^{\mp} := \tilde{\kappa}^{\mp}(\hat{c})$. *Admissibility.* If $\tilde{k}^- = 0$ then $\Pr\left[\mathsf{N}_{\hat{c}} < \tilde{k}^-\right] = 0$; otherwise $\Pr\left[\mathsf{N}_{\hat{c}} < \tilde{k}^-\right] \leq U^-_{\tilde{k}^-} \leq \varepsilon^-(\hat{c})$, the last step being the defining condition at $j = \tilde{k}^-$. And $\Pr\left[\mathsf{N}_{\hat{c}} > \tilde{k}^+\right] \leq U^+_{\tilde{k}^+} \leq \varepsilon^+(\hat{c})$. Both endpoints lie in $\{0, \ldots, n\}$, so this is Definition 7.1 (a); it also places $\tilde{k}^-$ in the set of (8.1) and $\tilde{k}^+$ in that of (8.2), so $\tilde{k}^- \leq \kappa^-(\hat{c})$ and $\tilde{k}^+ \geq \kappa^+(\hat{c})$ by the extremality of (iii). *Zero budget.* Let $0 < p_{\hat{c}} < 1$. At $\varepsilon^-(\hat{c}) = 0$, (ii) gives $\kappa^-(\hat{c}) = 0$, so $0 \leq \tilde{k}^- \leq 0$; at $\varepsilon^+(\hat{c}) = 0$ it gives $\kappa^+(\hat{c}) = n$, so $n \geq \tilde{k}^+ \geq n$. This is Definition 7.1 (b). *Equality with the exact rule.* If every $U_k^{\mp}$ equals its tail, the second set is that of (8.2); and since $k \mapsto \Pr[\mathsf{N}_{\hat{c}} < k]$ is non-decreasing (Lemma 7.2 (ii)), the condition holds at every $j \leq k$ exactly when it holds at $j = k$, so the first set is that of (8.1). Hence $\tilde{k}^{\mp} = \kappa^{\mp}(\hat{c})$. *Degenerate mass.* At $p_{\hat{c}} \in \{0, 1\}$

every bound equals its tail, so the rule is the exact one there, and (ii) gives Definition 7.1 (c). *Monotonicity in the allocation.* The bounds do not depend on the allocation. Raising $\varepsilon^-(\hat{c})$ enlarges the first set, so its maximum does not fall; raising $\varepsilon^+(\hat{c})$ enlarges the second, so its minimum does not rise. The consequence is Lemma 7.2 applied to $R = \tilde{R}$. □

The guard buys honest containment for every budget and every allocation, and never raises the failure probability. Its cost is not bounded by one count against the raw quantile interval, yet an endpoint it moves stays within 1/n of the honest value.

Remark 8.3 (Benefit and cost of the honest-containment guard). This remark reads the honest-containment guard on the exact rule of Definition 8.1, whose rule endpoints are the quantiles $\kappa^{\pm}$; Lemma 7.2 states what holds for every rule. It is written for a single honest value $p_{\hat{c}} = p^{\mathrm{hon}}(\hat{c})$, the degenerate envelope; for a general envelope every statement below holds side by side, by the side-wise reading of Definition 7.1: what it says of the lower endpoint with $p_{\hat{c}} = p^-_{\hat{c}}$, and what it says of the upper endpoint with $p_{\hat{c}} = p^+_{\hat{c}}$.

What the guard buys is unconditional: the whole envelope interval $[p^-_{\hat{c}}, p^+_{\hat{c}}]$ — for a single law, $p^{\mathrm{hon}}(\hat{c})$ — lies in $[k^-(\hat{c})/n,\, k^+(\hat{c})/n]$ for *every* admissible budget and *every* admissible allocation, including allocations so lopsided that the raw quantile interval $[\kappa^-(\hat{c}), \kappa^+(\hat{c})]$ excludes the honest value. Theorem 9.2 (iii) and Theorem 11.1 rest on exactly that, and without the guard both would carry a side condition on how large $\varepsilon_{\mathrm{comp}}$ may be. By Proposition 8.2 (iii) the widening is always admissible: it can only lower the failure probability.

What the guard costs has two different and equally correct answers, both stated in Lemma 7.2 (iii)–(iv) and Proposition 8.2 (vi); the second alone would overstate how little the guard costs.

Against the raw quantile interval the cost is (7.5), and it is *not* bounded by one count. A concrete admissible input: $\widehat{\mathcal{C}}_{\mathrm{con}} = \{\hat{c}\}$, $n = 10$, $p_{\hat{c}} = \frac{1}{2}$, $\varepsilon^-(\hat{c}) = \frac{9}{10}$ and $\varepsilon^+(\hat{c}) = \frac{1}{20}$, the allocation summing to $\frac{19}{20}$ and so admissible for any $\varepsilon_{\mathrm{comp}} \in [\frac{19}{20}, 1)$. There $\Pr[\mathsf{N}_{\hat{c}} < 7] = \frac{53}{64} \le \frac{9}{10}$ while $\Pr[\mathsf{N}_{\hat{c}} < 8] = \frac{121}{128} > \frac{9}{10}$, so $\kappa^- = 7$ against $\lfloor np_{\hat{c}} \rfloor = 5$: the guard widens the interval by two counts. Proposition 8.2 (vi) shows the widening is unbounded in the same direction — it is at least $1 - p_{\hat{c}}$ in frequency, uniformly in n, once a single tail's allocation reaches $1 - p_{\hat{c}}^n$.

Against the honest value the quantity bounded is the emitted endpoint's *distance* from $p^{\mathrm{hon}}(\hat{c})$, not the enlargement, and that distance is at most $1/n$, unconditionally: by (7.6), on the side where the guard acts the emitted endpoint sits within $1/n$ of $p^{\mathrm{hon}}(\hat{c})$. This is the sense in which the guard is cheap, and it is the sense that survives the comparison with Theorem 12.4 (iv): the $O(n^{-1/2})$ half-width of (12.5) is proved for the *guarded* endpoints, with no side condition, so the guard never breaks that bound.

Uses Def. 7.1, Lem. 7.2, Def. 8.1, Prop. 8.2 · *Used by* Rem. 10.3 · newly derived · accepted

Four exact comparisons of the allocation with the binomial distribution function decide whether the guard acts and whether it costs more than one count. It acts even at ordinary budgets, on any outcome whose positive expected count is at most its tail budget.

Remark 8.4 (Regimes in which the guard acts). This remark reads the guard on the exact rule of Definition 8.1, written for a single honest value $p_{\hat{c}} = p^{\mathrm{hon}}(\hat{c})$, the degenerate envelope; for a general envelope every statement holds side by side, by the side-wise reading of Definition 7.1.

The four exact conditions (8.4)–(8.7) of Proposition 8.2 (vi) say where the guard acts and what it costs there. By (8.4) and (8.6) the guard is entirely inert as soon as each allocated tail budget is strictly below the mass the count law places at or below $\lfloor np_{\hat{c}} \rfloor$, respectively at or above $\lceil np_{\hat{c}} \rceil$; by (8.5) and (8.7) it costs at most one count under the correspondingly weaker conditions. All four are comparisons of an allocated budget against a value of the binomial distribution function, hence decidable exactly in the closed forms (2.5) of Corollary 2.2: each is the negation of a comparison of Lemma 2.3 (ii) at the count next to the one it names, or a boundary value of Theorem 2.1 where that count leaves $\{0, \dots, n\}$, with no machinery beyond what Definition 8.1 already requires. The two pairs fail in different regimes, however, and are not to be read together.

A cost of more than one count — failure of (8.5) or of (8.7) — requires a single outcome's single tail to carry a budget at least the mass the count law places up to one step beyond $np_{\hat{c}}$, that is $\varepsilon^-(\hat{c}) \geq \Pr[\mathsf{N}_{\hat{c}} \leq \lfloor np_{\hat{c}} \rfloor + 1]$ respectively $\varepsilon^+(\hat{c}) \geq \Pr[\mathsf{N}_{\hat{c}} \geq \lceil np_{\hat{c}} \rceil - 1]$. At the ends of the range that cannot happen at all: if $np_{\hat{c}} < 1$ then $\lceil np_{\hat{c}} \rceil - 1 \leq 0$ and $\Pr[\mathsf{N}_{\hat{c}} \geq \lceil np_{\hat{c}} \rceil - 1] = 1 > \varepsilon^+(\hat{c})$, and if $\lfloor np_{\hat{c}} \rfloor \geq n-1$ then $\lfloor np_{\hat{c}} \rfloor + 1 \geq n$ and $\Pr[\mathsf{N}_{\hat{c}} \leq \lfloor np_{\hat{c}} \rfloor + 1] = 1 > \varepsilon^-(\hat{c})$, both using $\varepsilon^{\pm}(\hat{c}) \leq \varepsilon_{\mathrm{comp}} < 1$. In the example of Remark 8.3 the budget is indeed near-unit: $\varepsilon^-(\hat{c}) = \frac{9}{10}$ against $\Pr[\mathsf{N}_{\hat{c}} \leq 6] = \frac{53}{64}$. That is also the regime of the $\varepsilon_{\mathrm{comp}} = \frac{4}{5}$-on-one-outcome counterexample of Theorem 12.4 (v), and under an allocation that concentrates a near-unit budget on one or two constrained outcomes the enlargement is to be read off (7.5), not estimated by a rule of thumb.

Activation of the guard at all — failure of (8.4) or of (8.6) — is a different matter: it is not confined to near-unit budgets. It occurs at ordinary budgets and under the engine's default uniform allocation rule, for any outcome whose expected count over the whole run is positive and at or below the tail budget allocated to it. Precisely, if $0 < np_{\hat{c}} \leq \varepsilon^+(\hat{c})$ — which forces $np_{\hat{c}} < 1$ and hence $\lceil np_{\hat{c}} \rceil = 1$, since $\varepsilon^+(\hat{c}) < 1$ — then Bernoulli's inequality $(1 - p_{\hat{c}})^n \geq 1 - np_{\hat{c}}$ gives

$$\Pr\left[\mathsf{N}_{\hat{c}} \geq \lceil np_{\hat{c}} \rceil\right] = 1 - (1 - p_{\hat{c}})^n \;\leq\; np_{\hat{c}} \;\leq\; \varepsilon^+(\hat{c}),$$

so (8.6) fails, $\kappa^+(\hat{c}) = 0$, and the guard raises the emitted endpoint to $k^+(\hat{c}) = 1$. Symmetrically, if $0 < n(1 - p_{\hat{c}}) \leq \varepsilon^-(\hat{c})$ then $\lfloor np_{\hat{c}} \rfloor = n - 1$ and $\Pr[\mathsf{N}_{\hat{c}} \leq n - 1] = 1 - p_{\hat{c}}^n \leq n(1 - p_{\hat{c}}) \leq \varepsilon^-(\hat{c})$ — the same inequality applied at $1 - p_{\hat{c}}$ — so (8.4) fails, $\kappa^-(\hat{c}) = n$, and the guard lowers the emitted endpoint to $k^-(\hat{c}) = n - 1$. A concrete instance is an admissible input with the engine's default rule and default allocation (Requirements 16.1 and 16.2): $\widehat{\mathcal{C}} = \{\hat{c}_0, \hat{c}_1\}$, $n = 10$, $p^{\mathrm{hon}} = (\frac{999}{1000}, \frac{1}{1000})$, $\widehat{\mathcal{C}}_{\mathrm{con}} = \widehat{\mathcal{C}}$, $\varepsilon_{\mathrm{comp}} = \frac{1}{10}$ and the uniform allocation $\varepsilon^{\pm}(\hat{c}) = \varepsilon_{\mathrm{comp}}/(2|\widehat{\mathcal{C}}_{\mathrm{con}}|) = \frac{1}{40}$; there $np_{\hat{c}_1} = n(1 - p_{\hat{c}_0}) = \frac{1}{100} < \frac{1}{40}$, so the guard is active on the upper side of $\hat{c}_1$ and on the lower side of $\hat{c}_0$ — at a budget nowhere near 1, and with the budget spread uniformly rather than concentrated. Nor is this a small-n artefact: at $n = 10^6$ with $\varepsilon_{\mathrm{comp}} = \frac{1}{10}$ and $|\widehat{\mathcal{C}}_{\mathrm{con}}| = 5$ the uniform rule allocates $\varepsilon^{\pm}(\hat{c}) = 10^{-2}$, and a constrained outcome

of honest probability 10^{-9} has $np_{\hat{c}} = 10^{-3} \leq \varepsilon^{+}(\hat{c})$, so the guard forces $q^{+}(\hat{c}) = 1/n$ where the raw quantile interval would have given 0. Under A-TSM (Assumption "Output-state test statistics" (reductions/ass:tsm)), this module's alphabet being that of the consuming declaration (Definition 1.3), the acceptance alphabet is the joint outcome, carrying every per-round public choice: one made at a measuring station as a component of the outcome by A-TSM, and one on the letter side as a component of the letter (Definition "Protocol specification" (protocol-compiler/def:spec) (a)), drawn as declared by A-COIN (Assumption "Public choices as private coins of declared bias" (protocol-compiler/ass:coin)); so outcomes of expected count at or below one are a routine feature of a fine-grained $\widehat{\mathcal{C}}$ rather than an exotic one. What the guard costs on them is nevertheless bounded, by both measures: exactly one count — the relevant condition of the previous paragraph holds in each case, $\lceil np_{\hat{c}} \rceil - 1 \leq 0$ giving (8.7) on the upper side and $\lfloor np_{\hat{c}} \rfloor + 1 \geq n$ giving (8.5) on the lower side — and at most $1/n$ from the honest value by (7.6).

Uses Def. 1.3, Thm. 2.1, Cor. 2.2, Lem. 2.3, Def. 7.1, Lem. 7.2, Def. 8.1, Prop. 8.2, Def. 9.1 · *Assumes* A-TSM, A-ALPHABET · newly derived · accepted

9 The construction map

The guarded counts divided by the round count bound the constrained outcomes, the unconstrained ones keep the bounds 0 and 1, and the box is intersected with the simplex. The main result, Theorem 9.2, is that for every admissible endpoint rule the output is a canonical box of exact rationals containing every law of the envelope set, a non-empty, compact and convex subset of the simplex by construction.

The construction map bounds each constrained outcome by its guarded count endpoints divided by the round count, gives every other outcome the bounds 0 and 1, and intersects the box with the probability simplex. Its output is a polytope in canonical box form built from the honest envelope.

The map is defined for every admissible endpoint rule; which rule is used, by default the exact binomial quantile rule of Definition 8.1, is Requirement 16.1.

Definition 9.1 (The construction map)**.** Let $(\widehat{\mathcal{C}}, p^{\mathrm{hon}-}, p^{\mathrm{hon}+}, n, \varepsilon_{\mathrm{comp}}, \widehat{\mathcal{C}}_{\mathrm{con}}, \varepsilon^{-}, \varepsilon^{+})$ be the data of an admissible input (Definition 7.1), where $(p^{\mathrm{hon}-}, p^{\mathrm{hon}+})$ is its honest envelope (Definition 4.1) — an input p^{hon} in the point form being read as the degenerate envelope $(p^{\mathrm{hon}}, p^{\mathrm{hon}})$ — and let R be an admissible endpoint rule (Definition 7.1). With the guarded endpoints $k^{\pm}$ of R, (7.2), define

$$q^{-}(\hat{c}) := \begin{cases} k^{-}(\hat{c})/n, & \hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}, \\ 0, & \hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{con}}, \end{cases} \qquad q^{+}(\hat{c}) := \begin{cases} k^{+}(\hat{c})/n, & \hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}, \\ 1, & \hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{con}}. \end{cases} \tag{9.1}$$

The *construction map* Acc_R — written Acc where the rule is clear or is the default — returns the polytope geometry in canonical box form with these

$(\mathbf{q}^-, \mathbf{q}^+)$, i.e. the data (1.4), and

$$\mathcal{S}_{\mathrm{acc}} \;\coloneqq\; \{\mathbf{q} \in \Delta(\widehat{\mathcal{C}}) :\; q^-(\hat{c}) \le q(\hat{c}) \le q^+(\hat{c}) \;\; \forall \hat{c} \in \widehat{\mathcal{C}}\}. \tag{9.2}$$

Uses Def. 1.4, Def. 4.1, Def. 7.1, Def. 8.1 · *Used by* Rem. 8.4, Thm. 9.2, Rem. 9.3, Thm. 10.2, Lem. 12.2, Thm. 12.4, Prop. 15.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.statistics.construction:construct

For any admissible endpoint rule (count bounds meeting binomial tail budgets), the acceptance box has exact rational bounds and contains every law within the honest envelope's bounds. It is non-empty and compact and, by construction, an admissible acceptance set (Definition "Admissible acceptance set" (meat-core/ def:admissible-acceptance-set)).

Whatever admissible rule produced the endpoints, the output is a valid acceptance set that accepts every honest law of the envelope: the guard, not the rule, places the envelope inside the box.

Theorem 9.2 (Properties of the constructed box). On every admissible input (Definition 7.1) and for every admissible endpoint rule R, under a conservative evaluation (Definition 4.3):

(i) Acc_R is well defined, and all of its output data are exact rationals: each $q^{\pm}(\hat{c})$ is either an integer multiple of $1/n$ or one of $0, 1$, and the entries of B lie in $\{0, \pm 1\}$;

(ii) $0 \le \mathbf{q}^- \le \mathbf{q}^+ \le \mathbf{1}$ componentwise, so the pair $(\mathbf{q}^-, \mathbf{q}^+)$ meets the conditions of the canonical box form of Definition 1.4;

(iii) $\mathsf{H} \subseteq \mathcal{S}_{\mathrm{acc}}$, where H is the envelope set (4.2) of the input envelope: every honest law the envelope admits is accepted, and for the degenerate envelope this reads $p^{\mathrm{hon}} \in \mathcal{S}_{\mathrm{acc}}$. The envelope set is non-empty; consequently $\mathcal{S}_{\mathrm{acc}} \neq \varnothing$ and, by Theorem 1.5 (iii), $\sum_{\hat{c}} q^-(\hat{c}) \le 1 \le \sum_{\hat{c}} q^+(\hat{c})$;

(iv) $\mathcal{S}_{\mathrm{acc}}$ is a non-empty, closed, convex, compact subset of $\Delta(\widehat{\mathcal{C}})$; Definition "Admissible acceptance set" (meat-core/def:admissible-acceptance-set) holds for the output of Acc_R by construction, for every admissible input and every admissible rule.

Uses Def. 1.4, Thm. 1.5, Def. 4.1, Def. 4.3, Def. 7.1, Lem. 7.2, Def. 9.1 · *Assumes* A-ALPHABET · *Used by* Rem. 10.5, Thm. 11.1, Thm. 12.4 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.statistics.construction: construct · *Oracles* registry_statistics_acceptance.py

Proof. *(i).* Well-definedness: R returns integers in $\{0, \dots, n\}$ (Definition 7.1 (a)), so the guarded endpoints (7.2) exist and are unique, and (9.1)–(9.2) are explicit formulas in them. The rationality statement is immediate: $k^{\pm}(\hat{c})$ are integers in $\{0, \dots, n\}$ and $n \ge 1$.

(ii). For $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$, (7.3) gives $0 \le k^-(\hat{c}) \le k^+(\hat{c}) \le n$, hence $0 \le q^-(\hat{c}) \le q^+(\hat{c}) \le 1$. For $\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{con}}$, $0 \le 0 \le 1 \le 1$.

(iii). Let $\mathbf{p} \in \mathsf{H}$, let $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$ and write $p^{\mp} = p^{\mp}_{\hat{c}}$. By (7.3) and (4.2), $q^{-}(\hat{c}) = k^{-}(\hat{c})/n \le \lfloor np^{-} \rfloor / n \le np^{-}/n = p^{-} \le p(\hat{c})$ and $q^{+}(\hat{c}) = k^{+}(\hat{c})/n \ge \lceil np^{+} \rceil / n \ge np^{+}/n = p^{+} \ge p(\hat{c})$, so $q^{-}(\hat{c}) \le p(\hat{c}) \le q^{+}(\hat{c})$. For $\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{con}}$ the bounds are $0 \le p(\hat{c}) \le 1$, which hold because $\mathbf{p} \in \Delta(\widehat{\mathcal{C}})$. Since also $\mathbf{p} \in \Delta(\widehat{\mathcal{C}})$, all defining conditions of (9.2) are met and $\mathbf{p} \in \mathcal{S}_{\mathrm{acc}}$; hence $\mathsf{H} \subseteq \mathcal{S}_{\mathrm{acc}}$. For the degenerate envelope $\mathsf{H} = \{p^{\mathrm{hon}}\}$ (Definition 4.1).

Non-emptiness of H: by Definition 4.1 the pair $(p^{\mathrm{hon}^{-}}, p^{\mathrm{hon}^{+}})$ is a canonical box form with denotation H whose bounds satisfy (1.6), so Theorem 1.5 (iii), by which (1.6) and non-emptiness are equivalent, gives $\mathsf{H} \neq \varnothing$. Hence $\mathcal{S}_{\mathrm{acc}} \supseteq \mathsf{H}$ is non-empty. The box sums follow from the same equivalence applied to the non-empty $\mathcal{S}_{\mathrm{acc}}$, a canonical box by (ii).

(iv). Theorem 1.5 (i) applied to the geometry of (ii), whose validation predicate holds by (iii). □

Constraining fewer outcomes narrows each interval but leaves more coordinates bounded only by the simplex, so no choice of the constrained outcomes dominates. Completeness holds for every choice, and the unconstrained outcomes keep the bounds 0 and 1, so the box always has two rows per outcome.

Which outcomes are constrained is part of the input; the default, all of them, is Requirement 16.2.

Remark 9.3 (The role of the constrained outcome set). Constraining fewer outcomes concentrates the budget on those that are constrained, which narrows each interval; it also weakens the set, because the unconstrained coordinates are then bounded only by the simplex. Both effects are real and pull in opposite directions, so no choice of $\widehat{\mathcal{C}}_{\mathrm{con}}$ dominates. Completeness holds for whatever $\widehat{\mathcal{C}}_{\mathrm{con}}$ is supplied (Theorem 10.2), and the unconstrained coordinates receive the vacuous bounds $[0, 1]$ of (9.1) rather than being omitted from B — so the row count $m_{\mathrm{row}} = 2|\widehat{\mathcal{C}}|$ does not depend on $\widehat{\mathcal{C}}_{\mathrm{con}}$ and a consumer never has to infer a missing row.

Uses Def. 9.1 · *Used by* Req. 16.2 · newly derived · accepted

10 Completeness

A union bound combines the allocated tails into a bound on honest rejection, with no independence between outcome counts. A count whose per-round conditional success probability stays between two bounds has tails no heavier than the binomial laws at those bounds (Lemma 10.1), so independent rounds are not needed either. The main result, Theorem 10.2, bounds the honest run's probability of failing the test by ε^{*}, the sum of those tails, and ε^{*} by the budget; the remarks say how ε^{*} is evaluated and reported, why soundness does not depend on it, and what an envelope buys and costs.

A count of successes whose conditional success probability in each round, given the past, lies in [a, b] has an upper tail no heavier than the binomial law at b and a lower tail no heavier than that at a. The box can therefore be built from binomial tails without independent rounds.

This is the classical stochastic dominance of a sum of adapted Bernoulli variables by a binomial law, derived here in full by a backward induction on the conditional tail. Under Assumption 5.2 the indicators of one outcome over the rounds of an honest run are such a sequence, with a and b the envelope's two values at that outcome (Theorem 10.2).

Lemma 10.1 (Binomial dominance under bounded conditional success probabilities)**.** Let $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}_n$ be a filtration on a probability space, $n \geq 1$, and let $\chi_1, \ldots, \chi_n$ be $\{0,1\}$-valued random variables with χ_j $\mathcal{F}_j$-measurable. Let $0 \leq a \leq b \leq 1$ and suppose that, for every $j \in \{1, \ldots, n\}$, almost surely

$$a \;\leq\; P_j \;:=\; \Pr\left[\chi_j = 1 \;\middle|\; \mathcal{F}_{j-1}\right] \;\leq\; b. \tag{10.1}$$

Put $S := \sum_{j=1}^{n} \chi_j$, and for $m \in \{0, \ldots, n\}$, $r \in [0,1]$ and an integer k write $G_m^r(k) := \Pr[\mathrm{B}_m^r \geq k]$, where B_m^r is binomial with parameters (m, r) ($\mathrm{B}_0^r = 0$). Then for every integer k

$$\Pr[S \geq k] \;\leq\; \Pr\left[\mathrm{B}_n^b \geq k\right], \qquad \Pr[S \leq k] \;\leq\; \Pr\left[\mathrm{B}_n^a \leq k\right], \tag{10.2}$$

and equivalently

$$\Pr[S > k] \;\leq\; \Pr\left[\mathrm{B}_n^b > k\right], \qquad \Pr[S < k] \;\leq\; \Pr\left[\mathrm{B}_n^a < k\right]. \tag{10.3}$$

Used by Thm. 10.2 · newly derived · accepted

Proof. Two facts about the binomial tail. (1) For $m \geq 0$ and every integer k, $G_m^r(k-1) - G_m^r(k) = \Pr[\mathrm{B}_m^r = k-1] \geq 0$. (2) The *Pascal recursion of the binomial tail*: for $m \geq 0$ and every integer k,

$$G_{m+1}^r(k) \;=\; r\, G_m^r(k-1) + (1-r)\, G_m^r(k). \tag{10.4}$$

Indeed B_{m+1}^r has the law of $X+Y$ with Y binomial (m, r) and X Bernoulli(r) independent of Y, and conditioning on X gives $\Pr[X+Y \geq k] = r \Pr[Y \geq k-1] + (1-r)\Pr[Y \geq k]$.

The upper tail, in conditional form. For $j \in \{0, \ldots, n\}$ put $S_{>j} := \sum_{i=j+1}^{n} \chi_i$ (so $S_{>0} = S$ and $S_{>n} = 0$). We show by backward induction on j that for every integer k, almost surely,

$$\Pr\left[S_{>j} \geq k \;\middle|\; \mathcal{F}_j\right] \;\leq\; G_{n-j}^b(k). \tag{10.5}$$

Base $j = n$. $S_{>n} = 0$, so the left side is $\mathbf{1}\{k \leq 0\} = G_0^b(k)$. *Step from j to $j-1$*, for $j \in \{1, \ldots, n\}$, with $m := n - j$. Since $S_{>j-1} = \chi_j + S_{>j}$ and $\chi_j \in \{0,1\}$,

$$\mathbf{1}\{S_{>j-1} \geq k\} = \chi_j\, \mathbf{1}\{S_{>j} \geq k-1\} + (1-\chi_j)\, \mathbf{1}\{S_{>j} \geq k\}.$$

Conditioning on $\mathcal{F}_j$, with χ_j $\mathcal{F}_j$-measurable and bounded, and using the induction hypothesis at $k-1$ and at k together with $\chi_j \geq 0$ and $1-\chi_j \geq 0$, gives almost surely

$$\Pr\left[S_{>j-1} \geq k \,\middle|\, \mathcal{F}_j\right] \leq \chi_j\, G_m^b(k-1) + (1-\chi_j)\, G_m^b(k).$$

Conditioning further on $\mathcal{F}_{j-1} \subseteq \mathcal{F}_j$ (the tower property, and monotonicity of conditional expectation) gives almost surely

$$\Pr\left[S_{>j-1} \geq k \,\middle|\, \mathcal{F}_{j-1}\right] \leq G_m^b(k) + P_j\left(G_m^b(k-1) - G_m^b(k)\right) \leq G_m^b(k) + b\left(G_m^b(k-1) - G_m^b(k)\right) = G_{m+1}^b(k),$$

the second inequality by $P_j \leq b$ from (10.1) and fact (1), the equality by (10.4). As $m+1 = n-(j-1)$, this is (10.5) at $j-1$. At $j=0$, taking expectations gives $\Pr[S \geq k] \leq G_n^b(k)$, the first inequality of (10.2). Only the upper bound b was used.

The lower tail. Put $\chi'_j := 1-\chi_j$. These are $\{0,1\}$-valued and adapted, and $\Pr\left[\chi'_j = 1 \,\middle|\, \mathcal{F}_{j-1}\right] = 1-P_j \leq 1-a$ almost surely by (10.1). The upper-tail statement applied to them with the bound $1-a$ gives, for every integer k, $\Pr[S \leq k] = \Pr[\,n-S \geq n-k\,] \leq \Pr\left[\mathrm{B}_n^{1-a} \geq n-k\right]$. Since $n-\mathrm{B}_n^a$ is binomial with parameters $(n, 1-a)$ (it counts the failures), the right side is $\Pr[n-\mathrm{B}_n^a \geq n-k] = \Pr[\mathrm{B}_n^a \leq k]$, the second inequality of (10.2).

The strict forms. S and the binomial variables are integer-valued, so $\{S > k\} = \{S \geq k+1\}$ and $\{S < k\} = \{S \leq k-1\}$, and likewise for B_n^b and B_n^a; (10.2) at $k+1$ and at $k-1$ is (10.3), and conversely. □

If each honest round's conditional outcome law stays inside the envelope, an honest run fails the acceptance test with probability at most ε^*, the sum of the binomial tails at the envelope's ends beyond the box's count bounds, and ε^* is at most the completeness budget. No independence is needed.

The theorem covers a single honest law as a special case: Assumption A-HON, the acceptance statistics being independent and identically distributed with law p^{hon}, is Assumption 5.2 for the degenerate envelope $(p^{\mathrm{hon}}, p^{\mathrm{hon}})$ (Lemma 4.2). On an evaluation that is not conservative, an upward-rounded k^- or a downward-rounded k^+ can break the bound by the reported completeness figure $\varepsilon^\star$: a figure is reported that the set does not meet (Lemma 12.2 (v)).

Theorem 10.2 (The completeness bound). Assume a conservative evaluation (Definition 4.3), and assume A-HON-ENV (Assumption 5.2) for the honest envelope $(p^{\mathrm{hon}^-}, p^{\mathrm{hon}^+})$ of the input: on an honest run there is a filtration $\mathcal{F}_0 \subseteq \cdots \subseteq \mathcal{F}_n$ with $\widehat{C}_j$ $\mathcal{F}_j$-measurable and $p^{\mathrm{hon}^-}(\hat{c}) \leq \Pr\left[\widehat{C}_j = \hat{c} \,\middle|\, \mathcal{F}_{j-1}\right] \leq p^{\mathrm{hon}^+}(\hat{c})$ almost surely, for every j and $\hat{c}$. Let $\mathcal{S}_{\mathrm{acc}}$ be the output of Acc_R (Definition 9.1) for an admissible endpoint rule R

(Definition 7.1). Then

$$\Pr\left[\,\mathrm{freq}_{\widehat{C}_1^n} \notin \mathcal{S}_{\mathrm{acc}}\right] \;\le\; \varepsilon^\star \;:=\; \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{con}}} \Big(\Pr\left[\mathsf{N}^-_{\hat{c}} < k^-(\hat{c})\right] + \Pr\left[\mathsf{N}^+_{\hat{c}} > k^+(\hat{c})\right]\Big) \;\le\; \varepsilon_{\mathrm{comp}}, \tag{10.6}$$

where $\mathsf{N}^-_{\hat{c}}$ and $\mathsf{N}^+_{\hat{c}}$ are binomial with parameters $(n, {p^{\mathrm{hon}}}^-(\hat{c}))$ and $(n, {p^{\mathrm{hon}}}^+(\hat{c}))$ respectively (Definition 7.1). Equivalently, the honest run passes the acceptance test, the event Ω_{AT} of Definition "Fixed-length post-processing model" (meat-core/def:postprocessing-model), with probability $\Pr[\Omega_{\mathrm{AT}}] = \Pr\left[\mathrm{freq}_{\widehat{C}_1^n} \in \mathcal{S}_{\mathrm{acc}}\right] \ge 1 - \varepsilon^\star \ge 1 - \varepsilon_{\mathrm{comp}}$. For the degenerate envelope both $\mathsf{N}^{\mp}_{\hat{c}}$ are binomial $(n, p^{\mathrm{hon}}(\hat{c}))$.

Uses Def. 1.1, Lem. 1.2, Def. 4.1, Def. 4.3, Def. 7.1, Lem. 7.2, Def. 9.1, Lem. 10.1 · *Assumes* A-HON-ENV, A-ALPHABET · *Used by* Rem. 10.3, Rem. 10.4, Rem. 10.5, Thm. 12.4, Rem. 14.3, Req. 16.4, Req. 16.5, Rem. 18.2 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.statistics.construction: report_completeness · *Oracles* registry_statistics_acceptance.py

Proof. Idea. Each constrained outcome's count has tails no heavier than the binomial laws at the envelope's two values (Lemma 10.1), the test fails exactly when some count leaves its guarded endpoints, and the union bound adds the probabilities of those events with no independence; the guard keeps each tail within its allocation.

Per-outcome tails. Fix $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$ and let χ_j be the random variable that equals 1 if $\widehat{C}_j = \hat{c}$ and 0 otherwise. Each χ_j is $\mathcal{F}_j$-measurable, because $\widehat{C}_j$ is, and $\Pr[\chi_j = 1 \,|\, \mathcal{F}_{j-1}] = \Pr\left[\widehat{C}_j = \hat{c} \,|\, \mathcal{F}_{j-1}\right]$ lies in $[{p^{\mathrm{hon}}}^-(\hat{c}), {p^{\mathrm{hon}}}^+(\hat{c})]$ almost surely by A-HON-ENV. By Definition 1.1, $\mathsf{N}_{\hat{c}}(\widehat{C}_1^n) = \sum_{j=1}^n \chi_j$. Lemma 10.1 with $a = {p^{\mathrm{hon}}}^-(\hat{c})$ and $b = {p^{\mathrm{hon}}}^+(\hat{c})$, in its strict form (10.3), gives for every integer k

$$\Pr\left[\mathsf{N}_{\hat{c}}(\widehat{C}_1^n) < k\right] \le \Pr\left[\mathsf{N}^-_{\hat{c}} < k\right], \qquad \Pr\left[\mathsf{N}_{\hat{c}}(\widehat{C}_1^n) > k\right] \le \Pr\left[\mathsf{N}^+_{\hat{c}} > k\right].$$

No independence *across* different $\hat{c}$ is claimed, and none is used: the counts are in fact negatively related through $\sum_{\hat{c}} \mathsf{N}_{\hat{c}} = n$. Nor is independence *across rounds* used: the per-round conditional laws may change with j and with the past.

Event inclusion. By Lemma 1.2 (i), $\mathrm{freq}_{\widehat{C}_1^n} \in \Delta(\widehat{\mathcal{C}})$ always, so by (9.2) the event $\{\mathrm{freq}_{\widehat{C}_1^n} \notin \mathcal{S}_{\mathrm{acc}}\}$ occurs exactly when some coordinate leaves its box bounds. For $\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{con}}$ the bounds are $[0,1]$ by (9.1) and can never be violated. For $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$, and using $n > 0$, $\mathrm{freq}_{\widehat{C}_1^n}(\hat{c}) < q^-(\hat{c})$ is equivalent to $\mathsf{N}_{\hat{c}}(\widehat{C}_1^n) < k^-(\hat{c})$, and $\mathrm{freq}_{\widehat{C}_1^n}(\hat{c}) > q^+(\hat{c})$ to $\mathsf{N}_{\hat{c}}(\widehat{C}_1^n) > k^+(\hat{c})$. Therefore

$$\left\{\,\mathrm{freq}_{\widehat{C}_1^n} \notin \mathcal{S}_{\mathrm{acc}}\right\} \;=\; \bigcup_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{con}}} \Big(\left\{\mathsf{N}_{\hat{c}}(\widehat{C}_1^n) < k^-(\hat{c})\right\} \cup \left\{\mathsf{N}_{\hat{c}}(\widehat{C}_1^n) > k^+(\hat{c})\right\}\Big).$$

Aggregation. A probability measure is finitely subadditive: for events $A_1, \dots, A_M$, $\Pr[\bigcup_i A_i] \le \sum_i \Pr[A_i]$, with no independence hypothesis. (Equivalently, in complement form, $\Pr[\bigcap_i A_i^c] \ge 1 - \sum_i \Pr[A_i]$.) Applying it to the $2|\widehat{\mathcal{C}}_{\mathrm{con}}|$ events above, and bounding each by the per-outcome tails at $k = k^-(\hat{c})$ and $k = k^+(\hat{c})$, gives the first inequality of (10.6).

Budget. By Lemma 7.2 (ii) each term of $\varepsilon^\star$ is at most the corresponding $\varepsilon^{\mp}(\hat{c})$, and the allocation constraint of an admissible input (Definition 7.1) sums these to at most $\varepsilon_{\mathrm{comp}}$. The reformulation in terms of $\Pr[\Omega_{\mathrm{AT}}]$ is the complement, using $\Omega_{\mathrm{AT}} = \{\mathrm{freq}_{\hat{c}_1^n} \in \mathcal{S}_{\mathrm{acc}}\}$, the acceptance event of Definition "Fixed-length post-processing model" (meat-core/def:postprocessing-model) on the frequency vector of Definition 1.1. For the degenerate envelope, $\mathsf{N}_{\hat{c}}^-$ and $\mathsf{N}_{\hat{c}}^+$ have the one law binomial $(n, p^{\mathrm{hon}}(\hat{c}))$, and it is the law of $\mathsf{N}_{\hat{c}}(\widehat{C}_1^n)$ itself: with B binomial $(n, p^{\mathrm{hon}}(\hat{c}))$, the per-outcome tails, which are Lemma 10.1 at $a = b = p^{\mathrm{hon}}(\hat{c})$, give for every integer k $\Pr\left[\mathsf{N}_{\hat{c}}(\widehat{C}_1^n) < k\right] \le \Pr[\mathrm{B} < k]$, that is $\Pr\left[\mathsf{N}_{\hat{c}}(\widehat{C}_1^n) \ge k\right] \ge \Pr[\mathrm{B} \ge k]$, and $\Pr\left[\mathsf{N}_{\hat{c}}(\widehat{C}_1^n) > k-1\right] \le \Pr[\mathrm{B} > k-1]$, that is $\Pr\left[\mathsf{N}_{\hat{c}}(\widehat{C}_1^n) \ge k\right] \le \Pr[\mathrm{B} \ge k]$; the two upper tails agree at every k. So the terms of $\varepsilon^\star$ are the tails of $\mathsf{N}_{\hat{c}}(\widehat{C}_1^n)$ itself. □

The figure ε^* is exactly computable at a rational envelope and generally below the budget, as integrality, the guard and degenerate coordinates leave budget unspent. An upward-rounded evaluation clamped at the budget still bounds the honest failure probability; a downward-rounded one need not.

Which figure is reported on which rule and evaluation is Requirement 16.4; the remark gives the mathematics it rests on, and Remark 14.3 the figures a closed-form rule can afford.

Remark 10.3 (Exact evaluation of $\varepsilon^\star$ and its conservative report). Every term of $\varepsilon^\star$ is one of the closed forms (2.5) of Corollary 2.2, at the envelope's lower value for a lower tail and at its upper value for an upper tail, so $\varepsilon^\star$ is exactly computable at a rational envelope (for a single law, at rational p^{hon}; Lemma 2.3) and is generally strictly smaller than $\varepsilon_{\mathrm{comp}}$ — the integrality of $k^\pm$ and the guard of Remark 8.3 both leave budget unspent, and the degenerate-mass convention of Definition 7.1 (c) leaves a side's budget entirely unspent where that side's envelope value is 0 or 1 (Proposition 8.2 (v) for the exact rule, read side-wise). Where the tails cannot be evaluated exactly they are rounded *upward* (a conservative evaluation, Definition 4.3): by (10.6) an upward-rounded evaluation of $\varepsilon^\star$ is still at least the honest failure probability, whereas a downward-rounded one need not be, and would claim a completeness the set does not have.

Upward rounding can carry the figure above $\varepsilon_{\mathrm{comp}}$ — in the degenerate case where every tail meets its allocation exactly and the allocation sums to exactly $\varepsilon_{\mathrm{comp}}$, any strict over-estimate exceeds the budget and leaves the interval $[0, \varepsilon_{\mathrm{comp}}]$. Clamping closes this without weakening anything: for every upward-rounded evaluation $\tilde{\varepsilon} \ge \varepsilon^\star$, the figure $\min\{\varepsilon_{\mathrm{comp}}, \tilde{\varepsilon}\}$ is again a valid bound, since both $\varepsilon_{\mathrm{comp}}$ and $\tilde{\varepsilon}$ are at least $\varepsilon^\star$, which by (10.6) is at least the honest failure probability; and it lies in $[0, \varepsilon_{\mathrm{comp}}]$ by construction. Exact evaluation needs no clamp, because $\varepsilon^\star \le \varepsilon_{\mathrm{comp}}$

is part of (10.6).

Uses Cor. 2.2, Lem. 2.3, Def. 4.3, Def. 7.1, Prop. 8.2, Rem. 8.3, Thm. 10.2 · *Used by* Rem. 14.3, Req. 16.4 · newly derived · accepted

The honest-round assumptions enter only the completeness statements: if honest behaviour leaves the declared envelope, ε^* is void and aborts may be more frequent. Keys from accepted runs are unaffected, since the acceptance set enters the security statement only as a fixed set.

Remark 10.4 (Soundness does not depend on the honest-round assumptions). Assumption 5.2, and Assumption 5.1 which is its degenerate case (Lemma 4.2), are statements about the honest channel and are used only in the completeness statements, such as Theorems 10.2 and 12.4, which read A-HON-ENV, and Theorem 18.1, which reads A-HON. Under the output-state model of the test statistics (Assumption "Output-state test statistics" (reductions/ass:tsm)) the round statistics are state-dependent in general, so no such concentration is available against an adversary — and none is needed, because completeness is a usability property. If the true honest behaviour leaves the declared envelope — for a single declared law, if the true honest channel differs from the declared p^{hon} — the figure $\varepsilon^\star$ is void and the protocol may abort more often than advertised; the key produced on the runs that *do* accept is unaffected, because $\mathcal{S}_{\mathrm{acc}}$ enters the security statement only as the fixed set $\underline{h}_\alpha$ is an infimum over. Soundness is unaffected in either case.

Uses Thm. 10.2 · *Assumes* A-HON-ENV, A-HON, A-TSM · *Used by* Rem. 14.3 · newly derived · accepted

An envelope certifies completeness for an honest channel that drifts within and between runs, and a finite family of honest laws enters through its entrywise minimum and maximum, which is already tight. The price is box width, which costs rate and never soundness.

Remark 10.5 (Benefit and cost of an honest envelope). *What it covers.* Under Assumption 5.2 the conditional law of each honest round, given the past, may be any law inside the envelope, and may change from round to round. The construction therefore certifies completeness for an honest channel that drifts within a run, that differs from run to run (each run's conditional laws lying in the envelope), and whose drift depends on past outcomes. A *set* of candidate honest statistics $\{\mathbf{p}_1, \dots, \mathbf{p}_M\} \subseteq \Delta(\widehat{\mathcal{C}}) \cap \mathbb{Q}^{\widehat{\mathcal{C}}}$ — several calibrations, the members of a simulated channel family, a central value with a band around it — enters through its entrywise minimum and maximum, $p^{\mathrm{hon}^-}(\hat{c}) := \min_i p_i(\hat{c})$ and $p^{\mathrm{hon}^+}(\hat{c}) := \max_i p_i(\hat{c})$, which satisfy Definition 4.1 because each $\mathbf{p}_i$ does (for the sums, $\sum_{\hat{c}} p^{\mathrm{hon}^-}(\hat{c}) \le \sum_{\hat{c}} p_1(\hat{c}) = 1 \le \sum_{\hat{c}} p^{\mathrm{hon}^+}(\hat{c})$). The box then certifies every law of the envelope set H (Theorem 9.2 (iii)), which contains the convex hull of the family and in general strictly more; the convex hull itself, as the set of certified laws, would need a joint concentration argument over several coordinates at once, which this submodule's per-outcome construction does not make. With the degenerate envelope nothing changes (Lemma 4.2).

What it costs. The box is built from the lower endpoint at $p^-_{\hat{c}}$ and the upper endpoint at $p^+_{\hat{c}}$, so it widens by the envelope's width on each coordinate: by Lemma 7.2 (i) the interval $[k^-(\hat{c}), k^+(\hat{c})]$ contains $[\lfloor np^-_{\hat{c}} \rfloor, \lceil np^+_{\hat{c}} \rceil]$, whatever the bud-

get. A wider box lowers the consumer's $\underline{h}_\alpha$, since an infimum over a superset is smaller; the price is paid in rate, never in soundness (Remark 10.4).

No tightening step is needed. An envelope formed as the entrywise minimum and maximum of a finite family is already tight in the sense that matters here: for every $\hat{c}$, $p^{\mathrm{hon}^-}(\hat{c}) = p_i(\hat{c})$ for some member i, and then $p^{\mathrm{hon}^-}(\hat{c}) = 1 - \sum_{\hat{c}' \neq \hat{c}} p_i(\hat{c}') \geq 1 - \sum_{\hat{c}' \neq \hat{c}} p^{\mathrm{hon}^+}(\hat{c}')$; symmetrically $p^{\mathrm{hon}^+}(\hat{c}) \leq 1 - \sum_{\hat{c}' \neq \hat{c}} p^{\mathrm{hon}^-}(\hat{c}')$. So no coordinate bound can be moved inwards using the simplex constraint alone; indeed each bound is attained by a member of the family, which lies in H, so $\min_{\mathbf{p} \in \mathsf{H}} p(\hat{c}) = p^{\mathrm{hon}^-}(\hat{c})$ and $\max_{\mathbf{p} \in \mathsf{H}} p(\hat{c}) = p^{\mathrm{hon}^+}(\hat{c})$, and the family's envelope is its own tightening. A declared envelope that is looser than it could be — one whose bounds could be moved inwards without losing a law the channel may have — is still valid; it is only conservative, costing box width and hence rate, and the reported $\varepsilon^\star$ remains a bound on the honest failure probability by Theorem 10.2. If the true honest behaviour leaves the declared envelope, $\varepsilon^\star$ is void and soundness is unaffected (Remark 10.4).

Uses Def. 4.1, Lem. 7.2, Thm. 9.2, Thm. 10.2 · newly derived · accepted

11 Honest containment and infeasibility

The box contains every law of the envelope set for every budget and allocation, so a law of the envelope that the consuming declaration reaches gives a compatible pair and the consuming engine's infeasibility condition cannot arise (Theorem 11.1). The same pair makes the consuming engine's minimal divergence zero, so a positive lower bound on it shows that no law of the envelope is reached (Remark 11.2). The construction never fails on an admissible input, so every refusal comes from the input: a violated condition of an admissible input, or data not of the input's form.

Every law of the envelope set lies in the constructed box for every budget and allocation, so a reachable one gives a compatible pair and rules out the consuming engine's infeasibility condition. The construction never fails on an admissible input, so every refusal comes from the input.

Theorem 9.2 (iii) places every law of the envelope set in the box; this theorem draws two consequences, one for the consuming engine's infeasibility condition and one for refusals. The one clause of an admissible input that this module cannot decide, that the alphabet, its enumeration and the round count are those of the consuming declaration, is checked where both artefacts are available, in bundle validation (`spec/architecture.md#36-bundle-validation-is-input-validation-not-a-module`; Requirement 6.4); how the engine refuses on the other clauses is Requirement 16.3.

Theorem 11.1 (Honest containment and infeasibility)**.** Let $\mathcal{S}_{\mathrm{acc}}$ be the output of Acc_R on an admissible input (Definition 7.1), for any admissible endpoint rule R, under a conservative evaluation (Definition 4.3).

(i) $\mathsf{H} \subseteq \mathcal{S}_{\mathrm{acc}}$, where H is the envelope set (4.2) of the input envelope, for every admissible $\varepsilon_{\mathrm{comp}} \in [0,1)$, every $\widehat{\mathcal{C}}_{\mathrm{con}}$ and every admissible allocation; for the degenerate envelope, $p^{\mathrm{hon}} \in \mathcal{S}_{\mathrm{acc}}$.

(ii) Assume A-REACH (Assumption 5.4): there exists $\omega^{\mathrm{hon}} \in \mathcal{W}$, the feasible set of the consuming declaration $\mathfrak{D}$ (Definition "Feasible set" (protocol-compiler/def:feasible)), with $\mathbf{p}^{\mathrm{hon}} := \mathbf{\Phi}[\omega^{\mathrm{hon}}] \in \mathsf{H}$ (for the degenerate envelope, $\mathbf{\Phi}[\omega^{\mathrm{hon}}] = p^{\mathrm{hon}}$). Then the pair $(\mathbf{p}^{\mathrm{hon}}, \omega^{\mathrm{hon}})$ satisfies $\mathbf{p}^{\mathrm{hon}} \in \mathcal{S}_{\mathrm{acc}}$, $\omega^{\mathrm{hon}} \in \mathcal{W}$ and $\mathbf{p}^{\mathrm{hon}} \ll \mathbf{\Phi}[\omega^{\mathrm{hon}}]$. Consequently the condition

$$\textit{no } \mathbf{q} \in \mathcal{S}_{\mathrm{acc}} \text{ is absolutely continuous with respect to } \mathbf{\Phi}[\omega] \text{ for any } \omega \in \mathcal{W} \tag{11.1}$$

is *false* for this $\mathcal{S}_{\mathrm{acc}}$.

(iii) *(refusals come from the input)* The validation predicate of Theorem 1.5 (iii) is decidable in closed form, and on every admissible input it is satisfied by the output of Acc_R unconditionally, for every admissible rule. Hence the construction never fails on an admissible input, and a refusal can arise only from an input that is not admissible. On data of the form of an admissible input (Definition 7.1), the violations of its conditions that are decidable from this module's own inputs are exactly:

(1) $|\widehat{\mathcal{C}}| = 0$;
(2) $|\widehat{\mathcal{C}}|$ not finite;
(3) $n < 1$;
(4) an honest input in the point form with $p^{\mathrm{hon}} \notin \Delta(\widehat{\mathcal{C}})$, including a coordinate outside $\widehat{\mathcal{C}}$;
(5) an honest input in the envelope form violating (4.1): some entry of p^{hon^-} or p^{hon^+} outside $[0,1]$, some $p^{\mathrm{hon}^-}(\hat{c}) > p^{\mathrm{hon}^+}(\hat{c})$, $\sum_{\hat{c}} p^{\mathrm{hon}^-}(\hat{c}) > 1$, $\sum_{\hat{c}} p^{\mathrm{hon}^+}(\hat{c}) < 1$, or a coordinate outside $\widehat{\mathcal{C}}$;
(6) $\varepsilon_{\mathrm{comp}} \notin [0,1)$;
(7) some $\varepsilon^{\pm}(\hat{c}) < 0$;
(8) $\sum_{\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}} (\varepsilon^{-}(\hat{c}) + \varepsilon^{+}(\hat{c})) > \varepsilon_{\mathrm{comp}}$;
(9) $\widehat{\mathcal{C}}_{\mathrm{con}} \not\subseteq \widehat{\mathcal{C}}$;
(10) some coordinate of p^{hon} or of $p^{\mathrm{hon}^{\mp}}$, or $\varepsilon_{\mathrm{comp}}$, or some $\varepsilon^{\pm}(\hat{c})$ is not an exact rational, violating Definition 4.3;
(11) the requested endpoint rule, or the requested evaluation, is none of those of clause (viii) of Definition 7.1.

Exactly one clause of an admissible input is *not* decidable from this module's inputs: clause (ix) of Definition 7.1, the condition of Definition 1.3 that $\widehat{\mathcal{C}}$, its enumeration and n are those of the same declaration

$\mathfrak{D}$ the consumer solves. That clause is a relation between artefacts, not a property of an input.

Uses Def. 1.3, Thm. 1.5, Def. 4.1, Def. 4.3, Def. 7.1, Thm. 9.2 · *Assumes* A-REACH, A-ALPHABET · *Used by* Rem. 11.2, Req. 16.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.statistics.construction: validate_request · *Oracles* registry_statistics_acceptance.py

Proof. (i). Theorem 9.2 (iii).

(ii). By (i), $\mathbf{p}^{\mathrm{hon}} \in \mathsf{H} \subseteq \mathcal{S}_{\mathrm{acc}}$. Absolute continuity $\mathbf{p}^{\mathrm{hon}} \ll \mathbf{\Phi}[\omega^{\mathrm{hon}}]$ reads: $p^{\mathrm{hon}}(\hat{c}) > 0$ implies $\mathbf{\Phi}[\omega^{\mathrm{hon}}](\hat{c}) > 0$. The two vectors are equal by definition of $\mathbf{p}^{\mathrm{hon}}$, so the implication is trivially true. This exhibits a witness $(\mathbf{q}, \omega) = (\mathbf{p}^{\mathrm{hon}}, \omega^{\mathrm{hon}})$ to the negation of the universally quantified statement (11.1); for the degenerate envelope the witness is $(p^{\mathrm{hon}}, \omega^{\mathrm{hon}})$.

(iii). Decidability in closed form is (1.6); unconditional satisfaction on every admissible input is Theorem 9.2 (iii). The enumerated list is the negation, clause by clause, of the conditions the clauses of an admissible input (Definition 7.1) place on data of its form, with the single stated exception of the cross-artefact condition of Definition 1.3: an input of that form for which none of the eleven violations (1)–(11) holds, and whose alphabet and round count are matched to the consuming declaration, is admissible, and then (i)–(ii) apply. □

Under A-REACH a box built from the honest envelope contains the statistics of a feasible state, which rules out an infinite value and a positive minimal divergence in the consuming engine. The assumption is never verified here; a positive bound on that divergence refutes it.

Remark 11.2 (Scope of the statements on infeasibility and minimal divergence). The consuming engine's contract identifies (11.1) as the source of $\underline{h}_\alpha = +\infty$ (Remark "Reading an infinite value" (meat-core/rem:hlow-range)) and requires that value to propagate rather than be clamped (Requirement "Propagation of an infinite value" (meat-core/req:infinite-value)). Theorem 11.1 (ii) rules that condition out for a set built by Acc from an honest envelope one of whose laws is reachable — for a single law, from a reachable honest distribution — which is the precise sense in which this module *surfaces* infeasibility rather than passing it through: the construction is ruled out as a source of (11.1), so when the consuming engine meets that condition on a set built by Acc, the contrapositive of Theorem 11.1 (ii) identifies its cause as a failure of A-REACH, an incompatibility of the declared honest envelope with the declaration, instead of leaving an extended-real value with no diagnosis.

The same pair has a finite counterpart. Under A-REACH, Theorem 11.1 (ii) gives $\mathbf{p}^{\mathrm{hon}} \in \mathcal{S}_{\mathrm{acc}}$ with $\mathbf{p}^{\mathrm{hon}} = \mathbf{\Phi}[\omega^{\mathrm{hon}}]$ and $\omega^{\mathrm{hon}} \in \mathcal{W}$, so the minimal divergence between the accepted distributions and the statistics of the feasible set, the quantity $D_{\min}$ of Proposition "The minimal divergence as the value with every pair dropped" (meat-core/prop:minimal-divergence), is zero for a set built by Acc, and the consuming engine's value has no positive penalty floor. Contrapositively, a positive lower bound on $D_{\min}$, which the consuming engine forms on the runs it

checks (Requirement "The check of the acceptance set against the feasible statistics" (meat-core/req:unreachable-acceptance-set)), shows that A-REACH fails for the declared envelope and the declaration. The converse does not hold: the box contains the envelope set and may meet the statistics of the feasible set where no law of the envelope set does, so a bound that is not positive says nothing about A-REACH.

The statement has two limits. First, it is conditional on A-REACH (Assumption 5.4), which this module never verifies, and which the consuming engine can refute but never establish: whether some law of the declared honest envelope is producible by the declared protocol is a joint property of whoever declares the honest model and of the declaration, and asserting it here would assert something this module cannot see. Second, it excludes the condition (11.1); whether $\underline{h}_\alpha$ is then finite is a statement about the consuming engine's objective and belongs to that contract, which is why it is quoted here and used nowhere in a proof. At $\underline{h}_\alpha = +\infty$ the consuming engine emits no key-length claim and reports the mismatch between the declaration and the acceptance set (Requirement "Propagation of an infinite value" (meat-core/req:infinite-value)), and nothing here changes that.

Uses Thm. 11.1 · *Assumes* A-REACH · newly derived · accepted

12 Monotonicity and limiting cases

Lowering the box's lower bounds and raising its upper ones gives a superset that is failed no more often and has no larger infimum, while the reverse move can break the completeness figure (Lemma 12.2). Under a monotone allocation rule and an endpoint rule monotone in the allocation, a larger budget gives a smaller or equal set and hence a larger or equal infimum for any consumer (Theorem 12.4). A zero budget gives the whole simplex away from degenerate coordinates, a one-letter alphabet a constant set, and for the exact rule at a fixed positive allocation the box closes in on the envelope's box at rate n^(−1/2) without being nested in the round count.

A monotone allocation rule allocates each completeness budget within that budget, every allocated tail budget non-decreasing in it; the uniform rule is one. It is the hypothesis on the allocation under which Theorem 12.4 (i) compares the acceptance sets at two budgets.

Comparing the acceptance sets at two completeness budgets needs the allocation at the larger budget to give every tail at least as much as the allocation at the smaller one. Where a request names no allocation, the uniform rule is the default (Requirement 16.2).

Definition 12.1 (Monotone allocation rule)**.** A *monotone allocation rule* assigns to each budget $e \in [0,1)$ an allocation $\varepsilon^{\pm}_{(e)}$ on $\widehat{\mathcal{C}}_{\mathrm{con}}$ such that $\sum_{\hat{c}}(\varepsilon^{-}_{(e)}(\hat{c}) + \varepsilon^{+}_{(e)}(\hat{c})) \le e$ and each $e \mapsto \varepsilon^{\pm}_{(e)}(\hat{c})$ is non-decreasing. The *uniform rule* $\varepsilon^{\pm}_{(e)}(\hat{c}) = e/(2|\widehat{\mathcal{C}}_{\mathrm{con}}|)$ is monotone.

Used by Thm. 12.4, Req. 16.2 · newly derived · implemented · *Code* certiqs_key.statistics.construction:uniform_allocation

Lowering the box's lower bounds and raising its upper ones gives a superset that is failed no more often and has no larger infimum, and integer endpoints so moved still meet their tail budgets. Moving them the other way can break the tail budgets and the reported completeness figure ε^*.

A conservative evaluation (Definition 4.3) can only return a smaller lower and a larger upper endpoint than an exact one, and the consuming engine's outward rounding of the emitted bounds moves them the same way. The lemma says that every such move keeps what the construction guarantees and costs only rate, and that the reverse move does not; Proposition 8.2 (iii) and Lemma 7.2 (ii) are its counterparts for a rule's own interval.

Lemma 12.2 (Widening the box keeps its guarantees)**.** Let $\mathcal{S}_{\mathrm{acc}}$ be the output of Acc_R (Definition 9.1) on an admissible input (Definition 7.1) for an admissible endpoint rule R, with box bounds $\mathbf{q}^-, \mathbf{q}^+$ (9.1) and guarded endpoints $k^{\mp}$ (7.2). Let $\tilde{\mathbf{q}}^-, \tilde{\mathbf{q}}^+ \in \mathbb{R}^{\widehat{\mathcal{C}}}$ satisfy $\tilde{q}^-(\hat{c}) \le q^-(\hat{c})$ and $\tilde{q}^+(\hat{c}) \ge q^+(\hat{c})$ for every $\hat{c} \in \widehat{\mathcal{C}}$, and put

$$\mathcal{S}'_{\mathrm{acc}} \;:=\; \{\mathbf{q} \in \Delta(\widehat{\mathcal{C}}) :\; \tilde{q}^-(\hat{c}) \le q(\hat{c}) \le \tilde{q}^+(\hat{c}) \;\; \forall \hat{c} \in \widehat{\mathcal{C}}\}. \tag{12.1}$$

(i) *(containment)* $\mathcal{S}_{\mathrm{acc}} \subseteq \mathcal{S}'_{\mathrm{acc}}$.

(ii) *(failure probability)* For all $\widehat{\mathcal{C}}$-valued random variables $\widehat{C}_1, \dots, \widehat{C}_n$ on one probability space,

$$\Pr\left[\mathrm{freq}_{\widehat{C}_1^n} \notin \mathcal{S}'_{\mathrm{acc}}\right] \;\le\; \Pr\left[\mathrm{freq}_{\widehat{C}_1^n} \notin \mathcal{S}_{\mathrm{acc}}\right].$$

(iii) *(infima)* For every function $\Psi : \Delta(\widehat{\mathcal{C}}) \to [-\infty, +\infty]$,

$$\inf_{\mathbf{q} \in \mathcal{S}'_{\mathrm{acc}}} \Psi(\mathbf{q}) \;\le\; \inf_{\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}} \Psi(\mathbf{q}). \tag{12.2}$$

(iv) *(integer endpoints)* Let $\tilde{k}^-, \tilde{k}^+ : \widehat{\mathcal{C}}_{\mathrm{con}} \to \mathbb{Z}$ satisfy $0 \le \tilde{k}^-(\hat{c}) \le k^-(\hat{c})$ and $k^+(\hat{c}) \le \tilde{k}^+(\hat{c}) \le n$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$, and let $\tilde{\mathbf{q}}^{\mp}$ be given by (9.1) with $\tilde{k}^{\mp}$ in place of $k^{\mp}$. Then $\tilde{\mathbf{q}}^{\mp}$ satisfy the hypothesis above, so (i)–(iii) hold; the pair $(\tilde{\mathbf{q}}^-, \tilde{\mathbf{q}}^+)$ meets the conditions of the canonical box form of Definition 1.4, and its denotation (1.5) is $\mathcal{S}'_{\mathrm{acc}}$; and for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$, with $\mathsf{N}^{\mp}_{\hat{c}}$ as in Definition 7.1,

$$\Pr\left[\mathsf{N}^-_{\hat{c}} < \tilde{k}^-(\hat{c})\right] \le \Pr\left[\mathsf{N}^-_{\hat{c}} < k^-(\hat{c})\right] \le \varepsilon^-(\hat{c}), \qquad \Pr\left[\mathsf{N}^+_{\hat{c}} > \tilde{k}^+(\hat{c})\right] \le \Pr\left[\mathsf{N}^+_{\hat{c}} > k^+(\hat{c})\right] \le \varepsilon^+(\hat{c}). \tag{12.3}$$

Summed over $\widehat{\mathcal{C}}_{\mathrm{con}}$, the tails at $\tilde{k}^{\mp}$ therefore total at most the tails at $k^{\mp}$, and at most $\varepsilon_{\mathrm{comp}}$.

(v) *(not the reverse)* Replacing $k^-(\hat{c})$ by a larger integer or $k^+(\hat{c})$ by a smaller one keeps, in general, neither the budgets of (12.3) nor the tail sum at $k^\mp$ as a bound on the probability that the frequency vector leaves the set. Take $\widehat{\mathcal{C}} = \{\hat{c}_0, \hat{c}_1\}$, the degenerate envelope at $p^{\mathrm{hon}} = (\frac{1}{2}, \frac{1}{2})$ (Definition 4.1), $n = 3$, $\widehat{\mathcal{C}}_{\mathrm{con}} = \{\hat{c}_0\}$, $\varepsilon^-(\hat{c}_0) = \varepsilon^+(\hat{c}_0) = \frac{2}{5}$, $\varepsilon_{\mathrm{comp}} = \frac{4}{5}$ and the exact binomial quantile rule of Definition 8.1. Then $k^-(\hat{c}_0) = 1$, $k^+(\hat{c}_0) = 2$, and the tails at these endpoints sum to $\frac{1}{4}$. Replacing $k^-(\hat{c}_0)$ by 2, or $k^+(\hat{c}_0)$ by 1, gives a tail of $\frac{1}{2} > \frac{2}{5}$ on the replaced side; and for $\widehat{C}_1, \widehat{C}_2, \widehat{C}_3$ independent with law p^{hon}, the frequency vector lies outside the set (9.2) built from the replaced endpoints with probability $\frac{5}{8} > \frac{1}{4}$, while it lies outside $\mathcal{S}_{\mathrm{acc}}$ with probability $\frac{1}{4}$.

Uses Def. 1.1, Def. 1.4, Def. 4.1, Def. 7.1, Lem. 7.2, Def. 8.1, Def. 9.1 · *Used by* Thm. 12.4 · newly derived · accepted

Proof. (i). Let $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ and $\hat{c} \in \widehat{\mathcal{C}}$. By (9.2) and the hypothesis, $\tilde{q}^-(\hat{c}) \le q^-(\hat{c}) \le q(\hat{c}) \le q^+(\hat{c}) \le \tilde{q}^+(\hat{c})$. Since also $\mathbf{q} \in \Delta(\widehat{\mathcal{C}})$, every defining condition of (12.1) holds, and $\mathbf{q} \in \mathcal{S}'_{\mathrm{acc}}$.

(ii). The frequency vector $\mathrm{freq}_{\widehat{C}_1^n}$ of Definition 1.1 is a function of the finitely valued string $\widehat{C}_1^n$, so $\{\mathrm{freq}_{\widehat{C}_1^n} \in S\}$ is the finite union of the events $\{\widehat{C}_1^n = \hat{c}_1^n\}$ over the strings with $\mathrm{freq}_{\hat{c}_1^n} \in S$, an event for every set S. By (i), $\{\mathrm{freq}_{\widehat{C}_1^n} \in \mathcal{S}_{\mathrm{acc}}\} \subseteq \{\mathrm{freq}_{\widehat{C}_1^n} \in \mathcal{S}'_{\mathrm{acc}}\}$; taking complements, $\{\mathrm{freq}_{\widehat{C}_1^n} \notin \mathcal{S}'_{\mathrm{acc}}\} \subseteq \{\mathrm{freq}_{\widehat{C}_1^n} \notin \mathcal{S}_{\mathrm{acc}}\}$, and a probability measure is monotone under inclusion.

(iii). For every $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$, (i) gives $\mathbf{q} \in \mathcal{S}'_{\mathrm{acc}}$, so the infimum over $\mathcal{S}'_{\mathrm{acc}}$ is at most $\Psi(\mathbf{q})$. The infimum over $\mathcal{S}'_{\mathrm{acc}}$ is therefore a lower bound of $\{\Psi(\mathbf{q}) : \mathbf{q} \in \mathcal{S}_{\mathrm{acc}}\}$, hence at most its greatest lower bound, which is (12.2). This holds in the extended reals for every Ψ, with no continuity or measurability required, and also when $\mathcal{S}_{\mathrm{acc}}$ is empty, its infimum then being $+\infty$.

(iv). The hypothesis. For $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$, (9.1) gives $\tilde{q}^-(\hat{c}) = \tilde{k}^-(\hat{c})/n \le k^-(\hat{c})/n = q^-(\hat{c})$ and $\tilde{q}^+(\hat{c}) = \tilde{k}^+(\hat{c})/n \ge k^+(\hat{c})/n = q^+(\hat{c})$, using $n \ge 1$; for $\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{con}}$ both pairs of bounds are 0 and 1. So $\tilde{\mathbf{q}}^\mp$ satisfy the hypothesis, and (i)–(iii) apply.

The canonical box form. Every $\tilde{q}^\mp(\hat{c})$ is an integer divided by n, or 0 or 1, hence rational. For $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$, (7.3) of Lemma 7.2 (i) gives $k^-(\hat{c}) \le k^+(\hat{c})$, so $0 \le \tilde{k}^-(\hat{c}) \le k^-(\hat{c}) \le k^+(\hat{c}) \le \tilde{k}^+(\hat{c})$ and $0 \le \tilde{q}^-(\hat{c}) \le \tilde{q}^+(\hat{c})$; for $\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{con}}$, $0 \le 0 \le 1$. These are the conditions of the canonical box form, and its denotation (1.5) with these bounds is (12.1).

The tails. By Lemma 7.2 (ii) the map $k \mapsto \Pr\left[\mathsf{N}_{\hat{c}}^- < k\right]$ is non-decreasing and the map $k \mapsto \Pr\left[\mathsf{N}_{\hat{c}}^+ > k\right]$ is non-increasing on the integers, so $\tilde{k}^-(\hat{c}) \le k^-(\hat{c})$ and $\tilde{k}^+(\hat{c}) \ge k^+(\hat{c})$ give the first inequality of each half of (12.3); the second is (7.4) of the same lemma. Summing the first inequalities over $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$ compares the two tail sums, and summing the second ones bounds the sum at $k^\mp$ by $\sum_{\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}} (\varepsilon^-(\hat{c}) + \varepsilon^+(\hat{c}))$, which the allocation of an admissible

input keeps at most $\varepsilon_{\mathrm{comp}}$ (Definition 7.1 (vi)).

(v). With the degenerate envelope at $\frac{1}{2}$, the two counts $\mathsf{N}^{-}_{\hat{c}_0}$ and $\mathsf{N}^{+}_{\hat{c}_0}$ have one law; write N for a count binomial with parameters $(3, \frac{1}{2})$, whose law is $(\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8})$ on $\{0, 1, 2, 3\}$. The exact rule takes (8.1): $\Pr[\mathsf{N} < 0] = 0$ and $\Pr[\mathsf{N} < 1] = \frac{1}{8}$ are at most $\frac{2}{5}$, while $\Pr[\mathsf{N} < k] \geq \Pr[\mathsf{N} < 2] = \frac{1}{2} > \frac{2}{5}$ for every $k \geq 2$, so $\kappa^{-}(\hat{c}_0) = 1$. It takes (8.2): $\Pr[\mathsf{N} > 2] = \frac{1}{8} \leq \frac{2}{5}$ and $\Pr[\mathsf{N} > 3] = 0$, while $\Pr[\mathsf{N} > k] \geq \Pr[\mathsf{N} > 1] = \frac{1}{2} > \frac{2}{5}$ for every $k \leq 1$, so $\kappa^{+}(\hat{c}_0) = 2$. The guard (7.2) is inert, since $\lfloor \frac{3}{2} \rfloor = 1$ and $\lceil \frac{3}{2} \rceil = 2$, so $k^{-}(\hat{c}_0) = 1$ and $k^{+}(\hat{c}_0) = 2$, and the tails at them are $\Pr[\mathsf{N} < 1] + \Pr[\mathsf{N} > 2] = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$. The replaced tails are $\Pr[\mathsf{N} < 2] = \frac{1}{2}$ and $\Pr[\mathsf{N} > 1] = \frac{1}{2}$, each above $\frac{2}{5}$.

For the failure probabilities, let $\widehat{C}_1, \widehat{C}_2, \widehat{C}_3$ be independent with law p^{hon}. The count of $\hat{c}_0$ in $\widehat{C}_1^3$ is then binomial with parameters $(3, \frac{1}{2})$, so it has the law of N. On a two-letter alphabet a vector of $\Delta(\widehat{\mathcal{C}})$ is fixed by its coordinate at $\hat{c}_0$, and the bounds at $\hat{c}_1$ are 0 and 1 by (9.1), so a frequency vector lies in a set (9.2) built from endpoints (a, b) at $\hat{c}_0$ exactly when the count of $\hat{c}_0$ lies in $\{a, \ldots, b\}$. For $\mathcal{S}_{\mathrm{acc}}$, with $(a, b) = (1, 2)$, the failure probability is $\Pr[\mathsf{N} \in \{0, 3\}] = \frac{1}{4}$. With k^{-} replaced by 2, $(a, b) = (2, 2)$ and the failure probability is $1 - \Pr[\mathsf{N} = 2] = \frac{5}{8}$; with k^{+} replaced by 1, $(a, b) = (1, 1)$ and it is $1 - \Pr[\mathsf{N} = 1] = \frac{5}{8}$. Both exceed $\frac{1}{4}$. □

A binomial (n, p) count deviates from np by at least t > 0 with probability at most np(1 − p)/t². This gives the finite-sample width bound of the exact rule in Theorem 12.4 (iv).

Lemma 12.3 (Chebyshev's inequality for a binomial count). Let N be binomial with parameters (n, p) and let $t > 0$. Then $\Pr\big[|\mathsf{N} - np| \geq t\big] \leq np(1-p)/t^2$.

Used by Thm. 12.4 · newly derived · accepted

Proof. Write $\mathsf{N} = \sum_{j=1}^{n} \chi_j$ with χ_j independent Bernoulli(p); then $\mathbb{E}[\mathsf{N}] = np$ and, by independence, $\mathrm{Var}(\mathsf{N}) = \sum_j \mathrm{Var}(\chi_j) = np(1-p)$. Consider the non-negative random variable $(\mathsf{N} - np)^2$. Pointwise it is at least t^2 wherever $|\mathsf{N} - np| \geq t$ and at least 0 elsewhere, so it dominates the random variable taking the value t^2 wherever $|\mathsf{N} - np| \geq t$ and 0 elsewhere. Taking expectations, $t^2 \Pr\big[|\mathsf{N} - np| \geq t\big] \leq \mathbb{E}\big[(\mathsf{N} - np)^2\big] = \mathrm{Var}(\mathsf{N}) = np(1-p)$, which is the claim. □

The acceptance set shrinks as the budget grows under monotone allocation and endpoint rules, so no infimum over it, h_low included, decreases. Zero-budget and one-letter limits are explicit; at a fixed positive allocation the exact rule's set tends to the envelope's box as n^(−1/2), unnested in n.

Parts (i) and (ii) need the endpoint rule itself to be monotone in the allocation, and which rules are is each rule's own statement: the exact rule with its sweep evaluation (Proposition 8.2 (ii), and (vii) where a tail is decided against an upper bound) and the Bernstein rule (Proposition 14.2 (ii)) are,

while the enclosure evaluation of the exact rule (Proposition 15.3) does not claim it, so (i) and (ii) are not asserted for it. The monotonicity of $\underline{h}_\alpha$ in (ii) is a trade-off, not an improvement: the same change of budget makes the honest run abort no less often. A one-letter acceptance alphabet, the case of (vi), imposes no constraint on the consumer and carries no information; it is a degenerate case to be recognised, not a regime to be used.

Theorem 12.4 (Monotonicity in the budget and limiting cases)**.** Fix $\widehat{\mathcal{C}}$, an honest envelope $(p^{\mathrm{hon}-}, p^{\mathrm{hon}+})$ with envelope set H (Definition 4.1; a single law p^{hon} is the degenerate envelope, $\mathsf{H} = \{p^{\mathrm{hon}}\}$), n, $\widehat{\mathcal{C}}_{\mathrm{con}}$ and an admissible endpoint rule R (Definition 7.1) under a conservative evaluation (Definition 4.3), write $p^{\mp}_{\hat{c}}$ for the envelope's values as there, and write $\mathcal{S}_{\mathrm{acc}}(e)$ for the output of Acc_R at budget e under a fixed monotone allocation rule (Definition 12.1).

(i) *(budget monotonicity: the set shrinks as the budget grows)* If R is monotone in the allocation and $0 \le e_1 \le e_2 < 1$, then $\mathcal{S}_{\mathrm{acc}}(e_2) \subseteq \mathcal{S}_{\mathrm{acc}}(e_1)$.

(ii) *(consequence for any consumer infimum)* Under the hypothesis of (i), for every function $\Psi : \Delta(\widehat{\mathcal{C}}) \to [-\infty, +\infty]$ and $e_1 \le e_2$,

$$\inf_{\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}(e_1)} \Psi(\mathbf{q}) \;\le\; \inf_{\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}(e_2)} \Psi(\mathbf{q}). \tag{12.4}$$

In particular, taking $\Psi(\mathbf{q}) = \inf_{\omega \in \mathcal{W}} O_\alpha(\mathbf{q}, \omega)$, the quantity $\underline{h}_\alpha$, which Definition "Feasible set and computable objective" (meat-core/def:feasible) defines as $\inf_{\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}} \inf_{\omega \in \mathcal{W}} O_\alpha$, is *non-decreasing* in $\varepsilon_{\mathrm{comp}}$: a weaker completeness guarantee buys a larger or equal entropy bound.

(iii) *(limit $\varepsilon_{\mathrm{comp}} = 0$, every rule)* At $e = 0$ every $\varepsilon^{\pm}(\hat{c}) = 0$ and

$$\mathcal{S}_{\mathrm{acc}}(0) = \big\{ \mathbf{q} \in \Delta(\widehat{\mathcal{C}}) : q(\hat{c}) = 0 \text{ for } \hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}} \text{ with } p^{+}_{\hat{c}} = 0, \\ q(\hat{c}) = 1 \text{ for } \hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}} \text{ with } p^{-}_{\hat{c}} = 1 \big\},$$

with $\varepsilon^\star = 0$. If $p^{+}_{\hat{c}} > 0$ and $p^{-}_{\hat{c}} < 1$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$ (for the degenerate envelope: if $0 < p^{\mathrm{hon}}(\hat{c}) < 1$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$), then $\mathcal{S}_{\mathrm{acc}}(0) = \Delta(\widehat{\mathcal{C}})$: a zero completeness budget buys the whole simplex and no constraint at all.

(iv) *(limit $n \to \infty$, the exact rule)* Let R be the exact rule of Definition 8.1, let $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$ and write $p^{\mp} = p^{\mp}_{\hat{c}}$. Then

$$0 \le q^{+}(\hat{c}) - p^{+} \le \sqrt{\frac{p^{+}(1-p^{+})}{n\,\varepsilon^{+}(\hat{c})}} + \frac{1}{n}, \qquad 0 \le p^{-} - q^{-}(\hat{c}) \le \sqrt{\frac{p^{-}(1-p^{-})}{n\,\varepsilon^{-}(\hat{c})}} + \frac{1}{n}, \tag{12.5}$$

the first whenever $0 < p^{+} < 1$ and $\varepsilon^{+}(\hat{c}) > 0$, the second whenever $0 < p^{-} < 1$ and $\varepsilon^{-}(\hat{c}) > 0$. Consequently, if $\widehat{\mathcal{C}}_{\mathrm{con}} = \widehat{\mathcal{C}}$, $0 < p^{\mp}_{\hat{c}} < 1$

and $\varepsilon^{\pm}(\hat{c}) > 0$ for all $\hat{c}$, and the allocation is held fixed as n grows (for instance the uniform rule at a fixed $e > 0$), then every $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ satisfies $p^{-}_{\hat{c}} - O(n^{-1/2}) \le q(\hat{c}) \le p^{+}_{\hat{c}} + O(n^{-1/2})$ uniformly in $\hat{c}$: $\mathcal{S}_{\mathrm{acc}}$ lies coordinatewise within $O(n^{-1/2})$ of the box $\prod_{\hat{c}}[p^{-}_{\hat{c}}, p^{+}_{\hat{c}}]$, and it contains H by Theorem 9.2 (iii). For the degenerate envelope, $\sup_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}} \|\mathbf{q} - p^{\mathrm{hon}}\|_{\infty} = O(n^{-1/2})$ and $\mathcal{S}_{\mathrm{acc}}$ converges to $\{p^{\mathrm{hon}}\}$ in Hausdorff distance.

(v) *(no monotonicity in n, with a counterexample; the exact rule)* For the exact rule, $\mathcal{S}_{\mathrm{acc}}$ is *not* monotone in n, already for a degenerate envelope. Take $\widehat{\mathcal{C}} = \{\hat{c}_0, \hat{c}_1\}$, the degenerate envelope at $p^{\mathrm{hon}} = (\frac{1}{2}, \frac{1}{2})$, $\widehat{\mathcal{C}}_{\mathrm{con}} = \{\hat{c}_0\}$ and $\varepsilon^{-}(\hat{c}_0) = \varepsilon^{+}(\hat{c}_0) = \frac{2}{5}$ (so $\varepsilon_{\mathrm{comp}} = \frac{4}{5}$). Then

$$\begin{aligned} n = 2: &\quad [k^{-}, k^{+}] = [1, 1], \quad \mathcal{S}_{\mathrm{acc}} = \{(\tfrac{1}{2}, \tfrac{1}{2})\}; \\ n = 3: &\quad [k^{-}, k^{+}] = [1, 2], \quad \mathcal{S}_{\mathrm{acc}} = \{\mathbf{q} \in \Delta(\widehat{\mathcal{C}}) : \tfrac{1}{3} \le q(\hat{c}_0) \le \tfrac{2}{3}\}, \end{aligned}$$

so $\mathcal{S}_{\mathrm{acc}}(n{=}2) \subsetneq \mathcal{S}_{\mathrm{acc}}(n{=}3)$: the set is strictly larger at the larger round count.

(vi) *(degenerate alphabet $|\widehat{\mathcal{C}}| = 1$, every rule)* If $\widehat{\mathcal{C}} = \{\hat{c}_0\}$ then, for every $n \ge 1$, every $\varepsilon_{\mathrm{comp}} \in [0, 1)$, every $\widehat{\mathcal{C}}_{\mathrm{con}} \subseteq \widehat{\mathcal{C}}$ and every admissible allocation,

$$p^{\mathrm{hon}^{+}}(\hat{c}_0) = 1, \qquad \mathsf{H} = \mathcal{S}_{\mathrm{acc}} = \Delta(\widehat{\mathcal{C}}) = \{\delta_{\hat{c}_0}\},$$

and $\varepsilon^{\star} = 0$ whenever $p^{\mathrm{hon}^{-}}(\hat{c}_0) = 1$, in particular for the degenerate envelope, which is then $p^{\mathrm{hon}} = \delta_{\hat{c}_0}$. The emitted set is therefore constant in all of the construction's parameters on a one-letter alphabet, being at once the whole simplex and a single point; for the degenerate envelope so are its box data, and completeness is perfect. For a non-degenerate envelope ($p^{\mathrm{hon}^{-}}(\hat{c}_0) < 1$, which (4.1) permits here) the lower box bound $q^{-}(\hat{c}_0)$ and the reported $\varepsilon^{\star}$ may be positive and may vary with the parameters; the reported figure is still a valid bound (by Theorem 10.2), merely a loose one, the true honest failure probability being 0.

Uses Def. 4.1, Def. 4.3, Def. 7.1, Lem. 7.2, Def. 8.1, Def. 9.1, Thm. 9.2, Thm. 10.2, Def. 12.1, Lem. 12.2, Lem. 12.3 · *Assumes* A-ALPHABET · *Used by* Rem. 12.5, Rem. 18.2 · newly derived · accepted · *Oracles* registry_statistics_acceptance.py

Proof. (i). Fix $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$, write $p^{\mp} = p^{\mp}_{\hat{c}}$ and $\varepsilon^{\pm}_{i} := \varepsilon^{\pm}_{(e_i)}(\hat{c})$, so $\varepsilon^{\pm}_{1} \le \varepsilon^{\pm}_{2}$ by Definition 12.1. Since R is monotone in the allocation (Definition 7.1), $R^{-}(n, p^{-}, \varepsilon^{-}_{1}) \le R^{-}(n, p^{-}, \varepsilon^{-}_{2})$ and $R^{+}(n, p^{+}, \varepsilon^{+}_{1}) \ge R^{+}(n, p^{+}, \varepsilon^{+}_{2})$. Both guards in (7.2) are monotone in their first argument, their second arguments $\lfloor np^{-} \rfloor$ and $\lceil np^{+} \rceil$ not depending on the budget, whence $k^{-}_{1}(\hat{c}) \le k^{-}_{2}(\hat{c})$ and $k^{+}_{1}(\hat{c}) \ge k^{+}_{2}(\hat{c})$, that is, $[k^{-}_{2}, k^{+}_{2}] \subseteq [k^{-}_{1}, k^{+}_{1}]$. The endpoints $k^{\mp}_{1}$ at the smaller budget are therefore integer endpoints of the kind Lemma 12.2 (iv) admits for the output $\mathcal{S}_{\mathrm{acc}}(e_2)$: $0 \le k^{-}_{1}(\hat{c}) \le k^{-}_{2}(\hat{c})$ and $k^{+}_{2}(\hat{c}) \le k^{+}_{1}(\hat{c}) \le n$, the

outer bounds by (7.3) at e_1 and the inner ones as just shown. The set (12.1) they give is $\mathcal{S}_{\mathrm{acc}}(e_1)$, since both sets are built by (9.1) on the same $\widehat{\mathcal{C}}_{\mathrm{con}}$, with the bounds $[0,1]$ off $\widehat{\mathcal{C}}_{\mathrm{con}}$ at both budgets. Lemma 12.2 (i) gives $\mathcal{S}_{\mathrm{acc}}(e_2) \subseteq \mathcal{S}_{\mathrm{acc}}(e_1)$.

(ii). (12.4) is (12.2) of Lemma 12.2 (iii) applied to the inclusion of (i), with $\mathcal{S}_{\mathrm{acc}}(e_2)$ in the role of $\mathcal{S}_{\mathrm{acc}}$ and $\mathcal{S}_{\mathrm{acc}}(e_1)$ in that of $\mathcal{S}'_{\mathrm{acc}}$, and it holds for arbitrary Ψ with values in the extended reals, no continuity or measurability required. The specialization substitutes the definition of $\underline{h}_\alpha$ as an infimum over $\mathcal{S}_{\mathrm{acc}}$ and changes nothing else.

(iii). The allocation constraint at $e = 0$ forces every $\varepsilon^{\pm}(\hat{c}) = 0$, these being non-negative with sum at most 0. Fix $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$ and write $p^{\mp} = p^{\mp}_{\hat{c}}$; by (7.2) each side is computed from its own envelope value alone.

Lower side, $R^- = R^-(n, p^-, 0)$. If $p^- \in (0,1)$: Definition 7.1 (b) gives $R^- = 0$, so $k^-(\hat{c}) = \min\{0, \lfloor np^- \rfloor\} = 0$. If $p^- = 0$: Definition 7.1 (c) gives $R^- = 0$ and $k^-(\hat{c}) = \min\{0, 0\} = 0$. If $p^- = 1$: (c) gives $R^- = n$ and $k^-(\hat{c}) = \min\{n, n\} = n$. So the lower bound is $q^-(\hat{c}) = 0$ unless $p^- = 1$, where it is $q^-(\hat{c}) = 1$.

Upper side, $R^+ = R^+(n, p^+, 0)$. If $p^+ \in (0,1)$: (b) gives $R^+ = n$, so $k^+(\hat{c}) = \max\{n, \lceil np^+ \rceil\} = n$. If $p^+ = 0$: (c) gives $R^+ = 0$ and $k^+(\hat{c}) = \max\{0, 0\} = 0$. If $p^+ = 1$: (c) gives $R^+ = n$ and $k^+(\hat{c}) = \max\{n, n\} = n$. So the upper bound is $q^+(\hat{c}) = 1$ unless $p^+ = 0$, where it is $q^+(\hat{c}) = 0$.

On $\Delta(\widehat{\mathcal{C}})$ the bound $q(\hat{c}) \geq 1$ is $q(\hat{c}) = 1$ and the bound $q(\hat{c}) \leq 0$ is $q(\hat{c}) = 0$, and the bounds $[0,1]$ are vacuous; assembling the cases in (9.2) gives the displayed set, and it is the whole simplex when no $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$ has $p^+_{\hat{c}} = 0$ or $p^-_{\hat{c}} = 1$. Finally $\varepsilon^\star = 0$. A lower term of (10.6) is $\Pr\big[\mathsf{N}^-_{\hat{c}} < 0\big] = 0$ when $k^-(\hat{c}) = 0$, and when $p^- = 1$ it is $\Pr\big[\mathsf{N}^-_{\hat{c}} < n\big] = 0$ because $\mathsf{N}^-_{\hat{c}}$, binomial $(n, 1)$, equals n almost surely. An upper term is $\Pr\big[\mathsf{N}^+_{\hat{c}} > n\big] = 0$ when $k^+(\hat{c}) = n$, and when $p^+ = 0$ it is $\Pr\big[\mathsf{N}^+_{\hat{c}} > 0\big] = 0$ because $\mathsf{N}^+_{\hat{c}}$, binomial $(n, 0)$, equals 0 almost surely.

(iv). The left inequalities of (12.5) are (7.3) (Lemma 7.2 (i)): $q^+(\hat{c}) = k^+(\hat{c})/n \geq \lceil np^+ \rceil / n \geq p^+$ and $q^-(\hat{c}) = k^-(\hat{c})/n \leq \lfloor np^- \rfloor / n \leq p^-$.

For the upper one, $\kappa^+(\hat{c})$ is the quantile (8.2) at p^+ and $\mathsf{N}^+_{\hat{c}}$ is binomial (n, p^+) (the side-wise reading of Definition 7.1). Set $t := \sqrt{np^+(1-p^+)/\varepsilon^+(\hat{c})} > 0$ and $k := \lceil np^+ + t \rceil$. Any integer $j > k$ satisfies $j > np^+ + t$, so

$$\Pr\big[\mathsf{N}^+_{\hat{c}} > k\big] \;\leq\; \Pr\big[\mathsf{N}^+_{\hat{c}} - np^+ > t\big] \;\leq\; \Pr\big[|\mathsf{N}^+_{\hat{c}} - np^+| \geq t\big] \;\leq\; \frac{np^+(1-p^+)}{t^2} \;=\; \varepsilon^+(\hat{c})$$

by Lemma 12.3. If $k \leq n$ this exhibits an admissible index in (8.2), so $\kappa^+(\hat{c}) \leq k$ by minimality; if $k > n$ then $\kappa^+(\hat{c}) \leq n < k$ anyway. In both cases $\kappa^+(\hat{c}) \leq \lceil np^+ + t \rceil$. Since also $\lceil np^+ \rceil \leq \lceil np^+ + t \rceil$, the guard gives $k^+(\hat{c}) \leq \lceil np^+ + t \rceil \leq np^+ + t + 1$, hence $q^+(\hat{c}) - p^+ \leq t/n + 1/n$, which is the stated bound because $t/n = \sqrt{p^+(1-p^+)/(n\varepsilon^+(\hat{c}))}$.

Symmetrically, $\kappa^-(\hat{c})$ is the quantile (8.1) at p^- and $\mathsf{N}^-_{\hat{c}}$ is binomial (n, p^-). Set $t' := \sqrt{np^-(1-p^-)/\varepsilon^-(\hat{c})} > 0$ and $k' := \max\{0, \lfloor np^- - t' \rfloor\}$.

If $\lfloor np^- - t' \rfloor \le 0$ then $k' = 0$ and $\Pr\big[\mathsf{N}^-_{\hat c} < 0\big] = 0 \le \varepsilon^-(\hat c)$. Otherwise $k' = \lfloor np^- - t' \rfloor \le np^- - t'$, so any integer $j \le k' - 1$ satisfies $j < np^- - t'$ and $\Pr\big[\mathsf{N}^-_{\hat c} < k'\big] \le \Pr\big[\mathsf{N}^-_{\hat c} - np^- < -t'\big] \le \Pr\big[|\mathsf{N}^-_{\hat c} - np^-| \ge t'\big] \le \varepsilon^-(\hat c)$ by Lemma 12.3. Either way k' is admissible in (8.1), so $\kappa^-(\hat c) \ge k'$. In the unclipped case $k' = \lfloor np^- - t' \rfloor > np^- - t' - 1$; in the clipped case $\lfloor np^- - t' \rfloor \le 0$ forces $np^- - t' < 1$, hence $np^- - t' - 1 < 0 = k'$. In both cases $\kappa^-(\hat c) \ge k' > np^- - t' - 1$. As $\lfloor np^- \rfloor > np^- - 1 \ge np^- - t' - 1$, the guard gives $k^-(\hat c) \ge np^- - t' - 1$ and hence $p^- - q^-(\hat c) \le t'/n + 1/n$.

For the consequence: $\mathcal{S}_{\mathrm{acc}} \subseteq \prod_{\hat c}[q^-(\hat c), q^+(\hat c)]$ by (9.2), so every $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ has $p^-_{\hat c} - q(\hat c) \le p^-_{\hat c} - q^-(\hat c)$ and $q(\hat c) - p^+_{\hat c} \le q^+(\hat c) - p^+_{\hat c}$, which (12.5) bounds by $O(n^{-1/2})$ uniformly over the finite alphabet, both bounds of (12.5) applying at every $\hat c$ because $0 < p^\mp_{\hat c} < 1$ and $\varepsilon^\pm(\hat c) > 0$ there, with fixed values; $\mathsf{H} \subseteq \mathcal{S}_{\mathrm{acc}}$ is Theorem 9.2 (iii). For the degenerate envelope $p^-_{\hat c} = p^+_{\hat c} = p^{\mathrm{hon}}(\hat c)$, so the same two bounds give $\|\mathbf{q} - p^{\mathrm{hon}}\|_\infty \le \max_{\hat c} \max\{q^+(\hat c) - p^{\mathrm{hon}}(\hat c),\, p^{\mathrm{hon}}(\hat c) - q^-(\hat c)\} = O(n^{-1/2})$ for every $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$, and $p^{\mathrm{hon}} \in \mathcal{S}_{\mathrm{acc}}$ by Theorem 9.2 (iii), so the other half of the Hausdorff distance is 0.

(v). Direct computation with the degenerate envelope at $p = \frac12$, so that $\mathsf{N} \coloneqq \mathsf{N}^-_{\hat c_0} = \mathsf{N}^+_{\hat c_0}$ is binomial $(n, \frac12)$, and the exact rule. At $n = 2$ the law of N is $(\frac14, \frac12, \frac14)$ on $\{0,1,2\}$: $\Pr[\mathsf{N} < 0] = 0 \le \frac25$ and $\Pr[\mathsf{N} < 1] = \frac14 \le \frac25$ but $\Pr[\mathsf{N} < 2] = \frac34 > \frac25$, so $\kappa^- = 1$; $\Pr[\mathsf{N} > 0] = \frac34 > \frac25$ and $\Pr[\mathsf{N} > 1] = \frac14 \le \frac25$, so $\kappa^+ = 1$. The guard is inert ($\lfloor 1 \rfloor = \lceil 1 \rceil = 1$), so $[k^-, k^+] = [1,1]$ and $q(\hat c_0) = \frac12$, which on a two-letter alphabet forces $\mathbf{q} = (\frac12, \frac12)$. At $n = 3$ the law is $(\frac18, \frac38, \frac38, \frac18)$ on $\{0,1,2,3\}$: $\Pr[\mathsf{N} < 1] = \frac18 \le \frac25$ but $\Pr[\mathsf{N} < 2] = \frac12 > \frac25$, so $\kappa^- = 1$; $\Pr[\mathsf{N} > 1] = \frac12 > \frac25$ and $\Pr[\mathsf{N} > 2] = \frac18 \le \frac25$, so $\kappa^+ = 2$. The guard is again inert ($\lfloor \frac32 \rfloor = 1$, $\lceil \frac32 \rceil = 2$), so $[k^-, k^+] = [1,2]$ and $q(\hat c_0) \in [\frac13, \frac23]$. The two sets are as displayed and the inclusion is strict.

(vi). On $\widehat{\mathcal{C}} = \{\hat c_0\}$ the only vector with non-negative coordinates summing to 1 is $\delta_{\hat c_0}$, so $\Delta(\widehat{\mathcal{C}}) = \{\delta_{\hat c_0}\}$. Condition (4.1) gives $p^{\mathrm{hon}^+}(\hat c_0) \le 1 \le p^{\mathrm{hon}^+}(\hat c_0)$, that is, $p^{\mathrm{hon}^+}(\hat c_0) = 1$, and $p^{\mathrm{hon}^-}(\hat c_0) \le 1$, so $\delta_{\hat c_0} \in \mathsf{H} \subseteq \Delta(\widehat{\mathcal{C}})$ and $\mathsf{H} = \{\delta_{\hat c_0}\}$; for the degenerate envelope also $p^{\mathrm{hon}^-}(\hat c_0) = 1$, that is, $p^{\mathrm{hon}} = \delta_{\hat c_0}$. Since $\mathcal{S}_{\mathrm{acc}} \subseteq \Delta(\widehat{\mathcal{C}})$ by (9.2) and $\mathcal{S}_{\mathrm{acc}} \supseteq \mathsf{H} \ne \varnothing$ by Theorem 9.2 (iii), $\mathcal{S}_{\mathrm{acc}} = \Delta(\widehat{\mathcal{C}}) = \{\delta_{\hat c_0}\}$ whatever the parameters. For $\varepsilon^\star$: if $\widehat{\mathcal{C}}_{\mathrm{con}} = \varnothing$, (9.1) gives the vacuous bounds $[0,1]$ and $\varepsilon^\star$ is an empty sum, hence 0. If $\widehat{\mathcal{C}}_{\mathrm{con}} = \{\hat c_0\}$, the upper side has $p^+_{\hat c_0} = 1$, so Definition 7.1 (c) gives $R^+ = n$, the guard gives $k^+(\hat c_0) = n$ and the upper term $\Pr\Big[\mathsf{N}^+_{\hat c_0} > n\Big]$ is 0. If moreover $p^{\mathrm{hon}^-}(\hat c_0) = 1$, the same convention gives $R^- = n$, $k^-(\hat c_0) = \min\{n, n\} = n$, the box bounds $q^-(\hat c_0) = q^+(\hat c_0) = 1$ independent of every parameter, and the lower term $\Pr\Big[\mathsf{N}^-_{\hat c_0} < n\Big]$ is 0 because $\mathsf{N}^-_{\hat c_0}$, binomial $(n,1)$, equals n almost surely; so $\varepsilon^\star = 0$. The two cases exhaust $\widehat{\mathcal{C}}_{\mathrm{con}} \subseteq \widehat{\mathcal{C}}$, and neither used n, $\varepsilon_{\mathrm{comp}}$, the allocation or anything of R beyond Definition 7.1 (c). When $p^{\mathrm{hon}^-}(\hat c_0) < 1$ the lower term is the tail of a binomial $(n, p^{\mathrm{hon}^-}(\hat c_0))$ below $k^-(\hat c_0)$, which Lemma 7.2 (ii) bounds by $\varepsilon^-(\hat c_0)$

and which need not vanish; the bound of Theorem 10.2 still holds, the true failure probability being $\Pr\left[\mathrm{freq}_{\widehat{C}_1^n} \notin \{\delta_{\hat{c}_0}\}\right] = 0$. □

The exact rule's endpoints are integer quantiles on a lattice that refines with n, so its interval can widen when n grows by one, as Theorem 12.4 (v) shows, while the n^(−1/2) width of (iv) stands. A consumer may rely on monotonicity in the budget, not in the round count.

Remark 12.5 (Non-monotonicity of the exact rule in the round count). The endpoints $\kappa^{\pm}$ are integer quantiles of a distribution whose support lattice refines with n, so the attainable tail levels move in jumps and the interval $[k^-/n, k^+/n]$ can widen when n increases by one, as Theorem 12.4 (v) exhibits. The asymptotic statement (12.5) is unaffected, the width still falling as $n^{-1/2}$, and no claim of this module needs monotonicity in n. A consumer that assumed "more rounds, tighter set" would be assuming something false. The monotonicity that *is* available, and that a consumer may rely on, is monotonicity in $\varepsilon_{\mathrm{comp}}$, Theorem 12.4 (i), with the direction stated there.

Uses Thm. 12.4 · newly derived · accepted

13 Elementary enclosures

Rational bounds on logarithms from integer square roots, and two-sided Stirling bounds on factorials with the constant enclosed by Wallis's product, are the only transcendental ingredients of the closed-form rule and of the local evaluation. Every bound has an explicit direction and an explicit width.

For x > 0 and m ≥ 1, m(1 − x^(−1/m)) ≤ ln x ≤ m(x^(1/m) − 1), with an explicit slack; for rational x and m = 2^j, j integer square roots give rational bounds of any prescribed width. The Bernstein rule and the enclosure evaluation take every logarithm this way.

No logarithm is taken from a library: every logarithm the Bernstein rule and the local evaluation of the exact rule need is bounded from both sides by rationals. The bounds are the tangent inequality $\ln t \le t - 1$ at $t = x^{\pm 1/m}$, the identity by which Briggs tabulated logarithms through repeated square roots before series methods, and their slack is the second-order remainder of the exponential series; both are derived in full here.

Lemma 13.1 (Rational bounds on the logarithm from integer square roots)**.** Let $x > 0$ be real and $m \ge 1$ an integer. Then

$$m\big(1 - x^{-1/m}\big) \;\le\; \ln x \;\le\; m\big(x^{1/m} - 1\big), \tag{13.1}$$

with equality on either side if and only if $x = 1$, and both slacks are bounded by

$$\max\Big\{m\big(x^{1/m}-1\big)-\ln x,\;\; \ln x - m\big(1-x^{-1/m}\big)\Big\} \;\le\; \frac{(\ln x)^2}{2m}\, e^{|\ln x|/m}. \tag{13.2}$$

Consequently, if l, u are rationals with $0 < l \le x^{1/m} \le u$, then

$$m\Big(1 - \frac{1}{l}\Big) \;\le\; \ln x \;\le\; m\,(u - 1), \tag{13.3}$$

and for rational $x = a/b$ (a, b positive integers) and $m = 2^j$, such l, u of any prescribed positive width are computed by j integer square roots: with $\mathrm{isqrt}(N) := \lfloor \sqrt{N} \rfloor$ for an integer $N \ge 0$, one has for every rational $y = c/d > 0$

$$\frac{\mathrm{isqrt}(c\,d\,4^s)}{d\,2^s} \;\le\; \sqrt{y} \;<\; \frac{\mathrm{isqrt}(c\,d\,4^s) + 1}{d\,2^s}, \tag{13.4}$$

the two bounds differing by $1/(d\,2^s) \le 2^{-s}$, and iterating (13.4) j times, the lower bound always taken from the previous lower bound and the upper from the previous upper, gives $l \le x^{1/2^j} \le u$ whose width $u - l$ is an explicit rational the caller reads off and drives below any prescribed bound by the precisions s of the steps.

Used by Lem. 13.2, Prop. 14.2, Lem. 15.1, Prop. 15.3, Rem. 15.4, Req. 17.2 · newly derived · implemented · *Code* certiqs_key.statistics.logs:log_bounds

Proof. The tangent inequality. For $t > 0$ put $f(t) := t - 1 - \ln t$. Then $f'(t) = 1 - 1/t$, which is negative on $(0, 1)$ and positive on $(1, \infty)$, so f attains its minimum at $t = 1$, where $f(1) = 0$. Hence $\ln t \le t - 1$ for every $t > 0$, with equality if and only if $t = 1$.

(13.1). Apply the tangent inequality at $t = x^{1/m}$: $\ln x^{1/m} = (\ln x)/m \le x^{1/m} - 1$, i.e. $\ln x \le m(x^{1/m} - 1)$. Apply it at $t = x^{-1/m}$: $-(\ln x)/m \le x^{-1/m} - 1$, i.e. $\ln x \ge m(1 - x^{-1/m})$. Equality holds on either side exactly when the argument t equals 1, that is, when $x = 1$.

(13.2). Put $y := (\ln x)/m$, so $x^{1/m} = e^{y}$ and $x^{-1/m} = e^{-y}$. The two slacks are $m(e^{y} - 1 - y)$ and $m(e^{-y} - 1 + y)$. From the exponential series, $e^{\pm y} - 1 \mp y = \sum_{k \ge 2} (\pm y)^k / k!$, and since $2/k! \le 1/(k-2)!$ for $k \ge 2$,

$$\big|e^{\pm y} - 1 \mp y\big| \;\le\; \sum_{k \ge 2} \frac{|y|^k}{k!} \;=\; \frac{y^2}{2} \sum_{k \ge 2} \frac{2\,|y|^{k-2}}{k!} \;\le\; \frac{y^2}{2} \sum_{i \ge 0} \frac{|y|^i}{i!} \;=\; \frac{y^2}{2}\, e^{|y|}.$$

Multiplying by m and substituting $y = (\ln x)/m$ gives (13.2).

(13.3). If $0 < l \le x^{1/m}$ then $x^{-1/m} \le 1/l$, so $m(1 - x^{-1/m}) \ge m(1 - 1/l)$; if $x^{1/m} \le u$ then $m(x^{1/m} - 1) \le m(u - 1)$. Combine with (13.1).

(13.4). For $y = c/d > 0$, $\sqrt{y} = \sqrt{c\,d\,4^s}/(d\,2^s)$, and for the integer $N := c\,d\,4^s$ the definition of isqrt gives $\mathrm{isqrt}(N) \le \sqrt{N} < \mathrm{isqrt}(N) + 1$; divide by $d\,2^s$. The difference of the two bounds is $1/(d\,2^s) \le 2^{-s}$ because $d \ge 1$.

The iteration. The square root is non-decreasing on $[0, \infty)$, so from $l_i \le x^{1/2^i} \le u_i$ follows $\sqrt{l_i} \le x^{1/2^{i+1}} \le \sqrt{u_i}$; applying the lower bound of (13.4) to $y = l_i$ and the upper bound to $y = u_i$, at a precision s_i of the

caller's choice, gives rationals $l_{i+1} \le x^{1/2^{i+1}} \le u_{i+1}$ with $l_{i+1} > 0$ whenever $l_i > 0$ and s_i is large enough that the lower bound is positive. Starting from $l_0 = u_0 = x$, after j steps $l := l_j \le x^{1/2^j} \le u_j =: u$. For the width $w_i := u_i - l_i$: each step's exact roots satisfy $\sqrt{u_i} - \sqrt{l_i} = w_i/(\sqrt{u_i} + \sqrt{l_i})$, and each of the two roundings moves its bound by at most $1/(d\,2^{s_i}) \le 2^{-s_i}$, with d the denominator of its own argument, because the two bounds of (13.4) enclose the root and differ by $1/(d\,2^{s_i})$: the lower bound satisfies $l_{i+1} > \sqrt{l_i} - 2^{-s_i}$ and the upper bound $u_{i+1} \le \sqrt{u_i} + 2^{-s_i}$. The two add at most $2 \cdot 2^{-s_i}$ in total, so

$$w_{i+1} \;\le\; \frac{w_i}{\sqrt{u_i} + \sqrt{l_i}} + 2 \cdot 2^{-s_i} \;\le\; \frac{w_i}{2\sqrt{l_i}} + 2 \cdot 2^{-s_i},$$

with $w_0 = 0$. To bound the factors $1/(2\sqrt{l_i})$ independently of the precisions, write $x_i := x^{1/2^i}$ and require of each s_i that $2^{-s_i} \le (1/\sqrt{2} - 1/2)\, x_{i+1}$, a condition on x alone that every larger s_i also meets. Then $l_i \ge x_i/2$ for every $i \le j$, by induction: $l_0 = x$, and $l_i \ge x_i/2$ gives $l_{i+1} > \sqrt{l_i} - 2^{-s_i} \ge x_{i+1}/\sqrt{2} - (1/\sqrt{2} - 1/2)\, x_{i+1} = x_{i+1}/2$, using $\sqrt{x_i} = x_{i+1}$. In particular every l_i is positive, and $1/(2\sqrt{l_i}) \le 1/(\sqrt{2}\, x_{i+1}) \le K$ for $i < j$, where $K := \max\{1, \max_{i<j} 1/(\sqrt{2}\, x_{i+1})\}$ depends on x and j alone. By induction $w_j \le \sum_{i<j} K^{\,j-1-i}\, 2 \cdot 2^{-s_i}$: finitely many terms, each driven below any prescribed bound by its own s_i without changing K. The achieved width is the explicit rational $u_j - l_j$ a caller reads off. □

Robbins' bounds enclose ln m! between Stirling's approximation plus 1/(12m + 1) and plus 1/(12m), and Wallis's partial products enclose 2π by rationals of any width. With rational bounds on ln m and ln(2π) this encloses ln m! by rationals of explicit width.

A binomial mass point is a ratio of factorials, and the local evaluation of the exact rule needs $\ln m!$ from both sides without a library logarithm. Part (i) is Stirling's formula with the two-sided error bounds of Robbins, and part (ii) is Wallis's product for $\pi/2$, which makes the constant $\frac{1}{2}\ln(2\pi)$ computable in exact rational arithmetic; both are derived in full here, the constant being identified through the central binomial coefficient.

Lemma 13.2 (Robbins' Stirling bounds with Wallis's product). Put $C := \frac{1}{2}\ln(2\pi)$.

(i) *(Robbins' bounds)* For every integer $m \ge 1$,

$$\Big(m+\frac{1}{2}\Big)\ln m - m + C + \frac{1}{12m+1} \;<\; \ln m! \;<\; \Big(m+\frac{1}{2}\Big)\ln m - m + C + \frac{1}{12m}, \tag{13.5}$$

equivalently $\sqrt{2\pi m}\,(m/e)^m\, e^{1/(12m+1)} < m! < \sqrt{2\pi m}\,(m/e)^m\, e^{1/(12m)}$.

(ii) *(Wallis's product)* For every integer $M \ge 1$, the rational

$$P_M \;:=\; \prod_{k=1}^{M} \frac{(2k)^2}{(2k-1)(2k+1)} \tag{13.6}$$

satisfies

$$P_M \;<\; \frac{\pi}{2} \;<\; P_M\,\frac{2M+1}{2M}, \qquad \text{hence} \qquad 4P_M \;<\; 2\pi \;<\; 4P_M\Big(1+\frac{1}{2M}\Big), \tag{13.7}$$

so that $2C = \ln(2\pi)$ is enclosed, through Lemma 13.1 applied to the two rationals in (13.7), by rationals of any prescribed width.

(iii) *(rational enclosure of* $\ln m!$*)* Let $m \geq 1$ and let $\lambda^- \leq \ln m \leq \lambda^+$ and $c^- \leq \ln(2\pi) \leq c^+$ be rationals. Then

$$\Big(m+\frac{1}{2}\Big)\lambda^- - m + \frac{c^-}{2} + \frac{1}{12m+1} \;<\; \ln m! \;<\; \Big(m+\frac{1}{2}\Big)\lambda^+ - m + \frac{c^+}{2} + \frac{1}{12m}, \tag{13.8}$$

an enclosure by rationals whose width is at most $(m+\frac{1}{2})(\lambda^+ - \lambda^-) + \frac{1}{2}(c^+ - c^-) + \frac{1}{12m(12m+1)}$. For $m = 0$, $\ln 0! = 0$ exactly.

Uses Lem. 13.1 · *Used by* Lem. 15.1, Rem. 15.4 · newly derived · implemented · *Code* certiqs_key.statistics.logs:log_factorial_bounds

Proof. Idea. The sequence $d_m = \ln m! - (m+\frac{1}{2})\ln m + m$ has differences given by a series in $1/(2m+1)$. Bounding that series from both sides makes $d_m - \frac{1}{12m}$ increasing and $d_m - \frac{1}{12m+1}$ decreasing, with a common limit, and Wallis's product, proved from the sine-power integrals, identifies the limit as $\frac{1}{2}\ln(2\pi)$ through the central binomial coefficient.

Step 1 (a series). For real $|x| < 1$,

$$\ln\frac{1+x}{1-x} \;=\; 2\sum_{k\geq 0}\frac{x^{2k+1}}{2k+1}. \tag{13.9}$$

Indeed both sides vanish at $x = 0$, and their derivatives agree: $\frac{\mathrm{d}}{\mathrm{d}x}\ln\frac{1+x}{1-x} = \frac{1}{1+x} + \frac{1}{1-x} = \frac{2}{1-x^2} = 2\sum_{k\geq 0} x^{2k}$, the geometric series converging uniformly on $|x| \leq \rho$ for every $\rho < 1$, so it may be integrated termwise from 0 to x.

Step 2 (the differences of d_m*).* For $m \geq 1$ put $d_m := \ln m! - (m+\frac{1}{2})\ln m + m$. Then

$$d_m - d_{m+1} = \ln m! - \ln(m+1)! - \Big(m+\frac{1}{2}\Big)\ln m + \Big(m+\frac{3}{2}\Big)\ln(m+1) - 1 = \Big(m+\frac{1}{2}\Big)\ln\frac{m+1}{m} - 1,$$

using $\ln(m+1)! - \ln m! = \ln(m+1)$. Put $x := 1/(2m+1) \in (0, \frac{1}{3}]$; then $(1+x)/(1-x) = (2m+2)/(2m) = (m+1)/m$ and $m + \frac{1}{2} = 1/(2x)$, so by (13.9)

$$d_m - d_{m+1} = \frac{1}{2x}\cdot 2\sum_{k\geq 0}\frac{x^{2k+1}}{2k+1} - 1 = \sum_{k\geq 0}\frac{x^{2k}}{2k+1} - 1 = \sum_{k\geq 1}\frac{x^{2k}}{2k+1}. \tag{13.10}$$

Step 3 (two bounds on the difference). Every term of (13.10) is positive, so keeping the first,

$$d_m - d_{m+1} \;>\; \frac{x^2}{3} = \frac{1}{3(2m+1)^2} \;>\; \frac{1}{12m+1} - \frac{1}{12m+13} = \frac{12}{(12m+1)(12m+13)},$$

where the last inequality is, after cross-multiplication, $(12m+1)(12m+13) > 36(2m+1)^2$, i.e. $144m^2 + 168m + 13 > 144m^2 + 144m + 36$, i.e. $24m > 23$, true for $m \geq 1$. Bounding each term by $1/3$ times the geometric series,

$$d_m - d_{m+1} \;<\; \frac{x^2}{3} \sum_{k \geq 0} x^{2k} = \frac{x^2}{3(1-x^2)} = \frac{1}{3\big((2m+1)^2 - 1\big)} = \frac{1}{12m(m+1)} = \frac{1}{12m} - \frac{1}{12(m+1)}.$$

Step 4 (two monotone sequences with a common limit). Put $a_m := d_m - \frac{1}{12m}$ and $b_m := d_m - \frac{1}{12m+1}$. By the upper bound of Step 3, $a_{m+1} - a_m = -(d_m - d_{m+1}) + \frac{1}{12m} - \frac{1}{12(m+1)} > 0$, so (a_m) is strictly increasing; by the lower bound, $b_{m+1} - b_m = -(d_m - d_{m+1}) + \frac{1}{12m+1} - \frac{1}{12m+13} < 0$, so (b_m) is strictly decreasing. Since $a_m < b_m$ for every m, (a_m) is bounded above by b_1 and (b_m) below by a_1, so both converge, and $b_m - a_m = \frac{1}{12m} - \frac{1}{12m+1} \to 0$ forces a common limit C_0. Strict monotonicity gives $a_m < C_0 < b_m$ for every $m \geq 1$, that is,

$$C_0 + \frac{1}{12m+1} \;<\; d_m \;<\; C_0 + \frac{1}{12m} \qquad (m \geq 1). \tag{13.11}$$

Step 5 (Wallis's product). For an integer $n \geq 0$ put $I_n := \int_0^{\pi/2} \sin^n x \, \mathrm{d}x$, so $I_0 = \pi/2$ and $I_1 = 1$. For $n \geq 2$, integration by parts with $u = \sin^{n-1} x$ and $\mathrm{d}v = \sin x \, \mathrm{d}x$ gives $I_n = \big[-\sin^{n-1} x \cos x\big]_0^{\pi/2} + (n-1) \int_0^{\pi/2} \sin^{n-2} x \cos^2 x \, \mathrm{d}x = (n-1)(I_{n-2} - I_n)$, whence $I_n = \frac{n-1}{n} I_{n-2}$. Iterating,

$$I_{2M} = \frac{\pi}{2} \prod_{k=1}^{M} \frac{2k-1}{2k}, \qquad I_{2M+1} = \prod_{k=1}^{M} \frac{2k}{2k+1} \qquad (M \geq 0).$$

On $(0, \pi/2)$ one has $0 < \sin x < 1$, so $\sin^{n+1} x < \sin^n x$ there and $0 < I_{n+1} < I_n$ for every n. Hence, for $M \geq 1$, $I_{2M+1} < I_{2M} < I_{2M-1}$. Now $I_{2M}/I_{2M+1} = \frac{\pi}{2} \prod_{k=1}^{M} \frac{(2k-1)(2k+1)}{(2k)^2} = \frac{\pi/2}{P_M}$, and $I_{2M+1} < I_{2M}$ gives $P_M < \pi/2$. Further $I_{2M-1} = \prod_{k=1}^{M-1} \frac{2k}{2k+1} = I_{2M+1} \cdot \frac{2M+1}{2M}$, so $I_{2M} < I_{2M-1}$ gives $\frac{\pi/2}{P_M} < \frac{2M+1}{2M}$, i.e. $\pi/2 < P_M \frac{2M+1}{2M}$. Multiplying by 4 gives the second display of (13.7); this is (ii), and Lemma 13.1 applied to the two rationals $4P_M$ and $4P_M(1 + 1/(2M))$ encloses $\ln(2\pi) = 2C$ by rationals, of width at most the sum of the root-bound widths and $\ln(1 + 1/(2M)) \leq 1/(2M)$, hence of any prescribed width for M and the root precision large enough.

Step 6 (the constant). We identify $C_0 = C$. From $\ln m! = d_m + (m + \frac{1}{2}) \ln m - m$,

$$\ln \binom{2M}{M} = \ln(2M)! - 2 \ln M! = d_{2M} - 2d_M + \Big(2M + \frac{1}{2}\Big) \ln(2M) - 2M - (2M+1) \ln M + 2M = d_{2M} - 2d_M$$

so $\binom{2M}{M} = e^{\,d_{2M}-2d_M}\,4^M\sqrt{2/M}$ and

$$\Big(\frac{4^M}{\binom{2M}{M}}\Big)^2 = e^{\,2(2d_M-d_{2M})}\,\frac{M}{2}. \tag{13.12}$$

On the other hand P_M is expressible through the same coefficient: $\prod_{k=1}^{M}(2k)^2 = 4^M(M!)^2$, $\prod_{k=1}^{M}(2k-1) = \frac{(2M)!}{2^M M!}$ and $\prod_{k=1}^{M}(2k+1) = \frac{(2M+1)!}{2^M M!}$, so

$$P_M = \frac{4^M(M!)^2\cdot 4^M(M!)^2}{(2M)!\,(2M+1)!} = \frac{16^M(M!)^4}{\big((2M)!\big)^2(2M+1)} = \frac{1}{2M+1}\Big(\frac{4^M}{\binom{2M}{M}}\Big)^2 = e^{\,2(2d_M-d_{2M})}\,\frac{M}{2(2M+1)}.$$

As $M\to\infty$, $2d_M - d_{2M} \to 2C_0 - C_0 = C_0$ by Step 4 and $M/(2(2M+1)) \to 1/4$, while $P_M \to \pi/2$ by (13.7). Hence $\pi/2 = e^{2C_0}/4$, i.e. $e^{2C_0} = 2\pi$ and $C_0 = \frac{1}{2}\ln(2\pi) = C$. With this, (13.11) is (13.5), and exponentiating gives the product form. This proves (i).

Step 7 (part (iii)). In (13.5) replace $\ln m$ by its bounds (the coefficient $m+\frac{1}{2}$ is positive) and $C = \frac{1}{2}\ln(2\pi)$ by $\frac{1}{2}c^{\mp}$; the width is the sum of the two widths so introduced plus $\frac{1}{12m} - \frac{1}{12m+1} = \frac{1}{12m(12m+1)}$. At $m=0$, $0!=1$. □

14 The Bernstein rule

Bernstein's tail inequality gives count endpoints in closed form, with no tail evaluated. With upward-rounded rational bounds the rule is admissible, monotone in the allocation when every rounding is, and its unrounded half-width decreases strictly with the round count, though nesting of its integer-rounded boxes in the round count is not claimed (Proposition 14.2). A remark says which completeness figure a rule that evaluates no tail can report.

Either tail of a binomial (n, p) count at distance t > 0 from np has probability at most exp(−t²/(2(v + t/3))) with v = np(1 − p), and every t at least a closed-form root t(L) brings both below e^(−L). The Bernstein endpoint rule evaluates that root.

The lemma is Bernstein's inequality for a sum of independent, bounded, centred random variables, specialised to a binomial count and derived in full here; its closed-form deviation $t(L)$, the positive root of $t^2 - (2L/3)\,t - 2vL = 0$, is what the Bernstein endpoint rule evaluates.

Lemma 14.1 (Bernstein's inequality for a binomial count). Let N be binomial with parameters (n,p), $n\ge 1$, $p\in[0,1]$, and put $v := np(1-p)$. For every real $t>0$,

$$\Pr\big[\mathsf{N}-np\ge t\big] \;\le\; \exp\Big(-\frac{t^2}{2\,(v+t/3)}\Big), \qquad \Pr\big[\mathsf{N}-np\le -t\big] \;\le\; \exp\Big(-\frac{t^2}{2\,(v+t/3)}\Big). \tag{14.1}$$

Consequently, for every real $L > 0$ and every

$$t \;\geq\; t(L) \;:=\; \frac{L}{3} + \sqrt{\Big(\frac{L}{3}\Big)^2 + 2vL}, \tag{14.2}$$

both tails in (14.1) are at most e^{-L}.

Uses Def. 1.1 · *Used by* Prop. 14.2 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.statistics.bernstein:bernstein_deviation

Proof. Idea. The exponential Markov inequality, with the moment-generating-function bound $\mathbb{E}[e^{\lambda X}] \leq \exp\big(\sigma^2(e^\lambda - 1 - \lambda)\big)$ for a centred $X \leq 1$ and the estimate $e^\lambda - 1 - \lambda \leq \lambda^2/(2(1-\lambda/3))$, evaluated at $\lambda = t/(v + t/3)$; the deviation $t(L)$ then solves a quadratic.

If $p \in \{0, 1\}$ then $\mathsf{N} = np$ with probability one, both probabilities in (14.1) are 0 for $t > 0$, and there is nothing to prove. Let $0 < p < 1$, so $v > 0$.

Step 1 (a moment-generating-function bound). Let X be a real random variable with $\mathbb{E}[X] = 0$, $\mathrm{Var}(X) = \sigma^2 < \infty$ and $X \leq 1$ almost surely, and let $\lambda > 0$. Put $g(\lambda) := e^\lambda - 1 - \lambda$. We claim

$$\mathbb{E}\big[e^{\lambda X}\big] \;\leq\; 1 + \sigma^2 g(\lambda) \;\leq\; \exp\big(\sigma^2 g(\lambda)\big). \tag{14.3}$$

The second inequality is $1 + z \leq e^z$. For the first, consider $\psi(u) := (e^u - 1 - u)/u^2$ for $u \neq 0$, with $\psi(0) := 1/2$ (its limit). We show that ψ is non-decreasing on $\mathbb{R}$. For $u \neq 0$, $\psi'(u) = N(u)/u^3$ with $N(u) := (e^u - 1)u - 2(e^u - 1 - u) = e^u(u-2) + u + 2$. Now $N(0) = 0$, $N'(u) = e^u(u-1) + 1$ with $N'(0) = 0$, and $N''(u) = u\,e^u$, which has the sign of u; hence N' decreases on $(-\infty, 0]$ and increases on $[0, \infty)$, so $N' \geq N'(0) = 0$ everywhere, so N is non-decreasing, and with $N(0) = 0$ the sign of $N(u)$ is the sign of u. Therefore $\psi'(u) = N(u)/u^3 \geq 0$ for every $u \neq 0$, and ψ is non-decreasing (continuity at 0 follows from the series $\psi(u) = \sum_{k \geq 2} u^{k-2}/k!$). For a real $x \leq 1$ and $\lambda > 0$ we have $\lambda x \leq \lambda$, hence $\psi(\lambda x) \leq \psi(\lambda)$, i.e. for $x \neq 0$

$$\frac{e^{\lambda x} - 1 - \lambda x}{\lambda^2 x^2} \;\leq\; \frac{e^\lambda - 1 - \lambda}{\lambda^2}, \qquad \text{so} \qquad e^{\lambda x} \;\leq\; 1 + \lambda x + x^2 g(\lambda),$$

and the last inequality holds at $x = 0$ too. Apply it to X pointwise and take expectations: $\mathbb{E}[e^{\lambda X}] \leq 1 + \lambda\,\mathbb{E}[X] + g(\lambda)\,\mathbb{E}[X^2] = 1 + \sigma^2 g(\lambda)$, which is (14.3).

Step 2 (exponential Markov). Write $\mathsf{N} = \sum_{j=1}^n \chi_j$ with χ_j independent Bernoulli(p) (Definition 1.1), and $X_j := \chi_j - p$. Then $\mathbb{E}[X_j] = 0$, $\mathrm{Var}(X_j) = p(1-p)$, and $X_j \leq 1 - p \leq 1$. For $\lambda > 0$, Markov's inequality applied to the non-negative variable $e^{\lambda \sum_j X_j}$ and independence give

$$\Pr\Big[\sum_j X_j \geq t\Big] \;\leq\; e^{-\lambda t}\,\mathbb{E}\Big[e^{\lambda \sum_j X_j}\Big] \;=\; e^{-\lambda t} \prod_{j=1}^n \mathbb{E}\big[e^{\lambda X_j}\big] \;\leq\; \exp\big(-\lambda t + v\,g(\lambda)\big),$$

by (14.3) with $\sigma^2 = p(1-p)$ and $np(1-p) = v$.

Step 3 (an elementary bound on g). For integers $k \geq 2$, $k! \geq 2 \cdot 3^{k-2}$ (induction: $2! = 2$, and $(k+1)! = (k+1)\,k! \geq 3 \cdot 2 \cdot 3^{k-2}$). Hence for $0 < \lambda < 3$,

$$g(\lambda) = \sum_{k\geq 2} \frac{\lambda^k}{k!} \;\leq\; \frac{\lambda^2}{2} \sum_{k\geq 2} \Big(\frac{\lambda}{3}\Big)^{k-2} \;=\; \frac{\lambda^2}{2\,(1-\lambda/3)}.$$

Step 4 (the choice of λ). Put $\lambda := t/(v+t/3)$, which lies in $(0,3)$ because $v > 0$. Then $1-\lambda/3 = v/(v+t/3)$, so Step 3 gives $g(\lambda) \leq \lambda^2(v+t/3)/(2v) = t^2/\big(2v(v+t/3)\big)$, and the exponent of Step 2 is at most

$$-\lambda t + v\,g(\lambda) \;\leq\; -\frac{t^2}{v+t/3} + \frac{t^2}{2(v+t/3)} \;=\; -\frac{t^2}{2(v+t/3)},$$

which is the first inequality of (14.1). For the second, apply Steps 1–4 to $-X_j = p - \chi_j$, which has the same mean and variance and satisfies $-X_j \leq p \leq 1$: $\Pr\Big[\sum_j(-X_j) \geq t\Big] = \Pr[\mathsf{N} - np \leq -t]$.

Step 5 (the closed-form deviation). For $t > 0$, the inequality $t^2/(2(v + t/3)) \geq L$ is equivalent to $t^2 - (2L/3)\,t - 2vL \geq 0$. The quadratic on the left has the two real roots $L/3 \pm \sqrt{(L/3)^2 + 2vL}$, of which $t(L)$ is the positive one and the other is ≤ 0; so the inequality holds for every $t \geq t(L)$, and then both bounds in (14.1) are at most e^{-L}. □

Bernstein's inequality with upward-rounded rational bounds on the logarithm and the root gives closed-form endpoints, an admissible rule, monotone in the allocation when every rounding is. Its unrounded half-width falls strictly in n as n^(−1/2), and it is never narrower than the exact rule.

The emitted interval of the exact rule is not monotone in n (Theorem 12.4 (v)). The Bernstein rule is the alternative whose unrounded half-width decreases strictly with n; that its integer-rounded boxes are nested in n is not claimed. Its endpoints come in closed form from Bernstein's inequality, with no tail evaluated, and every logarithm and root is bounded by rationals rounded in the direction that can only widen the interval, the direction a conservative evaluation (Definition 4.3) admits. A request selects it (Requirement 16.1), and j and s are recorded with the rule (Requirement 16.5).

Proposition 14.2 (The Bernstein endpoint rule)**.** Assume a conservative evaluation (Definition 4.3). Fix an integer $j \geq 1$ and a precision $s \geq 0$ (parameters of the evaluation). The *Bernstein rule* R_{B} assigns to $(n, p, \varepsilon^-, \varepsilon^+)$, with $v := np(1-p)$:

- if $p = 0$: $R_{\mathrm{B}}^- = R_{\mathrm{B}}^+ = 0$; if $p = 1$: $R_{\mathrm{B}}^- = R_{\mathrm{B}}^+ = n$;
- otherwise, for each sign separately: if $\varepsilon^- = 0$ then $R_{\mathrm{B}}^- = 0$, and if $\varepsilon^+ = 0$ then $R_{\mathrm{B}}^+ = n$; if $\varepsilon^{\mp} > 0$, let $L^{\mp}$ be the rational upper bound $m(u-1)$

of (13.3) on $\ln(1/\varepsilon^{\mp})$ with $m = 2^j$ and u the upper root enclosure of (13.4) at precision s, let

$$t^{\mp} \;\coloneqq\; \text{a rational} \;\geq\; \frac{L^{\mp}}{3} + \sqrt{\left(\frac{L^{\mp}}{3}\right)^2 + 2vL^{\mp}} \tag{14.4}$$

obtained by rounding the square root upward with (13.4), and put

$$R_{\mathrm{B}}^{-} \;\coloneqq\; \max\left\{0,\; \lfloor np - t^{-} \rfloor\right\}, \qquad R_{\mathrm{B}}^{+} \;\coloneqq\; \min\left\{n,\; \lceil np + t^{+} \rceil\right\}. \tag{14.5}$$

Then:

(i) R_{B} is an admissible endpoint rule in the sense of Definition 7.1 (a)–(c), computed from n, p and the allocation by finitely many operations on integers and rationals (integer square roots included) and no others; Lemma 7.2 therefore applies to it;

(ii) R_{B} is monotone in the allocation, provided every rounding in $L^{\mp}$ and $t^{\mp}$ is a non-decreasing function of its real argument. The integer-square-root rounding of (13.4) is *not* such a function of a rational argument as written (its excess over the root depends on the denominator), but it becomes one when the argument is first rounded up to the fixed dyadic grid $2^{-s}\mathbb{Z}$, i.e. replaced by $\lceil y\,2^s \rceil / 2^s$ before the root is taken; the evaluation takes u from (13.4) applied to that grid point (with $d = 2^s$ fixed) at every step of Lemma 13.1's iteration, rounds the square root in (14.4) the same way, and takes $L^{\mp}$ and $t^{\mp}$ themselves as the least multiples of 2^{-s} at or above the values so obtained. Every such step is a composition of non-decreasing maps, and every rounding is upward, so (i) is unaffected;

(iii) *(half-width)* for $0 < p < 1$ and $\varepsilon^{\mp} > 0$, the guarded endpoints of R_{B} satisfy

$$0 \;\leq\; q^{+}(\hat{c}) - p \;\leq\; \frac{t^{+}}{n} + \frac{1}{n}, \qquad 0 \;\leq\; p - q^{-}(\hat{c}) \;\leq\; \frac{t^{-}}{n} + \frac{1}{n}, \tag{14.6}$$

where, for the unrounded value of (14.4),

$$\frac{t^{\mp}}{n} \;=\; \frac{L^{\mp}}{3n} + \sqrt{\left(\frac{L^{\mp}}{3n}\right)^2 + \frac{2p(1-p)L^{\mp}}{n}} \tag{14.7}$$

is strictly decreasing in n and is $O(n^{-1/2})$ at fixed p and $\varepsilon^{\mp}$. The rule never emits an interval narrower than the raw quantile interval of the exact rule of Definition 8.1 (Proposition 8.2 (iii)).

Uses Def. 4.3, Def. 7.1, Lem. 7.2, Def. 8.1, Lem. 13.1, Lem. 14.1 · *Used by* Rem. 14.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.statistics.rules:BernsteinRule

Proof. (i). Definition 7.1 (c) and (b) hold by the first two clauses of the rule. For (a) at $p \in \{0,1\}$: the count is 0 respectively n almost surely, so with $R_{\mathrm{B}}^{\mp} = 0$ respectively n both tails of (7.1) vanish and the values lie in $\{0,\dots,n\}$. For (a) at $0 < p < 1$: the values (14.5) lie in $\{0,\dots,n\}$ by the clipping. If $\varepsilon^+ = 0$ then $R_{\mathrm{B}}^+ = n$ and $\Pr[\mathsf{N} > n] = 0 \le \varepsilon^+$. If $\varepsilon^+ > 0$: by Lemma 13.1, $L^+ \ge \ln(1/\varepsilon^+)$, so $e^{-L^+} \le \varepsilon^+$; by Lemma 14.1 with $L = L^+$, every $t \ge t(L^+)$ has $\Pr[\mathsf{N} - np \ge t] \le e^{-L^+}$, and $t^+ \ge t(L^+)$ by (14.4). If $R_{\mathrm{B}}^+ = n$ the tail is 0. Otherwise $R_{\mathrm{B}}^+ = \lceil np + t^+ \rceil \ge np + t^+$, so the event $\{\mathsf{N} > R_{\mathrm{B}}^+\}$ is contained in $\{\mathsf{N} - np > t^+\} \subseteq \{\mathsf{N} - np \ge t^+\}$ and $\Pr\big[\mathsf{N} > R_{\mathrm{B}}^+\big] \le e^{-L^+} \le \varepsilon^+$. The lower side is the mirror image: if $R_{\mathrm{B}}^- = 0$ the tail $\Pr[\mathsf{N} < 0]$ is 0; otherwise $R_{\mathrm{B}}^- = \lfloor np - t^- \rfloor \le np - t^-$, so $\{\mathsf{N} < R_{\mathrm{B}}^-\} \subseteq \{\mathsf{N} - np < -t^-\} \subseteq \{\mathsf{N} - np \le -t^-\}$, whose probability is at most $e^{-L^-} \le \varepsilon^-$ by the second inequality of (14.1). The computation uses the rational arithmetic of the inputs, the integer square roots of (13.4) for u and for the root in (14.4) (applied to the rational $(L^{\mp}/3)^2 + 2vL^{\mp}$), and floor, ceiling, maximum and minimum.

(ii). Fix all arguments but ε^+ and let $\varepsilon_1^+ \le \varepsilon_2^+$, both positive. Then $1/\varepsilon_1^+ \ge 1/\varepsilon_2^+$. With the argument first rounded up to the grid, $y \mapsto \lceil y\, 2^s \rceil$ is non-decreasing, and for a *fixed* denominator $d = 2^s$ the bound $u = (\mathrm{isqrt}(c\,d\,4^s) + 1)/(d\,2^s)$ of (13.4) is non-decreasing in the integer numerator c because isqrt is non-decreasing on the integers; so each step of the root iteration is non-decreasing in its input, and u is non-decreasing in $1/\varepsilon^+$. The final upward rounding to the grid is non-decreasing too, so $L_1^+ \ge L_2^+$; the map $L \mapsto L/3 + \sqrt{(L/3)^2 + 2vL}$ is increasing in $L \ge 0$ and the upward rounding of the root is monotone, so $t_1^+ \ge t_2^+$; hence $R_{\mathrm{B},1}^+ \le R_{\mathrm{B},2}^+$ by (14.5), which is the required non-increase of $\varepsilon^+ \mapsto R_{\mathrm{B}}^+$. At $\varepsilon_1^+ = 0$ the value n is the largest possible, so the monotonicity extends to the zero budget. The lower side is the mirror image (R_{B}^- non-decreasing in ε^-, with the value 0 at $\varepsilon^- = 0$ the smallest possible).

(iii). The left inequalities of (14.6) are (7.3). For the right ones: by (7.2) and (14.5), $k^+(\hat{c}) = \max\{R_{\mathrm{B}}^+, \lceil np \rceil\} \le \lceil np + t^+ \rceil \le np + t^+ + 1$ (using $\lceil np \rceil \le \lceil np + t^+ \rceil$ and $n \le \lceil np + t^+ \rceil$ when the clip at n is active), so $q^+(\hat{c}) - p \le (t^+ + 1)/n$; symmetrically $k^-(\hat{c}) = \min\{R_{\mathrm{B}}^-, \lfloor np \rfloor\} \ge \lfloor np - t^- \rfloor \ge np - t^- - 1$ when the clip at 0 is inactive, and $k^-(\hat{c}) \ge 0 > np - t^- - 1$ when it is active (then $\lfloor np - t^- \rfloor \le 0$, so $np - t^- < 1$); hence $p - q^-(\hat{c}) \le (t^- + 1)/n$. Dividing (14.4) by n gives (14.7); each of its two terms is strictly decreasing in n (the first as $1/n$, the second as the square root of a sum of a term in $1/n^2$ and a term in $1/n$, both decreasing, at fixed $p \in (0,1)$ and $L^{\mp} > 0$), and it is $O(n^{-1/2})$ because the first term is $O(n^{-1})$ and the radicand is $O(n^{-1})$. The comparison with the exact rule: the exact rule's κ^+ is the minimum of the set in (8.2) (Definition 8.1), and $R_{\mathrm{B}}^+ \in \{0,\dots,n\}$ lies in that set by (i), so $R_{\mathrm{B}}^+ \ge \kappa^+$; likewise κ^- is the maximum of the set in (8.1), in which R_{B}^- lies by (i), so $R_{\mathrm{B}}^- \le \kappa^-$; the guard preserves both inequalities. □

The tightest completeness figure this submodule certifies on any admissible rule is ε^*, but a rule that evaluates no tail can report the allocation sum or a clamped upward-rounded tail evaluation; both bound the honest failure probability. Neither choice changes the emitted set or soundness.

Remark 14.3 (Completeness figures on a closed-form rule). Theorem 10.2 holds for every admissible endpoint rule, and its $\varepsilon^\star$, the sum of the binomial tails at the guarded endpoints, is the tightest completeness figure this submodule certifies for the emitted set, whatever rule produced the endpoints: every figure this remark lists is at least $\varepsilon^\star$, while the honest failure probability itself, which $\varepsilon^\star$ bounds by a union bound, may be smaller. What differs between rules is the cost of *evaluating* it. On the exact rule of Definition 8.1 the tails are decided in the course of finding the quantiles, and $\varepsilon^\star$ is reported exactly wherever they are decided exactly; where they are decided against the upper bounds of Proposition 8.2 (vii) it is rounded upward and clamped (Remark 10.3). On a closed-form rule such as the Bernstein rule of Proposition 14.2, the endpoints are found without evaluating any tail, and two reports are admissible, both upward-rounded in the sense of a conservative evaluation (Definition 4.3) and Remark 10.3:

(a) *the allocation sum* $\sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{con}}}\big(\varepsilon^-(\hat{c})+\varepsilon^+(\hat{c})\big)$. By Lemma 7.2 (ii) each tail at a guarded endpoint is at most its allocation, so the sum is at least $\varepsilon^\star$ and hence, by (10.6), at least the honest failure probability; and it lies in $[0,\varepsilon_{\mathrm{comp}}]$ because the allocation of an admissible input is non-negative and sums to at most $\varepsilon_{\mathrm{comp}}$ (Definition 7.1), so no clamp is needed. It costs nothing beyond the inputs and is what a rule that evaluates no tail can always report;

(b) *an upward-rounded evaluation of the tails at the emitted endpoints*, $\min\{\varepsilon_{\mathrm{comp}},\tilde{\varepsilon}\}$ for any $\tilde{\varepsilon}\geq\varepsilon^\star$ obtained by bounding each tail from above (exactly, by the closed forms of the exact rule's evaluation, or by an enclosure), which is the clamped figure of Remark 10.3. At $\tilde{\varepsilon}=\varepsilon^\star$ it is tighter than (a) by the budget the guarded endpoints leave unspent, $\sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{con}}}(\varepsilon^-(\hat{c})+\varepsilon^+(\hat{c}))-\varepsilon^\star$, which on a closed-form rule includes the slack of the rule's own tail bound besides what integrality and the guard leave; an upward-rounded $\tilde{\varepsilon}>\varepsilon^\star$ need not be at most (a), since an upper bound on a tail at a guarded endpoint may exceed that tail's allocation.

Neither choice affects the emitted set, and neither affects soundness (Remark 10.4); the difference is the advertised abort probability of the honest run, which (a) overstates by the unspent budget. Requirement 16.4 fixes which of the two is reported on each route, and Requirement 16.5 what is recorded beside the figure.

Uses Def. 4.3, Def. 7.1, Lem. 7.2, Def. 8.1, Thm. 10.2, Rem. 10.3, Rem. 10.4, Prop. 14.2 · *Used by* Rem. 15.4, Req. 16.4, Req. 16.5 · newly derived · accepted

15 The exact rule's local evaluation

A binomial tail is enclosed in log space from a window of mass points and a geometric majorant of the rest, each mass point enclosed through Stirling's bounds. Endpoints bracketed by these enclosures form an admissible endpoint rule that relaxes the exact quantiles: equal to them where the bracket is decided, and otherwise, where the tail decays at the quantile, conservative

by a number of counts bounded through the enclosure's slack and that decay (Proposition 15.3).

Rational bounds on ln m!, ln p and ln(1 − p) enclose the logarithm of a binomial mass point, with width the sum of the component widths. On the Stirling branch no intermediate is of the size of the mass point, of p^j or of a binomial coefficient.

At large n a binomial mass point such as p^n is too small, and its exact rational too long, for an implementation to hold, while its logarithm is of moderate size. The logarithm of the law of Theorem 2.1 is a sum of three log-factorials and two logarithms, and each is replaced by its rational bound in the direction its sign requires.

Lemma 15.1 (Rational enclosure of the logarithm of a binomial mass point). Let $n \geq 1$, let $p \in (0,1)$ be rational, let N be binomial with parameters (n,p) and write $\mathrm{pmf}(j) \coloneqq \Pr[\mathsf{N} = j] = \binom{n}{j} p^j (1-p)^{n-j} > 0$ for $0 \leq j \leq n$ (Theorem 2.1), and $F(m) \coloneqq \ln m!$ for an integer $m \geq 0$. Then

$$\ln \mathrm{pmf}(j) \;=\; F(n) - F(j) - F(n-j) + j \ln p + (n-j)\ln(1-p). \tag{15.1}$$

Let $F^-(m) \leq F(m) \leq F^+(m)$ be rational bounds for $m \in \{n, j, n-j\}$ — those of Lemma 13.2 (iii) for $m \geq 1$, or $F^{\mp}(m)$ the root bounds of Lemma 13.1 on the integer $m!$ where that integer is held exactly, and $F^{\mp}(0) = 0$ — and let $\lambda_p^- \leq \ln p \leq \lambda_p^+$ and $\lambda_q^- \leq \ln(1-p) \leq \lambda_q^+$ be rational bounds (Lemma 13.1). Put

$$\begin{aligned} \ell^-(j) &\coloneqq F^-(n) - F^+(j) - F^+(n-j) + j\,\lambda_p^- + (n-j)\,\lambda_q^-, \\ \ell^+(j) &\coloneqq F^+(n) - F^-(j) - F^-(n-j) + j\,\lambda_p^+ + (n-j)\,\lambda_q^+. \end{aligned} \tag{15.2}$$

Then $\ell^-(j) \leq \ln \mathrm{pmf}(j) \leq \ell^+(j)$, both bounds are rationals, and the width is the sum of the component widths,

$$\ell^+(j) - \ell^-(j) = \sum_{m \in \{n,\, j,\, n-j\}} \big(F^+(m) - F^-(m)\big) + j\,(\lambda_p^+ - \lambda_p^-) + (n-j)\,(\lambda_q^+ - \lambda_q^-). \tag{15.3}$$

In particular, if every logarithm entering the bounds is enclosed to width δ and $\ln(2\pi)$ to width δ_C, the width is at most $(3n + \frac{3}{2})\,\delta + \frac{3}{2}\,\delta_C + \frac{3}{156}$. A log precision of 2^{-s} with s about $\log_2 n$ bits beyond the wanted precision brings the terms in δ and δ_C within it; the remaining term, at most $\frac{1}{12m(12m+1)}$ for each $m \in \{n, j, n-j\}$ taken on the Stirling branch, is fixed by m, and no choice of s reduces it. On the Stirling branch every quantity in (15.2) is a rational whose size is polynomial in the bit sizes of n, p and the precision, and no intermediate has the magnitude of $\mathrm{pmf}(j)$, p^j or $\binom{n}{j}$ themselves; the exact-factorial branch is for small m only and holds the integer $m!$ by construction.

Uses Thm. 2.1, Lem. 13.1, Lem. 13.2 · *Used by* Prop. 15.3, Rem. 15.4, Req. 17.2 · newly derived · implemented · *Code* certiqs_key.statistics.enclosure:log_pmf_bounds

Proof. (15.1) is the logarithm of the law, using $\ln\binom{n}{j} = \ln n! - \ln j! - \ln(n-j)!$ and $p, 1-p > 0$. For the bounds, replace each term of (15.1) by the bound in the direction its sign requires: $F(n)$, $j \ln p$ and $(n-j)\ln(1-p)$ enter with positive coefficients ($j \geq 0$, $n-j \geq 0$) and are bounded below by their lower and above by their upper bounds; $F(j)$ and $F(n-j)$ enter with a minus sign, so a lower bound on $\ln \mathrm{pmf}(j)$ uses their upper bounds and an upper bound uses their lower bounds. Subtracting the two displays of (15.2) gives (15.3). For the estimate: on the Stirling branch, by Lemma 13.2 (iii), the width of $F^{\mp}(m)$ is at most $(m+\frac{1}{2})\delta + \frac{1}{2}\delta_C + \frac{1}{12m(12m+1)}$, and $\frac{1}{12m(12m+1)} \leq \frac{1}{156}$ for $m \geq 1$; on the exact-factorial branch it is the width of the root bounds on $\ln m!$, a logarithm entering the bounds, so at most $\delta \leq (m+\frac{1}{2})\delta$ for $m \geq 1$; and for $m = 0$ it is 0. So each of the three widths is at most $(m+\frac{1}{2})\delta + \frac{1}{2}\delta_C + \frac{1}{156}$; summing over $m \in \{n, j, n-j\}$, whose values add up to $2n$, and adding $j\delta + (n-j)\delta = n\delta$ gives at most $(2n+\frac{3}{2})\delta + n\delta + \frac{3}{2}\delta_C + \frac{3}{156}$. The size statement holds because each term of (15.2) is a sum of products of the given rationals with the integers n, j, $n-j$ and $m+\frac{1}{2}$; the rational $\frac{1}{12m+1}$ and its like are of the bit size of m. □

A binomial upper tail lies between a window of exact mass points and that window plus a geometric majorant of the rest, because consecutive mass ratios decrease. The same ratios give log-space bounds in units of one mass point, a per-count decay of the tail and, by reflection, the lower tail.

The lemma is the classical geometric bound on a binomial tail through the decreasing ratio of consecutive mass points, Feller's tail estimate for the binomial law, derived in full here from the ratio identity, a telescoping product and a geometric series.

Lemma 15.2 (The binomial tail from a window and a geometric majorant). Let $n \geq 1$, let $p \in (0,1)$ be rational, let N be binomial with parameters (n,p), and write $\mathrm{pmf}(j) := \Pr[\mathsf{N} = j]$ for $0 \leq j \leq n$ (Theorem 2.1) and $\mathrm{pmf}(j) := 0$ for $j > n$.

(i) *(ratios)* For $0 \leq j \leq n-1$,

$$r_j \;:=\; \frac{\mathrm{pmf}(j+1)}{\mathrm{pmf}(j)} \;=\; \frac{(n-j)\,p}{(j+1)\,(1-p)}, \tag{15.4}$$

a positive rational, strictly decreasing in j; put $r_n := 0$.

(ii) *(window plus majorant)* Let $0 \leq k \leq n-1$ and let $T \geq 0$ be an integer with $k+T+1 \leq n$ and $r_{k+T+1} < 1$ (which holds in particular when $k+T+1 = n$). Then, with $S_T := \sum_{i=0}^{T-1} \mathrm{pmf}(k+1+i)$,

$$S_T \;\leq\; \Pr[\mathsf{N} > k] \;\leq\; S_T + \frac{\mathrm{pmf}(k+T+1)}{1-r_{k+T+1}}. \tag{15.5}$$

If instead $k+T+1 > n$, then $\Pr[\mathsf{N} > k] = S_T$ exactly.

(iii) *(in units of one mass point)* In the setting of (ii) with $T \geq 1$, put

$$A_T \coloneqq \sum_{i=0}^{T-1} \prod_{l=k+1}^{k+i} r_l, \qquad B_T \coloneqq \frac{\prod_{l=k+1}^{k+T} r_l}{1 - r_{k+T+1}} \tag{15.6}$$

(the empty product is 1), positive rationals computable from n, p, k and T alone. Then

$$\ln \mathrm{pmf}(k+1) + \ln A_T \;\leq\; \ln \Pr[\mathsf{N} > k] \;\leq\; \ln \mathrm{pmf}(k+1) + \ln(A_T + B_T). \tag{15.7}$$

(iv) *(per-count decay)* For $1 \leq k \leq n$ with $r_k < 1$,

$$\Pr[\mathsf{N} > k] \;\leq\; r_k \Pr[\mathsf{N} > k-1]. \tag{15.8}$$

(v) *(the lower tail)* $\mathsf{N}' \coloneqq n - \mathsf{N}$ is binomial with parameters $(n, 1-p)$, and $\Pr[\mathsf{N} < k] = \Pr[\mathsf{N}' > n-k]$ for every integer k; so (i)–(iv) applied to N' enclose the lower tail of N.

Uses Thm. 2.1 · *Used by* Prop. 15.3, Rem. 15.4, Req. 17.2 · newly derived · implemented · *Code* certiqs_key.statistics.enclosure:upper_tail_log_bounds

Proof. *(i).* From the law, $\frac{\mathrm{pmf}(j+1)}{\mathrm{pmf}(j)} = \frac{\binom{n}{j+1}}{\binom{n}{j}} \cdot \frac{p}{1-p} = \frac{n-j}{j+1} \cdot \frac{p}{1-p}$, using $\binom{n}{j+1} = \binom{n}{j}\frac{n-j}{j+1}$. The factor $\frac{n-j}{j+1}$ is strictly decreasing in j (numerator decreasing, denominator increasing, both positive for $0 \leq j \leq n-1$), and $\frac{p}{1-p} > 0$.

(ii). $\Pr[\mathsf{N} > k] = \sum_{j=k+1}^{n} \mathrm{pmf}(j) = S_T + \sum_{j=k+T+1}^{n} \mathrm{pmf}(j)$, and every term is non-negative, so $S_T \leq \Pr[\mathsf{N} > k]$; if $k+T+1 > n$ the second sum is empty and equality holds. For the upper bound let $k+T+1 \leq n$ and $j \geq k+T+1$. Telescoping, $\mathrm{pmf}(j) = \mathrm{pmf}(k+T+1) \prod_{l=k+T+1}^{j-1} r_l$, and each factor satisfies $r_l \leq r_{k+T+1}$ by (i) (for $l \leq n-1$; and if the product reaches $l = n$ the mass point is 0 anyway), so $\mathrm{pmf}(j) \leq \mathrm{pmf}(k+T+1)\, r_{k+T+1}^{j-k-T-1}$. Summing over $j \geq k+T+1$ and bounding the finite geometric sum by the full series with ratio $r_{k+T+1} < 1$, $\sum_{j \geq k+T+1} \mathrm{pmf}(j) \leq \mathrm{pmf}(k+T+1)/(1 - r_{k+T+1})$.

(iii). Telescoping again, $\mathrm{pmf}(k+1+i) = \mathrm{pmf}(k+1) \prod_{l=k+1}^{k+i} r_l$ for $0 \leq i \leq T$, so $S_T = \mathrm{pmf}(k+1)\, A_T$ and $\mathrm{pmf}(k+T+1)/(1 - r_{k+T+1}) = \mathrm{pmf}(k+1)\, B_T$. Substituting into (15.5) and taking logarithms (all quantities positive: $A_T \geq 1$ since its first term is 1, and $B_T > 0$) gives (15.7). The rationals A_T, B_T are finite sums and products of the rationals (15.4).

(iv). For every $j \geq k$ with $j \leq n-1$, (i) gives $\mathrm{pmf}(j+1) = r_j \,\mathrm{pmf}(j) \leq r_k \,\mathrm{pmf}(j)$, and the same inequality holds trivially for $j \geq n$ (both sides 0). Summing over $j \geq k$: $\Pr[\mathsf{N} > k] = \sum_{j \geq k} \mathrm{pmf}(j+1) \leq r_k \sum_{j \geq k} \mathrm{pmf}(j) = r_k \Pr[\mathsf{N} \geq k] = r_k \Pr[\mathsf{N} > k-1]$.

(v). $\Pr[n - \mathsf{N} = i] = \Pr[\mathsf{N} = n-i] = \binom{n}{n-i} p^{n-i}(1-p)^i = \binom{n}{i}(1-p)^i p^{n-i}$, the law of a binomial with parameters $(n, 1-p)$; and $\mathsf{N} < k$ if and only if $n - \mathsf{N} > n - k$. □

Endpoints bracketed by certified tail enclosures form an admissible rule whose box contains the exact rule's. Each equals the exact quantile where its bracket is decided; otherwise, where the tail decays at the quantile, the excess in counts is bounded through the enclosure's slack and that decay.

The exact rule compares binomial tails with their budgets. Where exact sums are too costly, the comparisons can be decided against enclosures of the tails instead, which is what a conservative evaluation (Definition 4.3) admits: a comparison decided against an upper bound on the tail. A request selects this evaluation (Requirement 16.1), and its endpoints are found by the one search procedure Requirement 17.1 fixes; Remark 15.4 discusses the search, the enclosures and their cost. Through the closing sentences of (iii), the width bound (12.5) of Theorem 12.4 (iv), proved for the exact rule, carries over to this evaluation wherever the bracket is decided and, where it is not, up to the excess (iii) bounds, divided by n; beyond that it is not asserted for this evaluation.

Proposition 15.3 (The enclosure evaluation of the exact rule)**.** Assume a conservative evaluation (Definition 4.3). Let $n \geq 1$, let $p \in (0,1)$ and $\varepsilon^+ \in (0,1)$ be rational, let N be binomial with parameters (n,p), and let κ^+ be the exact upper quantile of Definition 8.1 at (n,p,ε^+). Let

$$\tau^-(k) \;\leq\; \Pr[\mathsf{N} > k] \;\leq\; \tau^+(k), \qquad k = 0,\dots,n, \tag{15.9}$$

be an enclosure with $\tau^-(n) = \tau^+(n) = 0$, each bound either a rational or given through a rational bound on its logarithm, such as the one Lemma 15.2 (ii)–(iii) yields with the mass point enclosed by Lemma 15.1 and the logarithms by Lemma 13.1. Call k *certified admissible*, $\mathsf{C}^+(k)$, when the available upper bound proves $\Pr[\mathsf{N} > k] \leq \varepsilon^+$: $\tau^+(k) \leq \varepsilon^+$, or in log space $U(k) \leq \lambda^-$ for rationals $U(k) \geq \ln \Pr[\mathsf{N} > k]$ and $\lambda^- \leq \ln \varepsilon^+$; and *certified inadmissible*, $\mathsf{D}^+(k)$, when the lower bound proves $\Pr[\mathsf{N} > k] > \varepsilon^+$: $\tau^-(k) > \varepsilon^+$, or in log space $\Lambda(k) > \lambda^+$ for rationals $\Lambda(k) \leq \ln \Pr[\mathsf{N} > k]$ and $\lambda^+ \geq \ln \varepsilon^+$. An index that is neither is *undecided*. The predicate $\mathsf{C}^+(n)$ always holds.

A *bracketed upper endpoint* is an integer $\tilde{k}^+ \in \{0,\dots,n\}$ with $\mathsf{C}^+(\tilde{k}^+)$ and, if $\tilde{k}^+ \geq 1$, $\neg\mathsf{C}^+(\tilde{k}^+ - 1)$. The *enclosure evaluation of the exact rule* is the endpoint rule R_{encl} whose R^+_{encl} is the bracketed upper endpoint returned by a fixed deterministic search procedure that returns such an endpoint for every input, whose R^-_{encl} is obtained in the same way for the lower tail through the mirror $\mathsf{N}' = n - \mathsf{N}$ of Lemma 15.2 (v), and which returns the values of Definition 7.1 (b) and (c) at a zero budget and at $p \in \{0,1\}$. Then:

(i) *(admissible relaxation)* Every bracketed upper endpoint satisfies $\Pr\left[\mathsf{N} > \tilde{k}^+\right] \leq \varepsilon^+$ and $\kappa^+ \leq \tilde{k}^+$; mirror-wise every bracketed lower endpoint $\tilde{k}^-$ satisfies $\Pr\left[\mathsf{N} < \tilde{k}^-\right] \leq \varepsilon^-$ and $\tilde{k}^- \leq \kappa^-$. Hence R_{encl} is an admissible endpoint rule, Lemma 7.2 applies to it, and the box $\mathsf{Acc}_{R_{\mathrm{encl}}}$ emits

(Definition 9.1) contains the box the exact rule emits: every q^- is at most and every q^+ at least the exact rule's.

(ii) *(the bracket decided)* If $\tilde{k}^+ \geq 1$ and $\mathsf{D}^+(\tilde{k}^+ - 1)$ holds, then $\tilde{k}^+ = \kappa^+$; if $\tilde{k}^+ = 0$ then $\tilde{k}^+ = \kappa^+$. The mirror statement holds for the lower side.

(iii) *(conservatism in counts)* Suppose $\tilde{k}^+ > \kappa^+$, $\kappa^+ \leq n-2$ and $r_{\kappa^+ +1} < 1$, with r_j the ratios (15.4). Let $w \geq 0$ be the slack of the failed certificate at $\tilde{k}^+ - 1$: $w := \ln \tau^+(\tilde{k}^+ - 1) - \ln \Pr\left[\mathsf{N} > \tilde{k}^+ - 1\right]$, or in log space $w := U(\tilde{k}^+ - 1) - \ln \Pr\left[\mathsf{N} > \tilde{k}^+ - 1\right] + (\ln \varepsilon^+ - \lambda^-)$. Then

$$\tilde{k}^+ - \kappa^+ \;<\; 1 + \frac{w}{-\ln r_{\kappa^+ +1}}. \tag{15.10}$$

The emitted endpoint is therefore the exact quantile up to a number of counts bounded by the enclosure's slack divided by the per-count decay of the tail at the quantile. The mirror statement holds for the lower side, through N' (Lemma 15.2 (v)): with $\kappa^- = n - \kappa'^+$ and $\tilde{k}^- = n - \tilde{k}'^+$, where the primed quantities are those of N' at level ε^-, $\kappa^- - \tilde{k}^- < 1 + w'/(-\ln r'_{\kappa'^+ +1})$ under the mirrored hypotheses. Consequently, wherever the bracket is decided the emitted endpoint is the exact rule's, and so are its guarded endpoint and its box bound; wherever it is undecided the guarded endpoint exceeds the exact rule's by at most $\tilde{k}^+ - \kappa^+$ counts and the box bound q^+ the exact rule's by at most $(\tilde{k}^+ - \kappa^+)/n$, which (15.10) bounds under the hypotheses above; mirror-wise for k^- and q^-. Where the hypothesis $r_{\kappa^+ +1} < 1$ fails (the exact quantile at or below the mean, reachable at a near-unit tail budget, Proposition 8.2 (vi)) the endpoint is still admissible by (i) and only this bound is unavailable; at $\kappa^+ = n - 1$ the excess $\tilde{k}^+ - \kappa^+$ is at most one count.

Uses Def. 4.3, Def. 7.1, Lem. 7.2, Def. 8.1, Prop. 8.2, Def. 9.1, Lem. 13.1, Lem. 15.1, Lem. 15.2 · *Used by* Rem. 15.4, Req. 17.1, Req. 17.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.statistics.enclosure:bracketed_upper_endpoint

Proof. (i). $\mathsf{C}^+(\tilde{k}^+)$ gives $\Pr\left[\mathsf{N} > \tilde{k}^+\right] \leq \tau^+(\tilde{k}^+) \leq \varepsilon^+$, or in log space $\Pr\left[\mathsf{N} > \tilde{k}^+\right] \leq e^{U(\tilde{k}^+)} \leq e^{\lambda^-} \leq \varepsilon^+$. So $\tilde{k}^+$ belongs to the set in (8.2), and $\kappa^+ \leq \tilde{k}^+$ by the minimality that defines κ^+ (Proposition 8.2 (iii)). The lower side is the same argument for N': a certified index $\tilde{k}'$ for the upper tail of N' at level ε^- gives $\Pr\left[\mathsf{N} < n - \tilde{k}'\right] = \Pr\left[\mathsf{N}' > \tilde{k}'\right] \leq \varepsilon^-$ by Lemma 15.2 (v), so $\tilde{k}^- := n - \tilde{k}'$ lies in the set of (8.1) and $\tilde{k}^- \leq \kappa^-$ by maximality. Both values lie in $\{0, \ldots, n\}$; with the conventions at a zero budget and at $p \in \{0, 1\}$ imposed by definition, R_{encl} satisfies Definition 7.1 (a)–(c). For the containment: by (7.2) the guarded endpoints are monotone in the rule's

endpoints, so $\tilde{k}^- \le \kappa^-$ and $\tilde{k}^+ \ge \kappa^+$ give $k^-_{\mathrm{encl}}(\hat{c}) \le k^-_{\mathrm{ex}}(\hat{c})$ and $k^+_{\mathrm{encl}}(\hat{c}) \ge k^+_{\mathrm{ex}}(\hat{c})$ at every constrained outcome, hence the same order for the bounds (9.1) and the inclusion of the sets (9.2).

(ii). If $\tilde{k}^+ = 0$ then $\kappa^+ \le 0$ by (i), so $\kappa^+ = 0$. Otherwise $\mathsf{D}^+(\tilde{k}^+ - 1)$ gives $\Pr\big[\mathsf{N} > \tilde{k}^+ - 1\big] \ge \tau^-(\tilde{k}^+ - 1) > \varepsilon^+$ (in log space $\Pr\big[\mathsf{N} > \tilde{k}^+ - 1\big] \ge e^{\Lambda(\tilde{k}^+ - 1)} > e^{\lambda^+} \ge \varepsilon^+$), so $\tilde{k}^+ - 1$ is not in the set of (8.2). That set is an upset of $\{0, \dots, n\}$, because $k \mapsto \Pr[\mathsf{N} > k]$ is non-increasing (Lemma 7.2 (ii)); hence its minimum κ^+ exceeds $\tilde{k}^+ - 1$, i.e. $\kappa^+ \ge \tilde{k}^+$, and with (i) equality. The lower side is the mirror.

(iii). Since $\tilde{k}^+ > \kappa^+ \ge 0$, the index $\tilde{k}^+ - 1$ exists and is not certified admissible, so $\tau^+(\tilde{k}^+ - 1) > \varepsilon^+$ (in log space $U(\tilde{k}^+ - 1) > \lambda^-$), and by the definition of w,

$$\ln \Pr\big[\mathsf{N} > \tilde{k}^+ - 1\big] \;=\; \ln \tau^+(\tilde{k}^+ - 1) - w \;>\; \ln \varepsilon^+ - w$$

(in log space $\ln \Pr\big[\mathsf{N} > \tilde{k}^+ - 1\big] = U(\tilde{k}^+ - 1) - \big(U(\tilde{k}^+ - 1) - \ln \Pr\big[\mathsf{N} > \tilde{k}^+ - 1\big]\big) > \lambda^- - \big(w - (\ln \varepsilon^+ - \lambda^-)\big) = \ln \varepsilon^+ - w$). On the other hand $\tilde{k}^+ - 1 \ge \kappa^+$, and for every l with $\kappa^+ + 1 \le l \le \tilde{k}^+ - 1$ one has $r_l \le r_{\kappa^+ + 1} < 1$ by Lemma 15.2 (i), so (15.8) gives $\Pr[\mathsf{N} > l] \le r_l \Pr[\mathsf{N} > l - 1] \le r_{\kappa^+ + 1} \Pr[\mathsf{N} > l - 1]$; iterating from $l = \kappa^+ + 1$ to $l = \tilde{k}^+ - 1$,

$$\Pr\big[\mathsf{N} > \tilde{k}^+ - 1\big] \;\le\; r_{\kappa^+ + 1}^{\tilde{k}^+ - 1 - \kappa^+} \Pr\big[\mathsf{N} > \kappa^+\big] \;\le\; r_{\kappa^+ + 1}^{\tilde{k}^+ - 1 - \kappa^+} \, \varepsilon^+,$$

the last step by (8.3). Taking logarithms and combining with the previous display, $\ln \varepsilon^+ - w < \ln \varepsilon^+ + (\tilde{k}^+ - 1 - \kappa^+) \ln r_{\kappa^+ + 1}$, i.e. $(\tilde{k}^+ - 1 - \kappa^+)(-\ln r_{\kappa^+ + 1}) < w$, which is (15.10) since $-\ln r_{\kappa^+ + 1}$ is positive and finite: $r_{\kappa^+ + 1} < 1$ by hypothesis, and $r_{\kappa^+ + 1} > 0$ by Lemma 15.2 (i) because $\kappa^+ + 1 \le n - 1$. The mirror statement is the same argument for the upper tail of N' at level ε^-, translated by $k \mapsto n - k$: the exact lower quantile of N is n minus the exact upper quantile of N', because $\Pr[\mathsf{N} < k] = \Pr[\mathsf{N}' > n - k]$ turns the maximum in (8.1) into the minimum in (8.2) for N'.

The closing sentences of (iii). By (7.2) the guarded upper endpoint is $\max\{\tilde{k}^+, \lceil np \rceil\}$ for this evaluation and $\max\{\kappa^+, \lceil np \rceil\}$ for the exact rule, and by (9.1) the box bound is the guarded endpoint divided by n. Where the bracket is decided, (ii) gives $\tilde{k}^+ = \kappa^+$, so the two guarded endpoints and the two bounds coincide. Otherwise, the map $x \mapsto \max\{x, \lceil np \rceil\}$ is non-decreasing and changes by at most $x' - x$ between $x \le x'$, so with $\kappa^+ \le \tilde{k}^+$ from (i) the guarded endpoint of this evaluation exceeds the exact rule's by at least 0 and at most $\tilde{k}^+ - \kappa^+$, and dividing by n gives the bound on q^+; (15.10) bounds $\tilde{k}^+ - \kappa^+$ under the hypotheses of (iii). At $\kappa^+ = n - 1$, $\tilde{k}^+ \le n$ gives $\tilde{k}^+ - \kappa^+ \le 1$. The lower side is the mirror, with $x \mapsto \min\{x, \lfloor np \rfloor\}$ and $\tilde{k}^- \le \kappa^-$. □

Proposition 15.3 holds for every search, window and enclosure, so these are choices outside the proposition. A bisection finds a bracketed endpoint in about $\log_2 n$ evaluations, and the window trades evaluation time for the last counts of the endpoint.

Remark 15.4 (Evaluating the exact rule locally). Proposition 15.3 holds for every bracketed endpoint, whatever procedure found it, and for every enclosure of the form (15.9). Four choices therefore lie outside the proposition: the search, the window, the arithmetic substrate and the report of the bracket verdict. Requirements 17.1, 17.2 and 17.3 fix the engine's search, substrate and report.

The search. Certified admissibility C^+ is a predicate on $\{0,\dots,n\}$ that holds at n; a bisection that keeps C^+ true at its upper end and false at its lower end (starting from n and from the fictitious index -1), treating an undecided index as not certified, returns a bracketed upper endpoint after about $\log_2 n$ evaluations of the enclosure. A downward scan from any index known to be certified is the alternative; the endpoint of the Bernstein rule (Proposition 14.2) is admissible and usually certified, and is a natural start. Fixing the procedure, with its parameters, makes the evaluation a rule in the sense of Definition 7.1; Requirement 17.1 fixes it. Where the map $k \mapsto \tau^+(k)$ is non-increasing (which it is up to the rounding of the enclosure, since the tail itself decreases by the factor r_k per count, Lemma 15.2 (iv), and the enclosure's width varies slowly), every bracketed endpoint is the least certified index, and the evaluation is then monotone in the allocation as well (the set of certified indices grows with ε^+ when the logarithm of ε^+ is enclosed monotonically, as the fixed-grid root bounds of Proposition 14.2 (ii) are); the claims do not rest on this.

The window. In (15.7) the bracket's width is $\ln(1 + B_T/A_T)$, and B_T/A_T is about $r^T/(1-r)$ with r the ratio near k; at a deviation of t counts above the mean, $1 - r_k$ is of the order of $t/(np(1-p))$, so a window of T of the order of $(\ln \frac{1}{1-r} + c)/(1-r)$ terms leaves a relative slack of about e^{-c} in the tail, that is, about e^{-c} nats in w. Together with the per-count decay $-\ln r_{\kappa^+ +1}$, which is of the same order $t/(np(1-p))$, (15.10) says what the last counts cost in evaluation time: the window trades time for the last counts, and the exact quantile is reached exactly when the bracket is decided.

The substrate. The Stirling remainders and the root bounds are rationals of the size of a logarithm, while the products $\prod r_l$ inside the rationals A_T and B_T of (15.6), held as exact rationals, would have denominators that grow with the window. Every bound is used in the direction a conservative evaluation (Definition 4.3) admits: a tail comparison is decided against a bound that rounds the tail upward. Requirement 17.2 states the arithmetic of the evaluation.

The verdict. Whether the bracket was decided is Proposition 15.3 (ii)'s condition. The bound (15.10) is not computable as it stands: w involves the unknown $\ln \Pr\left[\mathsf{N} > \tilde{k}^+ - 1\right]$ and $r_{\kappa^+ +1}$ the unknown κ^+. With a certified-inadmissible index k_{D} below the bracket in hand and $0 < r_{k_{\mathsf{D}}+2} < 1$, and the enclosure in log space, a computable bound on the excess $\tilde{k}^+ - \kappa^+$ is (15.10) with $-\ln r_{k_{\mathsf{D}}+2}$ in place of $-\ln r_{\kappa^+ +1}$ and $W \coloneqq U(\tilde{k}^+ - 1) - \Lambda(\tilde{k}^+ - 1) + (\lambda^+ - \lambda^-)$ in place of w. Indeed $W \geq w$, because $\Lambda(\tilde{k}^+ - 1) \leq \ln \Pr\left[\mathsf{N} > \tilde{k}^+ - 1\right]$ and $\lambda^+ \geq \ln \varepsilon^+$; and since k_{D} is certified inadmissible, $\kappa^+ \geq k_{\mathsf{D}} + 1$, so the decreasing ratios give $r_{\kappa^+ +1} \leq r_{k_{\mathsf{D}}+2} < 1$ and $-\ln r_{\kappa^+ +1} \geq -\ln r_{k_{\mathsf{D}}+2} > 0$. Where $\tilde{k}^+ > \kappa^+$ and $\kappa^+ \leq n - 2$, the hypotheses of Proposition 15.3 (iii) then hold and the substituted bound is at least its right-hand side; where $\tilde{k}^+ > \kappa^+ = n - 1$, the excess is 1, below the substituted bound

because $W \geq w > 0$, the certificate at $\tilde{k}^+ - 1 \geq \kappa^+$ having failed although that tail is at most ε^+; and where $\tilde{k}^+ = \kappa^+$ the excess is 0. Requirement 17.3 says what is recorded. Completeness on this evaluation is the figure of Remark 14.3 (b), from the upper enclosures τ^+ at the emitted guarded endpoints, clamped by $\varepsilon_{\text{comp}}$ (Requirement 16.4); on the sweep evaluation no bracket is formed: the exact rule is evaluated by the closed forms of Definition 8.1, its tail comparisons decided exactly or against the upper bounds of Proposition 8.2 (vii), and none of this arises.

Uses Def. 4.3, Def. 7.1, Def. 8.1, Lem. 13.1, Lem. 13.2, Rem. 14.3, Lem. 15.1, Lem. 15.2, Prop. 15.3 · *Used by* Req. 17.1, Req. 17.2 · newly derived · accepted

16 Requirements: the request, its refusal and the reports

The engine builds the box with the endpoint rule and tail evaluation the request selects, and takes its defaults where the request is silent. Before constructing anything it refuses an input that violates a clause its own inputs decide, reports the completeness figure its route names and never a downward-rounded one, and records beside it the input, the rule and the evaluation it was computed under.

The engine builds the box with the endpoint rule and tail evaluation the request selects: the exact rule evaluated by its closed forms, which is the default, the enclosure evaluation of the exact rule, or the Bernstein rule. It runs no other rule.

Each rule of this submodule is an admissible endpoint rule, so every result stated for all such rules holds for the box the engine builds, whichever the request chooses; a result that needs more of the rule, such as monotonicity in the allocation, holds on the rules whose propositions prove it.

Requirement 16.1 (The rule the request selects). The engine builds the box of Definition 9.1 with the rule that the request's `endpoint_rule` and `tail_evaluation` select, and with no other: `endpoint_rule:exact` with `tail_evaluation:sweep`, the exact binomial quantile rule of Definition 8.1 with its tail comparisons decided on the closed forms, exactly or against upper bounds on the tails as Proposition 8.2 (vii) admits, which is the default; `endpoint_rule:exact` with `tail_evaluation:enclosure`, the enclosure evaluation R_{encl} of the exact rule (Proposition 15.3), an admissible relaxation of it; and `endpoint_rule:bernstein`, the Bernstein rule of Proposition 14.2, on which `tail_evaluation` selects only how the completeness figure is computed (Requirement 16.4).

Motivated by Def. 7.1, Def. 8.1, Prop. 8.2, Def. 9.1, Prop. 14.2, Prop. 15.3, Req. 16.4 · implemented (effective: accepted) · *Code* certiqs_key.statistics.rules:resolve_rule

Where a request names no constrained outcomes the engine constrains every outcome, and where it names no allocation it takes the uniform rule, which gives each tail of each constrained outcome the budget divided by twice the number of constrained outcomes.

No choice of the constrained outcomes dominates (Remark 9.3), and the uniform rule is a monotone allocation rule (Definition 12.1); the engine fixes both only as defaults.

REQUIREMENT 16.2 (Defaults for the constrained outcomes and the allocation). Where a request omits the constrained outcomes, the engine takes $\widehat{\mathcal{C}}_{\mathrm{con}} = \widehat{\mathcal{C}}$ (`constrained_outcomes`); where it omits the allocation, the engine takes the uniform rule of Definition 12.1 at the request's budget, $\varepsilon^{\pm}(\hat{c}) = \varepsilon_{\mathrm{comp}}/(2|\widehat{\mathcal{C}}_{\mathrm{con}}|)$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$ (`budget_allocation`).

Motivated by Rem. 9.3, Def. 12.1 · implemented (effective: accepted) · *Code* certiqs_key.statistics.construction:resolve

Before constructing anything the engine decides every condition of an admissible input its inputs determine; on a violation it reports the first by its code and emits no acceptance set, repairing nothing. Data not of the input's form, with the request defaults in place, are refused too.

By Theorem 11.1 (iii) the construction never fails on an admissible input, so every refusal comes from the input; this is Requirement "Exact decision by every consumer" (foundations/req:consumer-decides-exactly) for the module's own inputs. The clause relating the alphabet and the round count to the consuming declaration is not among those decided here: it is checked in bundle validation (`spec/architecture.md#36-bundle-validation-is-input-validation-not-a-module`).

REQUIREMENT 16.3 (Refusal of an input that violates a decidable clause). Before it constructs anything, the engine decides exactly every condition of an admissible input (Definition 7.1) that its own inputs determine on data of the form that definition fixes. On a violation it emits `infeasibility_report` naming the violated condition by its code, the first violation in the order the interface fixes, and it emits no `acceptance_set`; it repairs no input and puts no default in place of a violating value. The codes of the violations (1)–(11) of Theorem 11.1 (iii) are, in that order, `alphabet_empty`, `alphabet_not_finite`, `rounds_out_of_range`, `honest_not_a_distribution`, `honest_envelope_not_admissible`, `budget_out_of_range`, `allocation_negative`, `allocation_exceeds_budget`, `selector_not_a_subset`, `inputs_not_rational` and `endpoint_rule_unknown`. Data that, with the defaults of Requirements 16.2 and 16.1 in place, do not have that form are refused as well: the engine emits no `acceptance_set` for them and substitutes nothing for them.

Motivated by Def. 7.1, Thm. 11.1, Req. 16.1, Req. 16.2, foundations/req:consumer-decides-exactly · implemented (effective: accepted) · *Code* certiqs_key.statistics.construction:validate_request

The engine reports ε^* itself on the exact rule evaluated by its closed forms, and otherwise the figure its route's row names: the allocation sum or an upward-rounded evaluation of the tails clamped at the budget. It never reports a downward-rounded figure.

A completeness figure bounds how often an honest run aborts, so it may be loose but never low: it is rounded in the direction in which Requirement "Rounding direction of emitted bounds" (foundations/req:rounding-direction) rounds an emitted upper bound.

REQUIREMENT 16.4 (The reported completeness figure). The engine reports under `completeness_certified` the following figure. On `endpoint_rule:exact` with `tail_evaluation:sweep` it reports $\varepsilon^\star$ of Theorem 10.2 itself, and where it cannot evaluate a tail exactly it rounds the tail upward and reports $\min\{\varepsilon_{\text{comp}}, \tilde{\varepsilon}\}$ for the resulting $\tilde{\varepsilon} \geq \varepsilon^\star$ (Remark 10.3). On every other route it reports the admissible figure of Remark 14.3 that the route's row of `completeness_report` names: the allocation sum, or $\min\{\varepsilon_{\text{comp}}, \tilde{\varepsilon}\}$ for an upward-rounded evaluation $\tilde{\varepsilon} \geq \varepsilon^\star$ of the tails at the emitted endpoints. It never reports a downward-rounded figure.

Motivated by Thm. 10.2, Rem. 10.3, Rem. 14.3, foundations/req:rounding-direction · implemented (effective: accepted) · *Code* certiqs_key.statistics.construction:report_completeness

With every completeness figure it reports, the engine records what the figure was computed from: the honest model, n, the constrained outcomes, the allocation, the endpoint rule, the tail evaluation and the parameters in force. A completeness figure is meaningful only relative to that record.

The bound of Theorem 10.2 holds for every admissible endpoint rule, but the figure reported for it depends on the input, on the rule and on how its tails were evaluated (Remark 14.3).

REQUIREMENT 16.5 (The honest-model record beside every completeness figure). With every completeness figure it reports, the engine records, in the record `honest_model`, the input and the evaluation under which the figure was computed: the honest model (the law in the point form, or the honest envelope), the round count n, the constrained outcomes $\widehat{\mathcal{C}}_{\text{con}}$ and the allocation; the endpoint rule and the tail evaluation; and the parameters in force, for the Bernstein rule its j and s (Proposition 14.2) and, wherever the tails are evaluated by enclosures, the parameters of those enclosures (Proposition 15.3). A completeness figure is meaningful only relative to that record.

Motivated by Thm. 10.2, Prop. 14.2, Rem. 14.3, Prop. 15.3 · implemented (effective: accepted) · *Code* certiqs_key.statistics.construction:HonestModel

17 Requirements: the enclosure evaluation

On the enclosure evaluation of the exact rule the engine finds every bracketed endpoint by one fixed, total and deterministic search, and computes in log space with directed rounding, never forming a number of the size of a mass point. It records, per constrained outcome and side, whether the bracket was decided.

On the enclosure evaluation of the exact rule the engine finds every bracketed endpoint by one fixed, total and deterministic search with fixed parameters, the procedure the interface records. Fixing it is what makes the evaluation an endpoint rule.

Proposition 15.3 holds for every bracketed endpoint, whatever procedure found it; fixing the procedure, with its parameters, is what makes the evaluation a rule in the sense of Definition 7.1 (Remark 15.4).

REQUIREMENT 17.1 (One fixed search for the enclosure evaluation). On the enclosure evaluation of the exact rule, the engine finds every bracketed endpoint of Proposition 15.3, upper and lower, by one fixed search procedure, total and deterministic, with fixed parameters: the procedure `search_procedure` records.

Motivated by Def. 7.1, Prop. 15.3, Rem. 15.4 · implemented (effective: accepted) · *Code* certiqs_key.statistics.enclosure:bracketed_upper_endpoint

The enclosure evaluation computes in log space with directed rounding: the window's products are dyadic numbers rounded in the direction each bound requires, and the only numbers formed have the magnitude of logarithms. No mass point, power of p or binomial coefficient is ever formed.

A tail comparison of the enclosure evaluation is decided against a bound that rounds the tail upward, which is what a conservative evaluation (Definition 4.3) admits; exact rationals for the window would have denominators that grow with the window (Remark 15.4).

REQUIREMENT 17.2 (Log-space arithmetic with directed rounding). On the enclosure evaluation the engine computes everything in log space with directed rounding: the mass point through (15.2), the window through the rationals A_T and B_T of (15.6), whose logarithms Lemma 13.1 encloses, and the budget through the same lemma. It holds the products $\prod r_l$ inside A_T and B_T as dyadic numbers with an explicit binary exponent, rounded in the direction each bound requires (down for A_T in the lower bound, up for $A_T + B_T$ in the upper), not as exact rationals, and the Stirling remainders and the root bounds as rationals of the size of a logarithm; the only numbers it ever forms have the magnitude of logarithms.

Motivated by Def. 4.3, Lem. 13.1, Lem. 15.1, Lem. 15.2, Rem. 15.4 · implemented (effective: accepted) · *Code* certiqs_key.statistics.dyadic:round_dyadic

On the enclosure evaluation the engine records, per constrained outcome and side, whether the bracket was decided, that is whether the emitted endpoint is proved equal to the exact quantile. It may add the bound on the conservatism in counts.

The enclosure evaluation may emit an endpoint more conservative than the exact quantile, and nothing in the emitted box shows it; the bracket verdict of Proposition 15.3 (ii) does.

REQUIREMENT 17.3 (The bracket verdict per outcome and side). On the enclosure evaluation of the exact rule the engine records in the diagnostic `endpoint_exact`, per constrained outcome and per side, whether the bracket was decided, that is whether the emitted endpoint is proved equal to the exact quantile by Proposition 15.3 (ii). When it holds a certified-inadmissible index k_{D} below the bracket with $0 < r_{k_{\mathsf{D}}+2} < 1$, it may also report the bound (15.10) in counts with $-\ln r_{k_{\mathsf{D}}+2}$ in place of $-\ln r_{\kappa^+ +1}$ and the computable upper bound $U(\tilde{k}^+ - 1) - \Lambda(k^+ - 1) + (\lambda^+ - \lambda^-)$ on w in place of w (Remark 15.4).

Motivated by Prop. 15.3, Rem. 15.4 · implemented (effective: accepted) · *Code* certiqs_key.statistics.construction:EndpointExact

Part IV
Singleton acceptance set

This submodule treats the acceptance test that accepts exactly one declared frequency vector. Its main result is that under independent honest rounds with one law the test's accept probability is the multinomial probability of that vector (Theorem 18.1). A vector off the count grid, or one that puts weight on an outcome the honest law never produces, is never accepted, and then no completeness budget below one exists; otherwise the least budget is one minus the accept probability. That budget is zero only when the honest law and the vector are the same point mass, and at any realistic alphabet and round count it is close to one, which is why the construction map never emits a singleton. An instance that declares one records its least budget, or that none exists, and the singleton reads the honest model in point form only.

18 Completeness of a singleton acceptance set

A test that accepts exactly one frequency vector is far more restrictive than a box around the honest law. The main result, Theorem 18.1, computes its accept probability from the multinomial law, says when no completeness budget below one exists and when the least budget is zero; the remark explains why the construction map never emits a singleton.

Under i.i.d. honest rounds a singleton acceptance set accepts with its point's multinomial probability. That is 0, with no completeness budget below one, if n times a coordinate is no integer or the point leaves the law's support; else the least budget is one minus it, zero only for one point mass.

A test that accepts exactly one frequency vector is far more restrictive than a box around the honest law. Under independent honest rounds with one law its accept probability is a single multinomial mass, so its completeness can be computed exactly; an instance that declares such a set records the least budget of (ii), or that none exists (Requirement 19.1).

Theorem 18.1 (Completeness of a singleton acceptance set). Assume A-HON, let $\widehat{\mathcal{C}}$ and n be a matched alphabet and round count (Definition 1.3), and let $\mathcal{S}_{\mathrm{acc}} = \{\mathbf{q}^*\}$ be a singleton geometry with $\mathbf{q}^* \in \Delta(\widehat{\mathcal{C}})$. Call a completeness budget $\varepsilon_{\mathrm{comp}}$ *admissible for this geometry* when $\varepsilon_{\mathrm{comp}} \in [0, 1)$ and $\Pr\left[\mathrm{freq}_{\widehat{C}_1^n} \notin \mathcal{S}_{\mathrm{acc}}\right] \le \varepsilon_{\mathrm{comp}}$.

(i) *(exact accept probability)* With the convention $0^0 = 1$,

$$\Pr\left[\,\mathrm{freq}_{\widehat{C}_1^n} \in \mathcal{S}_{\mathrm{acc}}\right] = \begin{cases} \dfrac{n!}{\prod_{\hat{c}\in\widehat{\mathcal{C}}} \left(n\, q^*(\hat{c})\right)!} \displaystyle\prod_{\hat{c}\in\widehat{\mathcal{C}}} p^{\mathrm{hon}}(\hat{c})^{\,n\, q^*(\hat{c})}, & \text{if } n\, q^*(\hat{c}) \in \mathbb{Z}_{\geq 0}\ \forall \hat{c}, \\ 0, & \text{otherwise;} \end{cases} \tag{18.1}$$

(ii) *(the least admissible budget, and when there is none)* the accept probability (18.1) is 0 if and only if either $n\, q^*(\hat{c}) \notin \mathbb{Z}_{\geq 0}$ for some $\hat{c}$, or $\mathbf{q}^* \not\ll p^{\mathrm{hon}}$, that is, some $\hat{c}$ has $q^*(\hat{c}) > 0$ and $p^{\mathrm{hon}}(\hat{c}) = 0$. In that case *no* $\varepsilon_{\mathrm{comp}} \in [0, 1)$ is admissible for this geometry, and no completeness statement whatsoever can be made about it; a budget of 1 lies outside $[0, 1)$. Otherwise the accept probability is strictly positive and the least admissible completeness budget is

$$\varepsilon_{\mathrm{comp}} = 1 - \Pr\left[\,\mathrm{freq}_{\widehat{C}_1^n} \in \mathcal{S}_{\mathrm{acc}}\right] \ \in [0, 1), \tag{18.2}$$

which does lie in $[0, 1)$;

(iii) *(when the budget is zero)* the budget (18.2) equals 0 if and only if p^{hon} is a point mass $\delta_{\hat{c}_0}$ for some $\hat{c}_0 \in \widehat{\mathcal{C}}$ and $\mathbf{q}^* = \delta_{\hat{c}_0}$; in particular it equals 0 whenever $|\widehat{\mathcal{C}}| = 1$, and it is strictly positive whenever p^{hon} gives positive mass to two distinct outcomes.

Uses Def. 1.1, Def. 1.3 · *Assumes* A-HON, A-ALPHABET · *Used by* Rem. 18.2, Req. 19.1, Req. 19.2 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.statistics.singleton: singleton_completeness · *Oracles* registry_statistics_acceptance.py

Proof. (i). By Definition 1.1, $\mathrm{freq}_{\widehat{C}_1^n} = \mathbf{q}^*$ holds exactly when $\mathsf{N}_{\hat{c}} = n\, q^*(\hat{c})$ for every $\hat{c}$, which is impossible unless every $n\, q^*(\hat{c})$ is a non-negative integer; that gives the second branch. In the first branch write $n_{\hat{c}} \coloneqq n\, q^*(\hat{c})$, so $\sum_{\hat{c}} n_{\hat{c}} = n$. Under A-HON each string $\hat{c}_1^n$ has probability $\prod_{\hat{c}} p^{\mathrm{hon}}(\hat{c})^{\,n_{\hat{c}}}$ whenever its count vector is $(n_{\hat{c}})_{\hat{c}}$, and the number of such strings is the multinomial coefficient $n!/\prod_{\hat{c}} n_{\hat{c}}!$; summing over them gives (18.1).

(ii). If some $n\, q^*(\hat{c})$ is not a non-negative integer the second branch of (18.1) gives probability 0. Otherwise the first branch applies; its multinomial coefficient is a positive integer, and under the convention $0^0 = 1$ each factor $p^{\mathrm{hon}}(\hat{c})^{\,n_{\hat{c}}}$ with $n_{\hat{c}} = 0$ equals 1. The product therefore vanishes if and only if some $\hat{c}$ has $n_{\hat{c}} \geq 1$ and $p^{\mathrm{hon}}(\hat{c}) = 0$; and $n_{\hat{c}} \geq 1$ is equivalent to $q^*(\hat{c}) > 0$, so this says precisely that $\mathbf{q}^* \not\ll p^{\mathrm{hon}}$. That establishes the stated characterization of a zero accept probability.

By the definition in the statement, a budget $\varepsilon_{\mathrm{comp}}$ is admissible for this geometry exactly when it satisfies both the completeness requirement $\Pr\left[\mathrm{freq}_{\widehat{C}_1^n} \notin \mathcal{S}_{\mathrm{acc}}\right] \leq \varepsilon_{\mathrm{comp}}$, that is, $\varepsilon_{\mathrm{comp}} \geq 1 - \Pr\left[\mathrm{freq}_{\widehat{C}_1^n} \in \mathcal{S}_{\mathrm{acc}}\right]$, and the domain $\varepsilon_{\mathrm{comp}} \in [0, 1)$ of the completeness budget. The admissible budgets

are therefore the interval $[\,1 - \Pr\left[\mathrm{freq}_{\widehat{C}_1^n} \in \mathcal{S}_{\mathrm{acc}}\right], 1)$, which is non-empty if and only if $\Pr\left[\mathrm{freq}_{\widehat{C}_1^n} \in \mathcal{S}_{\mathrm{acc}}\right] > 0$, and which then has least element (18.2). When the accept probability is 0 the interval is empty: there is no admissible budget, and in particular $\varepsilon_{\mathrm{comp}} = 1$ is not one.

(iii). The budget (18.2) is 0 exactly when the accept probability is 1. That probability equals 1 if and only if $\mathrm{freq}_{\widehat{C}_1^n} = \mathbf{q}^*$ almost surely. Suppose p^{hon} gives positive mass to two distinct outcomes $\hat{c} \neq \hat{c}'$. The string $(\hat{c}, \hat{c}, \ldots, \hat{c})$ and the string $(\hat{c}', \hat{c}, \ldots, \hat{c})$ both have positive probability under A-HON, and their frequency vectors differ (they differ by $1/n$ in coordinate $\hat{c}$), so $\mathrm{freq}_{\widehat{C}_1^n}$ is not almost surely constant and the probability is strictly less than 1. Hence p^{hon} must be a point mass $\delta_{\hat{c}_0}$; then $\mathrm{freq}_{\widehat{C}_1^n} = \delta_{\hat{c}_0}$ almost surely and the probability is 1 exactly when $\mathbf{q}^* = \delta_{\hat{c}_0}$. If $|\widehat{\mathcal{C}}| = 1$ the only distribution on $\widehat{\mathcal{C}}$ is that point mass, and $\mathbf{q}^*$ equals it. □

The consuming engine certifies a singleton's acceptance-side minimum, but its completeness figure 1 − Pr[freq = q] is close to one at realistic alphabets and round counts, and no budget exists whenever n q* leaves the count grid. Its accept probability needs one honest law.*

Remark 18.2 (Certifiability and completeness of the singleton). The consuming engine certifies the acceptance-side minimum of a singleton geometry (Remark "Exact minimisation on the acceptance side" (meat-core/frank-wolfe/rem:acc-side) (a)), so a singleton acceptance set is certifiable. By Theorem 18.1 its completeness figure is $1 - \Pr[\mathrm{freq} = \mathbf{q}^*]$, which is close to 1 for any realistic alphabet and round count; that is why the construction map never emits it. The degenerate case of Theorem 18.1 (ii), where the accept probability is exactly 0 and *no* admissible completeness budget exists, is not exotic: it is reached whenever $n\,\mathbf{q}^*$ has a non-integer coordinate, which for a fixed $\mathbf{q}^*$ happens at most round counts. Requirement 19.1 says what an instance that declares a singleton set records.

A singleton is also the $n = 2$ output of the counterexample in Theorem 12.4 (v): the construction can produce a set with a single point, and when it does, (10.6) still bounds the honest failure probability. The two statements agree, because there the budget $\varepsilon_{\mathrm{comp}} = \frac{4}{5}$ was large enough to pay for it.

Unlike the box construction, the singleton reads its honest input in the point form only: Theorem 18.1 assumes A-HON, one law p^{hon} for independent rounds, because its accept probability (18.1) is the exact multinomial probability under that one law and has no counterpart for an envelope of laws. A single law is the degenerate envelope (Definition 4.1); Requirement 19.2 says how an envelope is read.

Uses Def. 4.1, Thm. 10.2, Thm. 12.4, Thm. 18.1 · newly derived · accepted

19 Requirements

An instance that declares a singleton acceptance set records its least completeness budget, or that no admissible budget exists, and claims no com-

pleteness the set does not have. The singleton reads the honest model in point form only: a degenerate envelope is read as its law, and a non-degenerate one is refused.

An instance that declares a singleton acceptance set records the least completeness budget of Theorem 18.1, or, where the honest run can never meet the set, that no admissible budget exists and never a budget of one. It claims no completeness the set does not have.

The construction map never emits a singleton (Remark 18.2), but an instance may declare one, and Theorem 18.1 gives its completeness exactly.

REQUIREMENT 19.1 (The completeness budget of a declared singleton). An instance that declares a singleton acceptance set $\{\mathbf{q}^*\}$ records the completeness budget $\varepsilon_{\text{comp}}$ of (18.2), or, in the degenerate case of Theorem 18.1 (ii) where the accept probability is exactly 0, records that *no* admissible completeness budget exists rather than a budget of 1, which is the honest reading of a geometry the honest run can never satisfy. In either case it claims no completeness the set does not have.

Motivated by Thm. 18.1, Rem. 18.2 · accepted

The singleton reads the honest model in the point form only: a degenerate envelope is read as its law, and a non-degenerate envelope is refused rather than reduced to one of its laws.

The accept probability of Theorem 18.1 is a multinomial mass under one law and has no counterpart for an envelope of laws (Remark 18.2).

REQUIREMENT 19.2 (Point-form honest input for the singleton). The singleton takes its honest input in the point form only: the engine reads a degenerate envelope as its law, and refuses a non-degenerate envelope with `honest_envelope_not_degenerate` rather than reducing it to one of its laws.

Motivated by Thm. 18.1,
Rem. 18.2 · accepted · *Code* certiqs_key.statistics.singleton:singleton_completeness

Interface

The declared inputs and outputs of the module and its submodules: each key with its symbol, when it is present and the claims that fix it. The domains, types and refusal rules are in the interface files themselves.

Inputs: Acceptance statistics (statistics)

From `interface.yaml`.

Key and symbol	Presence	Fixed by
`acceptance_alphabet` Chat	always	Def. 1.3, Req. 6.4, Def. 1.4, Thm. 12.4
`honest_distribution` p^hon	always, in this form or as 'honest_envelope' (exactly one of the two)	Def. 4.3, Ass. 5.1, Def. 4.1, Def. 7.1, Def. 9.1
`honest_envelope` (p^hon-, p^hon+)	instead of 'honest_distribution' (exactly one of the two)	Def. 4.1, Ass. 5.2, Def. 4.3, Def. 7.1, Def. 9.1, Thm. 11.1, Req. 16.3
`rounds` n	always	Def. 1.3, Ass. 5.3, Def. 1.1, Thm. 9.2
`completeness_budget` ecomp	always	Def. 8.1, Thm. 12.4, Thm. 18.1, Req. 19.1
`constrained_outcomes` Chat_con	always; DEFAULT Chat_con = Chat	Def. 9.1, Rem. 9.3, Req. 16.2, Thm. 12.4
`budget_allocation` (epsilon^-, epsilon^+)	always; DEFAULT the uniform rule epsilon^{+-}(chat) = ecomp / (2 |Chat_con|)	Def. 8.1, Def. 12.1, Req. 16.2, Thm. 12.4
`endpoint_rule` R	always; DEFAULT 'exact'	Def. 7.1, Def. 8.1, Prop. 8.2, Prop. 14.2, Req. 16.1, Thm. 11.1, Req. 16.3
`tail_evaluation`	always; DEFAULT 'sweep'	Prop. 15.3, Rem. 15.4, Prop. 8.2, Cor. 2.2, Lem. 2.3, Req. 16.1

Outputs: Acceptance statistics (statistics)

From `interface.yaml`.

Key and symbol	Presence	Fixed by
`acceptance_set` AcceptSet	on every run whose inputs lie in the input domain; absent – not empty, not defaulted – when ‘infeasibility_report‘ is non-null	Def. 1.4, Req. 6.1, Req. 6.2, Thm. 1.5, Def. 9.1, Thm. 9.2, Rem. 1.6
`acceptance_support` Chat_acc	always	Thm. 3.1, Req. 6.3, meat-core/frank-wolfe/def:first-order-setting
`completeness_certified` epsilon_star, or the upper bound on it the route reports	on every run that emits an acceptance_set	Thm. 10.2, Rem. 10.3, Rem. 14.3, Req. 16.4
`honest_model`	on every run that emits an acceptance_set	Req. 16.5, Ass. 5.1, Ass. 5.2, Lem. 4.2, Rem. 14.3, Rem. 15.4
`infeasibility_report`	always	Thm. 11.1, Req. 16.3, Thm. 1.5, Thm. 9.2
`endpoint_exact`	present exactly under ‘endpoint_rule = exact‘ with ‘tail_evaluation = enclosure‘ (binomial-box/req:endpoint-exact-record). Absent under ‘sweep‘, where no bracket is formed: the emitted endpoint is the exact quantile wherever the tails are evaluated exactly, and otherwise the endpoint of binomial-box/prop:rule-exact (vii); absent under ‘endpoint_rule = bernstein‘, where no bracket is formed.	Req. 17.3, Prop. 15.3, Rem. 15.4
`assumption_ledger` A-...	on every result that carries an acceptance_set; absent with it when ‘infeasibility_report‘ is non-null	Ass. 5.1, Ass. 5.2, Ass. 5.3, Ass. 5.4

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